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MAX-M10N

Standard precision GNSS module
Professional grade

Integration manual

Abstract

This document describes the features and application of the u-blox MAX-M10N module. The MAX-M10N module provides an ultra-low-power standard precision GNSS receiver for high-performance tracking applications with firmware upgradability.

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This document applies to the following products:

Product name	Type number	FW version	IN/PCN reference	RN reference
MAX-M10N	MAX-M10N-00B-00	EXT SPG 5.30	UBXDOC-304424225-20429	UBXDOC-304424225-20393
MAX-M10N	MAX-M10N-10B-00	EXT SPG 5.30	UBXDOC-304424225-20429	UBXDOC-304424225-20393

Contents

1 System description

This section gives an overview of the MAX-M10N receiver, and outlines the basics of operation with the receiver.

1.1 Overview

MAX-M10N features the u-blox M10 standard precision GNSS platform and provides exceptional sensitivity and acquisition time for L1 GNSS signals. MAX-M10N supports concurrent reception of three major GNSS constellations (GPS, Galileo, and BeiDou).

The firmware upgradeable MAX-M10N integrates a flash memory for the firmware image. The integrated flash memory also offers additional functionality including extended AssistNow™ data storage.

MAX-M10N grants lifetime access to AssistNow Live Orbits. This premium assistance service does not only offer improved time-to-first-fix (TTFF), but also improves position accuracy, especially in typical IoT applications with integrated antennas.

The u-blox low energy accurate position (LEAP) technology reduces the power consumption while retaining superior position accuracy.

The u-blox Super-Signal (Super-S) technology improves the dynamic position accuracy with small antennas.

The MAX-M10N product variants offer an optimized RF front-end either for the highest immunity or the best sensitivity in passive antenna designs. The MAX-M10N variants are summarized in Table 1.

Product name	RF front-end	Description
MAX-M10N-00B	LNA-SAW	Best sensitivity for designs with small antennas
MAX-M10N-10B	SAW-LNA-SAW	Highest immunity e.g. for designs with a cellular modem

Table 1: MAX-M10N product variants

MAX-M10N offers backwards pin-to-pin compatibility with products from the previous u-blox generations saving the designer's effort and reducing cost when upgrading designs to the advanced low-power u-blox M10 GNSS technology.

1.2 Architecture

The MAX-M10N receiver provides all the necessary RF and baseband processing to enable multi-constellation operation. The block diagram below shows the key functionality.

1.2.1 Block diagram

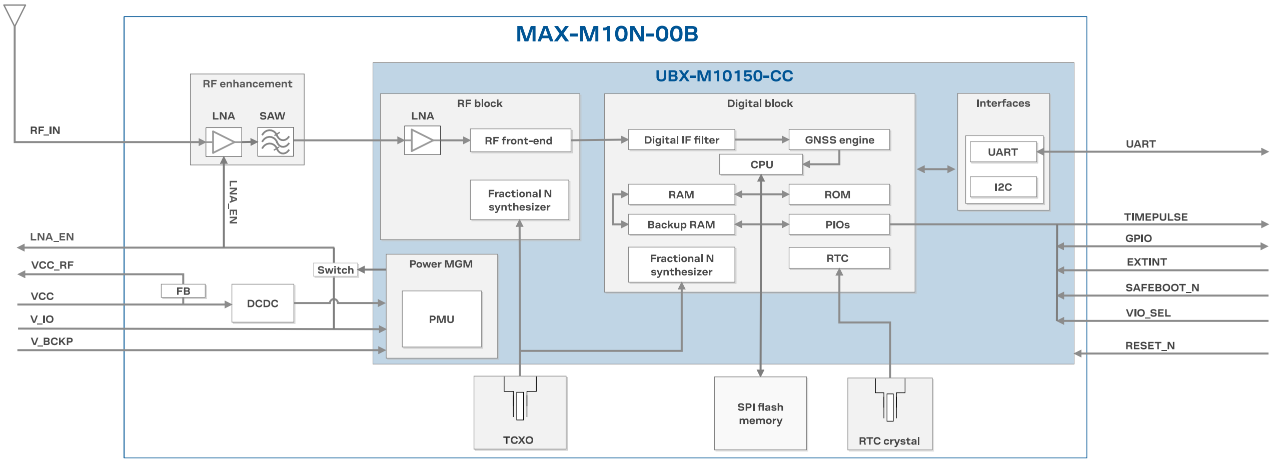


Figure 1: MAX-M10N-00B block diagram

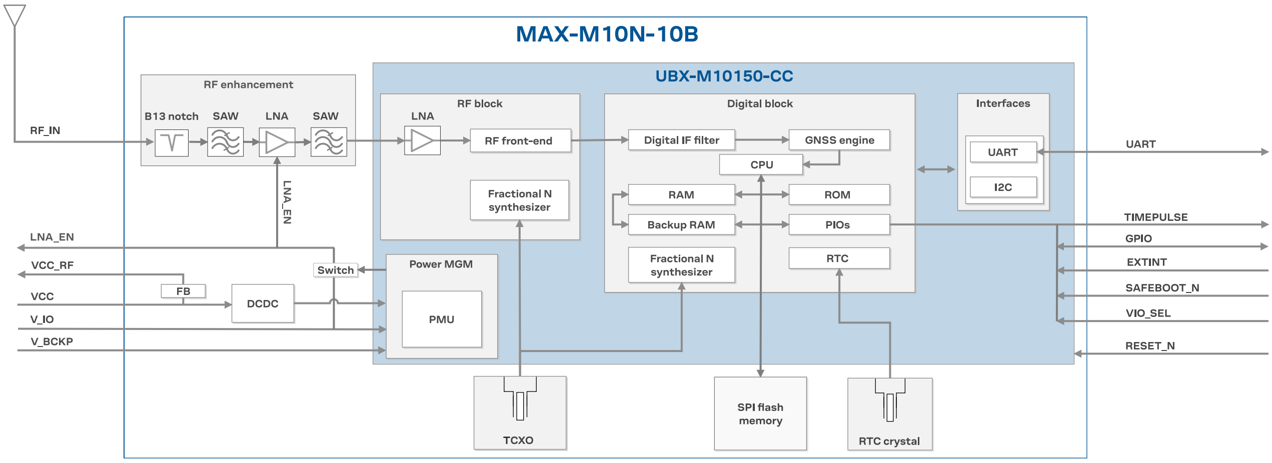


Figure 2: MAX-M10N-10B block diagram

1.3 Pin assignment

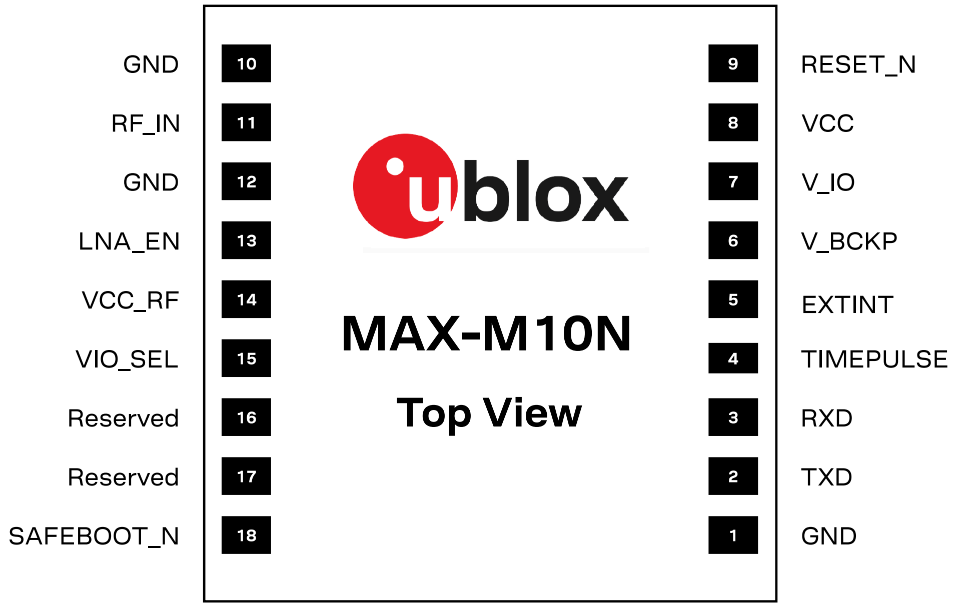


Figure 3: MAX-M10N pin assignment

Pin no.	Name	PIO no.	I/O	Description	Remarks
1	GND	-	-	-	Connect to GND
2	TXD	1	O	UART TX	If not used, leave open. Alternative functions[1].
3	RXD	0	I	UART RX	If not used, leave open. Alternative functions[1].
4	TIMEPULSE	4	O	Time pulse signal	See section TIMEPULSE for more information. Alternative functions[1].
5	EXTINT	5	I	External interrupt	See EXTINT for more information. Alternative functions[1].
6	V_BCKP	-	I	Backup voltage supply.	Leave open if no external backup supply. See V_BCKP for more information.
7	V_IO	-	I	IO voltage supply	See V_IO for more information.
8	VCC	-	I	Main voltage supply	See VCC for more information.
9	RESET_N	-	I	System reset (active low)	It has to be low for at least 1 ms to trigger a reset. Leave open if not used. See RESET_N section for more information.
10	GND	-	-	-	Connect to GND
11	RF_IN	-	I	GNSS signal input	The RF signal line is DC blocked internally. The line must match the 50 Ω impedance. See sections RF front-end and Layout for more information about the RF signal considerations.
12	GND	-	-	-	Connect to GND
13	LNA_EN	-	O	On/Off external LNA or active antenna	This pin cannot be used for another purpose as it also controls the internal LNA. See LNA_EN for more information.

1 Alternatively, this pin can be used for ANT_DETECT, ANT_SHORT_N, TX_READY, and Data batching. Care must be taken when the assigned function sets the pin as an output.

Pin no.	Name	PIO no.	I/O	Description	Remarks
14	VCC_RF	-	O	Output voltage RF section	This pin supplies a filtered voltage that can be used for optional external active antenna or LNA. This pin is internally connected to VCC through a ferrite bead.
15	VIO_SEL	-	I	Voltage selector for V_IO supply	Connect to GND for 1.8 V supply, or leave open for 3.3 V supply
16	Reserved			Not connected	
17	Reserved			Not connected	
18	SAFEBOOT_N	-	I	Safeboot mode	To enter safeboot mode, set this pin to low at receiver's startup. Otherwise, leave it open. The SAFEBOOT_N pin is internally connected to TIMEPULSE pin through a 1 kΩ series resistor.

Table 2: MAX-M10N pin assignment

2 Receiver configuration

The configuration determines all aspects of the GNSS receiver operation and therefore, information in this section is essential for the successful integration of MAX-M10N.

MAX-M10N is configured using UBX configuration interface keys. The configuration database in the receiver's RAM holds the current configuration, which is used by the receiver at runtime. It is constructed at the receiver startup from several sources of configuration. For more information on the receiver configuration, see the Interface description [3].

The configuration can be stored in the RAM, battery-backed RAM (BBR), and flash memory. It is recommended to store the configuration in the flash memory only one time during production, not at every start-up. Infrequent configuration changes on the run are permitted.

The permanence of the stored configuration and the actions to clear it in each memory are listed in Table 3.

Memory	Permanence of storage	Clearing actions
RAM	Settings remain effective until power-down	• Activating the RESET_N pin • A UBX-CFG-RST message excluding GNSS stop (resetMode 0x08) and GNSS start (resetMode 0x09) • Entering software standby mode • Using external control in power save mode (PSM) to enter the inactive state • Entering the off state of the PSM on/off operation (PSMOO) • Entering the inactive state of the PSM cyclic tracking operation (PSMCT)
BBR	The receiver retains the settings stored as long as the backup power supply remains	• Activating the RESET_N pin • A UBX-CFG-RST message with reset mode set to a hardware reset (resetMode 0x00 and 0x04)
Flash	Permanent storage of the configuration settings	• A UBX-CFG-CFG message to erase the configuration on the flash layer

Table 3: Permanence of storage and clearing actions for each memory

For more information about the UBX-CFG-RST message, refer to Forcing receiver reset.

⚠ **CAUTION** The configuration interface has changed from earlier u-blox positioning receivers. Users must adopt the configuration interface described in this document.

The configuration interface settings are stored in a database consisting of separate configuration items. An item is made up of a pair consisting of a key ID and a value. Related items are grouped together and identified under a common group name: CFG-GROUP-*; a convention used in u-center 2 and within this document. Within u-center 2, a configuration group is identified as "Group name" and the configuration item is identified as the "item name" in the "Device configuration" window.

The UBX messages available to change or poll the configurations are the UBX-CFG-VALSET, UBX-CFG-VALGET, and UBX-CFG-VALDEL messages. For more information about these messages and the configuration keys, see the configuration interface section in the Interface description [3].

2.1 Basic receiver configuration

This section summarizes the most commonly used, basic receiver configurations.

2.1.1 Basic hardware configuration

The MAX-M10N receiver is preconfigured in module production and is fully operational after connecting a proper power supply, the communication interfaces with the host application device, and a suitable antenna signal.

2.1.2 GNSS signal configuration

MAX-M10N supports GPS, Galileo, BeiDou, QZSS and SBAS GNSS constellations. The default configuration is concurrent reception of GPS, Galileo and BeiDou B1I with QZSS and SBAS enabled.

☞ BeiDou B1I signal cannot be used simultaneously with the BeiDou B1C.

GNSS constellations and signals can be configured using the CFG-SIGNAL-* configuration group. Each GNSS constellation can be enabled or disabled independently except for QZSS and SBAS, which are functional only with GPS. In addition to the configuration key for each constellation, there is a configuration key for each signal supported by the firmware.

For example, if CFG-SIGNAL-GPS_ENA is set to zero, all signals from the GPS constellation are disabled. Alternatively, if CFG-SIGNAL-GPS_L1CA_ENA is set to zero, only the GPS L1 C/A signal is disabled.

Unsupported combinations are rejected with a UBX-ACK-NAK message, and the warning "inv sig cfg" is sent via UBX-INF and NMEA-TXT messages (if enabled).

☞ Any change to the signal configuration items triggers a restart of the GNSS subsystem. During the restart, the host application should wait for message acknowledgement and a margin of 0.5 seconds prior to sending any further commands.

☞ To mitigate possible cross-correlation issues between the signals, it is recommended to enable also the QZSS L1C/A when the GPS L1C/A signal is enabled.

For more information on the CFG-SIGNAL-* configuration group, refer to the Interface description [3].

2.1.2.1 BeiDou B1I and B1C signals

BeiDou B1I and B1C signals differ in terms of the center frequency, bandwidth, and modulation. Therefore, there are differences in performance and supported GNSS signals and features.

Benefits of using BeiDou B1I signal:

BeiDou B1I offers superior GNSS performance. High start-up sensitivity and fast time-to-first-fix (TTFF) enable acquisition of a large number of BeiDou satellites. The tracking and reacquisition sensitivity for acquired signals is approximately at the same level for BeiDou B1I and B1C.

- **Faster TTFF and higher start-up sensitivity.** BeiDou B1I signals are acquired significantly faster and at a lower signal level than BeiDou B1C signals.
- **Better availability.** Higher start-up sensitivity results in a larger number of BeiDou satellites tracked and used in navigation solution especially at low signal level.
- **AssistNow support.** Enhanced start-up sensitivity and TTFF.

Benefits of using BeiDou B1C signal:

BeiDou B1C signal has the same center frequency as GPS L1 C/A. Multi-GNSS configurations with BeiDou B1C also have a lower power consumption compared to those with BeiDou B1I. Consequently, no additional frequency band is required for a multi-GNSS configuration, resulting in a lower power consumption during acquisition and tracking phases.

- **Lower power consumption.** No additional frequency band required for BeiDou B1C in multi-GNSS constellations resulting in a lower power consumption during acquisition and tracking phases.

2.1.3 Communication interface configuration

Several configuration groups allow configuring the operation mode of the communication interfaces. These include parameters for the data framing, transfer rate and enabled input/output protocols. See Communication interfaces and PIOs section for details. The configuration groups available for each interface are:

Interface	Configuration groups
UART	CFG-UART1-* CFG-UART1INPROT-* CFG-UART1OUTPROT-*

Table 4: Interface configuration

☞ The UART baudrate in MAX-M10N is configured to 9600 baud which is different to the firmware default. This ensures backwards compatibility with previous generations of u-blox MAX modules.

2.1.4 Message output configuration

The receiver supports two protocols for output messages: industry-standard NMEA and u-blox UBX. Any message type can be enabled or disabled individually and the output rate is configurable.

The message output rate is related to the frequency of an event. For example, the output message UBX-NAV-PVT (position, velocity, and time solution) is related to the navigation event, which generates a navigation epoch. In this case, the rate for each navigation epoch is defined by the configuration keys CFG-RATE-MEAS and CFG-RATE-NAV. For configuration examples of CFG-RATE-MEAS and CFG-RATE-NAV, see Table 5.

Set the navigation rate value higher than one when the raw measurement data output rate needs to be higher than the navigation data rate.

Configuration example	CFG-RATE-MEAS	CFG-RATE-NAV	Measurement interval	Navigation epoch interval	Description
Example 1	1000	1	1000 ms	1000 ms	Measurement every 1000 ms, navigation solution for each measurement.
Example 2	1000	2	1000 ms	2000 ms	Measurement every 1000 ms, navigation solution for every second measurement.
Example 3	2000	1	2000 ms	2000 ms	Measurement every 2000 ms, navigation solution for each measurement.

Configuration example	CFG-RATE-MEAS	CFG-RATE-NAV	Measurement interval	Navigation epoch interval	Description
Example 4	500	4	500 ms	2000 ms	Measurement every 500 ms, navigation solution for every fourth measurement.

Table 5: Measurement rate vs navigation rate configuration examples

☞ In the LEAP mode, CFG-RATE-NAV must be set to one 1.

The output rate for each message is defined in the CFG-MSGOUT-* configuration group. If the output rate of the message is set to one (1) on the UART interface, CFG-MSGOUT-UBX_NAV_PVT_UART1 = 1, the message is output for every navigation epoch. If the rate is set to two (2), the message is output for every other navigation epoch. If the rate is zero (0), then corresponding message is not output. As seen in this example, the rates of the output messages are individually configurable per communication interface.

Some messages, such as UBX-MON-VER, are non-periodic and are only output as an answer to a poll request.

The UBX-INF-* and NMEA-Standard-TXT information messages are non-periodic output messages that do not have a message rate configuration. Instead they can be enabled for each communication interface via the CFG-INFMSG-* configuration group.

☞ All message output is additionally subject to the protocol configuration of the communication interfaces. Messages of a given protocol are not output unless the protocol is enabled for output on the interface. See Communication interface configuration for details.

☞ The output rate of the NMEA-GxGSV message on the UART interface is configured to 5 in MAX-M10N. This differs from the firmware default to avoid overloading the communication interface and buffers.

2.1.5 Antenna supervisor configuration

This section gives an overview of the antenna supervisor configuration keys. The implementation of the antenna supervisor and a detailed description can be found in Antenna supervisor.

The antenna supervisor is used to control an active antenna. The configuration of the antenna supervisor allows the following:

- Control voltage supply to the antenna, which allows the antenna supervisor to cut power to the antenna in the event of a short circuit or optimize power to the antenna in power save modes.
- Detect a short circuit in the antenna and automatically recover the antenna supply after the short circuit is no longer present.
- Detect an open circuit, which can be used to indicate if the antenna has been disconnected.

☞ Using some antenna supervisor features may require disabling the UART or I2C interface and reconfiguring the PIOs as antenna supervisor pins.

Table 6 describes the configuration items.

Configuration item	Description	Comments
CFG-HW-ANT_CFG_VOLTCTRL	Enable active antenna voltage control	
CFG-HW-ANT_CFG_SHORTDET	Enable short circuit detection	
CFG-HW-ANT_CFG_SHORTDET_POL	Short antenna detection polarity	Set to 1 if the required logic polarity is active-low (default).
CFG-HW-ANT_CFG_OPENDET	Enable open circuit detection	

Configuration item	Description	Comments
CFG-HW-ANT_CFG_OPENDET_POL	Open antenna detection polarity	Set to 1 if the required logic polarity is active-low (default).
CFG-HW-ANT_CFG_PWRDOWN	Power down antenna supply if short circuit is detected	Requires CFG-HW-ANT_CFG_VOLTCTRL and CFG-HW-ANT_CFG_SHORTDET to be enabled.
CFG-HW-ANT_CFG_PWRDOWN_POL	Power down antenna logic polarity	Set to 1 if the required logic polarity is active-high (default).
CFG-HW-ANT_CFG_RECOVER	Enables auto-recovery in the event of a short circuit	To use this feature, enable short circuit detection and CFG-HW-ANT_CFG_PWRDOWN.
CFG-HW-ANT_SUP_SWITCH_PIN	PIO number of the pin used for switching antenna supply	PIO5 is recommended if available. This pin can be used as an LNA_EN signal to control an external LNA, especially if the software standby mode or the power save mode on/off (PSMOO) operation is used.
CFG-HW-ANT_SUP_SHORT_PIN	PIO number of the pin used for detecting a short circuit in the antenna supply	
CFG-HW-ANT_SUP_OPEN_PIN	PIO number of the pin used for detecting open/disconnected antenna	
CFG-HW-ANT_ON_SHORT_US	Time delay between the antenna supply being turned on and the short circuit detection being activated (in microseconds)	Increase the time delay to avoid a short circuit to be detected before the antenna supply voltage has stabilized. Recommended values: 500 us (default) to 5000 us.

Table 6: Antenna supervisor configuration

It is possible to obtain the status of the antenna supervisor from the UBX-MON-RF message. For information about the *antStatus* and *antPower* fields, refer to the Interface description [3]. In addition, any changes in the status of the antenna supervisor are reported to the host interface as ANTSTATUS in NMEA notice messages.

ANTSTATUS	Description
OFF	Antenna is off
ON	Antenna is on
DONTKNOW	Antenna power status is not known

Table 7: Antenna power status

2.2 Navigation configuration

This section presents various configuration options related to the navigation engine. These options can be configured through CFG-NAVSPG-* configuration keys.

2.2.1 Dynamic platform

The dynamic platform model can be configured through the CFG-NAVSPG-DYNMODEL configuration item. For the supported dynamic platform models and their details, see Table 8 and Table 9.

Platform	Description
Portable (default)	Applications with low acceleration, e.g. portable devices. Suitable for most situations.
Stationary	Used in timing applications (antenna must be stationary) or other stationary applications. Velocity restricted to 0 m/s. Zero dynamics assumed.
Pedestrian	Applications with low acceleration and speed, e.g. how a pedestrian would move. Low acceleration assumed.

Platform	Description
Automotive	Used for applications with equivalent dynamics to those of a passenger car. Low vertical acceleration assumed.
At sea	Recommended for applications at sea, with zero vertical velocity. Zero vertical velocity assumed. Sea level assumed.
Airborne <1g	Used for applications with a higher dynamic range and greater vertical acceleration than a passenger car. No 2D position fixes supported.
Airborne <2g	Recommended for typical airborne environments. No 2D position fixes supported.
Airborne <4g	Only recommended for extremely dynamic environments. No 2D position fixes supported.
Wrist	Only recommended for wrist-worn applications. Receiver will filter out arm motion.

Table 8: Dynamic platform models

Platform	Max altitude [m]	Max horizontal velocity [m/s]	Max vertical velocity [m/s]	Sanity check type	Max position deviation
Portable	12000	310	50	Altitude and velocity	Medium
Stationary	9000	10	6	Altitude and velocity	Small
Pedestrian	9000	30	20	Altitude and velocity	Small
Automotive	6000	100	15	Altitude and velocity	Medium
At sea	500	25	5	Altitude and velocity	Medium
Airborne <1g	80000	100	6400	Altitude	Large
Airborne <2g	80000	250	10000	Altitude	Large
Airborne <4g	80000	500	20000	Altitude	Large
Wrist	9000	30	20	Altitude and velocity	Medium

Table 9: Dynamic platform model details

Applying dynamic platform models designed for high acceleration systems (e.g. airborne <2g) can result in a higher standard deviation in the reported position.

If a sanity check against the limit of the dynamic platform model fails, the position solution becomes invalid. Table 9 shows the types of sanity checks which are applied for a particular dynamic platform model.

2.2.2 Ultra-high sensitivity mode

The ultra-high sensitivity mode allows tracking GNSS signals down to –167 dBm power level. By default, the ultra-high sensitivity mode is disabled in MAX-M10N excluding the weakest signals from the navigation solution.

To set the receiver in the ultra-high sensitivity mode, send the configuration string given in Table 10. The configuration can be applied at RAM, battery-backed RAM (BBR), and flash layers. The configuration applied at the RAM layer takes an immediate effect. To apply the configuration at the BBR and flash layers, send the UBX-CFG-RST message with resetMode 0x01.

To revert back to the default configuration, send the configuration string given in Table 11.

Configuration layer	Configuration string
RAM	B5 62 06 8A 09 00 00 01 00 00 01 00 C6 20 08 89 D0
BBR	B5 62 06 8A 09 00 00 02 00 00 01 00 C6 20 08 8A D8
FLASH	B5 62 06 8A 09 00 00 04 00 00 01 00 C6 20 08 8C E8

Table 10: UBX binary strings to enable ultra-high sensivity mode

Configuration layer	Configuration string
RAM	B5 62 06 8A 09 00 00 01 00 00 01 00 C6 20 09 8A D1
BBR	B5 62 06 8A 09 00 00 02 00 00 01 00 C6 20 09 8B D9
FLASH	B5 62 06 8A 09 00 00 04 00 00 01 00 C6 20 09 8D E9

Table 11: UBX binary strings to revert to the default sensitivity mode

2.2.3 Navigation input filters

The navigation input filters in the CFG-NAVSPG-* configuration group control how the navigation engine handles the input data that comes from the satellite signal.

Configuration item	Description
CFG-NAVSPG-FIXMODE	By default, the receiver calculates a 3D position fix if possible but it reverts to 2D position if necessary (auto 2D/3D). The receiver can be configured to only calculate 2D (2D only) or 3D (3D only) positions.
CFG-NAVSPG-CONSTR_ALT, CFG-NAVSPG-CONSTR_ALTVAR	The fixed altitude is used if fixMode is set to 2D only. A variance greater than zero must also be supplied.
CFG-NAVSPG-INFIL_MINELEV	Minimum elevation of a satellite above the horizon to be used in the navigation solution. Low-elevation satellites may provide degraded accuracy, due to the long signal path through the atmosphere.
CFG-NAVSPG-INFIL_MINSVS, CFG-NAVSPG-INFIL_MAXSVS	Minimum and maximum number of satellites to use in the navigation solution. There is an absolute maximum limit of 32 satellites that can be used for navigation.
CFG-NAVSPG-INFIL_NCNOTHRS, CFG-NAVSPG-INFIL_CNOTHRS	A navigation solution will only be attempted if there is at least the given number of satellites with signals at least as strong as the given threshold.

Table 12: Navigation input filter parameters

If the receiver has only three satellites for calculating a position, the navigation algorithm uses a constant altitude to compensate for the missing fourth satellite. This is called a 2D fix. The constant altitude value is taken from the last successful 3D fix using a minimum of four available satellites.

☞ u-blox receivers do not calculate any navigation solution with fewer than three satellites.

2.2.4 Navigation output filters

The result of a navigation solution is initially classified by the fix type (as detailed in the `fixType` field of the UBX-NAV-PVT message). This distinguishes between failures to obtain a fix ("No Fix") and cases where a fix has been achieved, which are further subdivided into specific types of fixes (for example, 2D, 3D).

Where a fix has been achieved, the fix is checked to determine whether it is valid or not. A fix is only valid if it passes the navigation output filters as defined in CFG-NAVSPG-OUTFIL. In particular, both PDOP and accuracy values must be below the respective limits.

☞ Important: Users are recommended to check the `gnssFixOK` flag in the UBX-NAV-PVT or the NMEA valid flag. Fixes not marked as valid should not be used.

UBX-NAV-STATUS message also reports whether a fix is valid in the `gpsFixOK` flag. These messages have only been retained for backwards compatibility and it is recommended to use the UBX-NAV-PVT message.

2.2.5 Odometer filters

2.2.5.1 Speed (3D) low-pass filter

The CFG-ODO-OUTLPVEL configuration item activates a speed (3D) low-pass filter. The output of the speed low-pass filter is available in the UBX-NAV-VELNED message (`speed` field). The filtering level can be set via the CFG-ODO-VELLPGAIN configuration item and must be between 0 (heavy low-pass filtering) and 255 (weak low-pass filtering).

☞ The internal filter gain is computed as a function of speed. Therefore, the level defines the nominal filtering level for speeds below 5 m/s, as defined in the CFG-ODO-VELLPGAIN configuration item.

2.2.5.2 Course over ground low-pass filter

The CFG-ODO-OUTLPCOG configuration item activates a course over ground low-pass filter when the speed is below 8 m/s. The output of the course over ground (also named heading of motion 2D) low-pass filter is available in the UBX-NAV-PVT message (`headMot` field), UBX-NAV-VELNED message (`heading` field), NMEA-RMC message (`cog` field), and NMEA-VTG message (`cogt` field). The filtering level can be set via the CFG-ODO-COGLPGAIN configuration item and must be between 0 (heavy low-pass filtering) and 255 (weak low-pass filtering).

☞ The filtering level defines the filter gain for speeds below 8 m/s, as defined in the CFG-ODO-COGLPGAIN configuration item. If the speed is 8 m/s or higher, no course over ground low-pass filtering is performed.

2.2.5.3 Low-speed course over ground filter

The CFG-ODO-USE_COG configuration item activates this feature and the CFG-ODO-COGMAXSPEED, CFG-ODO-COGMAXPOSACC configuration items are used to configure a low-speed course over ground filter (also named heading of motion 2D). This filter derives the course over ground from position at very low speed. The output of the low-speed course over ground filter is available in the UBX-NAV-PVT message (`headMot` field), UBX-NAV-VELNED message (`heading` field), NMEA-RMC message (`cog` field) and NMEA-VTG message (`cogt` field). If the low-speed course over ground filter is not configured, then the course over ground is computed as described in section Freezing the course over ground.

2.2.6 Static hold

The static hold mode allows the navigation algorithms to decrease the noise in the position output when the velocity is below a predefined "Static Hold Threshold" level. This reduces the position wander caused by environmental factors such as multi-path and improves position accuracy especially in stationary applications.

By default, the static hold mode is disabled.

The CFG-MOT-GNSSSPEED_THRS configuration item defines the static hold speed threshold. If the speed drops below the defined "Static Hold Threshold", static hold mode is activated. Once static hold mode is active, the position output is kept static and the velocity is set to zero until there is evidence of movement again. Such evidence can be velocity, acceleration, changes of the valid flag (for example, position accuracy estimate exceeding the position accuracy mask, see also section Navigation output filters), position displacement, etc.

The CFG-MOT-GNSSDIST_THRS configuration item defines the static hold distance threshold. If the distance between the estimated position and the static hold position exceeds the defined threshold, the static hold mode is suspended or deactivated until there is evidence of no movement.

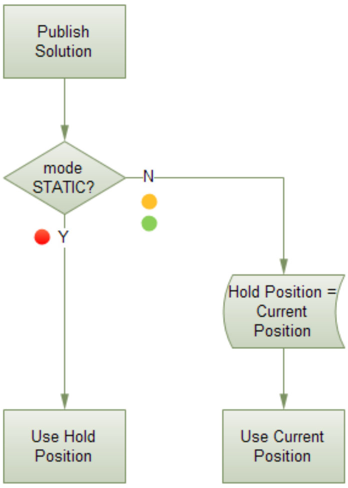


Figure 4: Position output in static hold mode

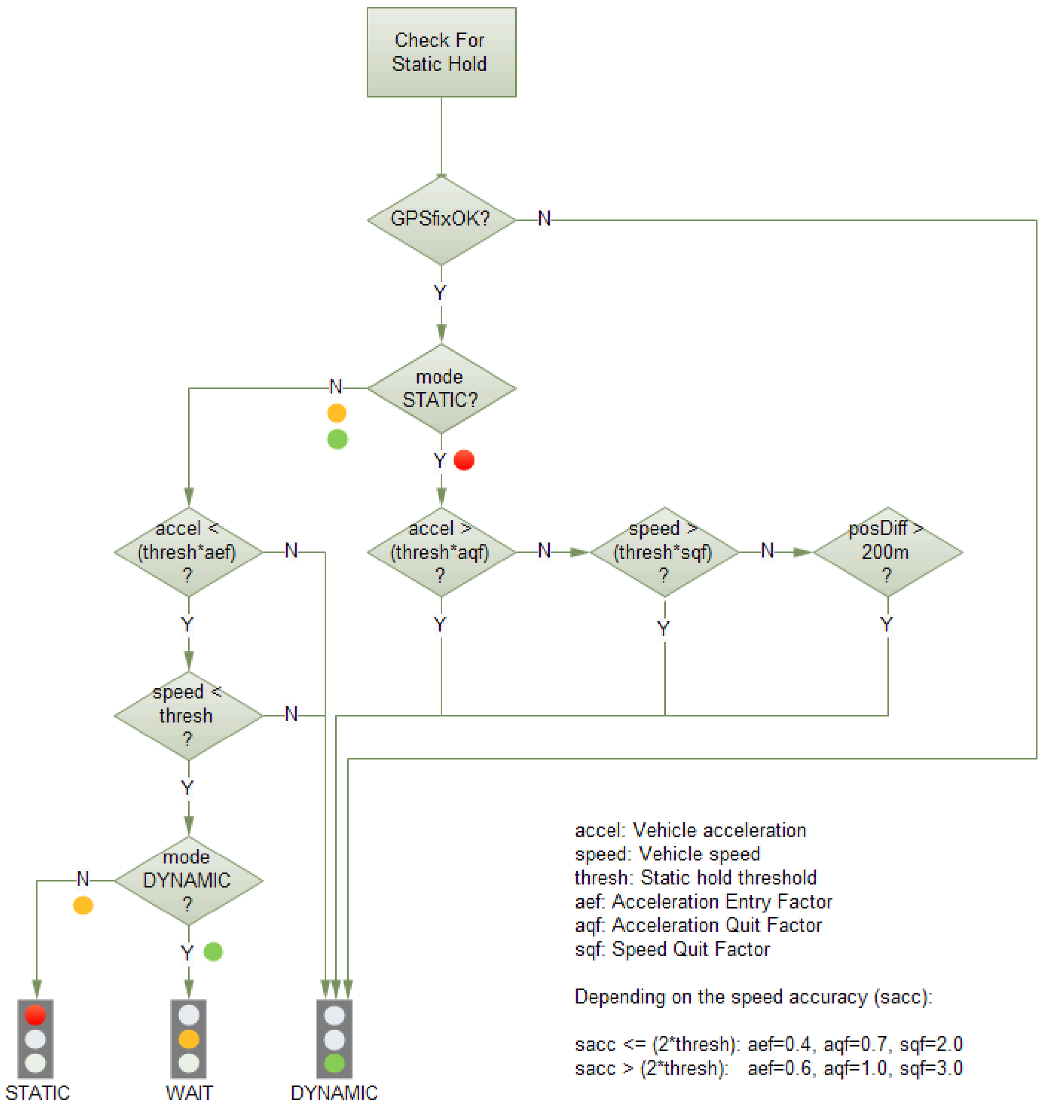


Figure 5: Flowchart of static hold mode

2.2.7 Freezing the course over ground

If the low-speed course over ground filter is deactivated or inactive (see section Low-speed course over ground filter), the receiver derives the course over ground from the GNSS velocity information. If the velocity cannot be calculated with sufficient accuracy (for example, with bad signals) or if the absolute speed value is very low (under 0.1 m/s), the course over ground value becomes inaccurate too. In this case the course over ground value is frozen, that is, the previous value is kept and its accuracy degrades over time. These frozen values will not be output in the NMEA messages NMEA-RMC and NMEA-VTG unless the NMEA protocol is explicitly configured to do so (see NMEA protocol configuration in the applicable Interface description [3]).

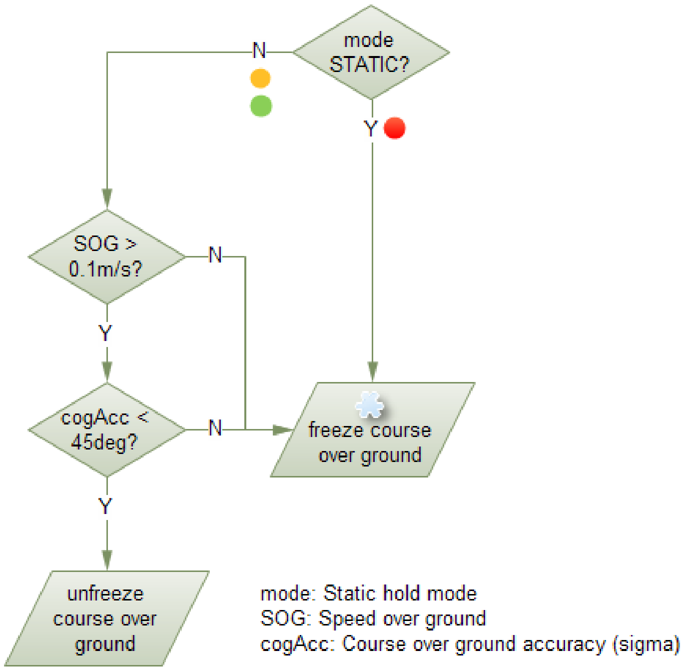


Figure 6: Flowchart of course over ground freezing

2.2.8 Super-Signal (Super-S) technology

This feature improves receiver performance when GNSS signals are weakened, for example, due to low-gain antennas or poor antenna placement.

Under normal conditions, the receiver down-weights or discards weak signals to reduce errors, such as those caused by multipath reflections. The Super-S feature modifies this behavior by adjusting the weighting of weak signals. By compensating for reduced signal strength, the receiver maintains reliable positioning in challenging conditions.

The weak signal compensation can be configured to three different modes with the CFG-NAVSPG-SIGATTCOMP configuration key as follows:

- Disabled: no weak signal compensation
- Automatic: the receiver automatically estimates and compensates for the weak signal
- Configured: the receiver compensates for the weak signal based on a configured value

When using the `configured` mode, set the maximum expected C/N0 observed in a clear-sky environment, excluding any outliers or unusually high values.

☞ When enabling this feature, due to multi-path contamination or interference rather than poor signal reception, the receiver may overtrust distorted signals. This can reduce the positioning accuracy.

2.2.9 Optimizing navigation performance in common GNSS challenges

Configurable receiver parameters can be used to optimize navigation performance when operating under common GNSS challenges. These challenges are commonly encountered in specific GNSS applications, such as asset, vehicle, and pet/people tracking.

Table 13 details which configuration parameters help balance navigation filter's responsiveness, fast TTFF and position fix rate with the need to prevent inaccurate or unreliable position outputs.

GNSS challenge	Configuration item	Application
Position fix outliers at startup. Large initial position errors may occur during the receiver startup, especially in challenging areas, and they typically require several epochs to converge.	CFG-NAVSPG-OUTFIL_PACC CFG-NAVSPG-INFIL_NCNOTHRS CFG-NAVSPG-INFIL_MINCNO CFG-NAVSPG-INFIL_MINSVS CFG-NAVSPG-INFIL_MINELEV	Asset tracker
Position fix outliers after leaving an obscured area. Significant position jumps that can occur immediately after signal reacquisition, such as when exiting tunnels. These typically require several epochs to converge.	CFG-NAVSPG-OUTFIL_PACC CFG-NAVSPG-INFIL_NCNOTHRS CFG-NAVSPG-INFIL_MINCNO CFG-NAVSPG-INFIL_MINSVS CFG-NAVSPG-INFIL_MINELEV	Vehicle tracker Pet/people tracker
Gradual position drift during temporary stops. Progressive drift may occur when a receiver that is typically in motion becomes briefly static, for example a vehicle pausing at a traffic light.	CFG-NAVSPG-OUTFIL_PACC CFG-MOT-GNSSSPEED_THRS CFG-MOT-GNSSDIST_THRS	Vehicle tracker Pet/people tracker
General low C/N0 ratio observed (non-optimal or covered antenna). This refers to the situation when GNSS signals are weakened, for example, due to low-gain antennas or poor antenna placement.	CFG-NAVSPG-SIGATTCOMP	Asset tracker Vehicle tracker
General navigation accuracy below expectations. These are cases in which the overall navigation performance degrades because the navigation filter's dynamic model does not match the application scenario. The solution is to select the dynamic model corresponding to the scenario (e.g., use the AUTO dynamic model for typical car-tracking use cases).	CFG-NAVSPG-DYNMODEL	All

Table 13: Configuration parameters for typicall GNSS challenges

☞ Detailed descriptions of the configuration parameters are provided in the Interface description [3] and in the following subchapters:

- The navigation input filters (CFG-NAVSPG-INFIL) are described in Navigation input filters.
- The navigation output filters (CFG-NAVSPG-OUTFIL) described in Navigation output filters.
- CFG-MOT-GNSSSPEED_THRS and CFG-MOT-GNSSDIST_THRS are descibed in Static hold.
- The CFG-NAVSPG-SIGATTCOMP configuration is described in Super-Signal (Super-S) technology.

3 Receiver functionality

This chapter describes MAX-M10N operational features and their configuration.

3.1 Augmentation systems

3.1.1 SBAS

MAX-M10N is capable of receiving multiple Satellite Based Augmentation System (SBAS) signals concurrently, even from different SBAS systems (WAAS, EGNOS, etc.). SBAS signals are recommended to be used only for correction data. These signals can also be used for navigation, but with their low weighting, they only have a minor impact on the navigation solution.

For receiving correction data, MAX-M10N automatically chooses the best SBAS satellite as its primary source. It selects only one satellite since the information received from other SBAS satellites is redundant and could be inconsistent. The selection strategy is determined by the proximity of the satellites, the services offered by the satellite, the configuration of the receiver (test mode allowed/disallowed, integrity enabled/disabled) and the signal link quality to the satellite.

If corrections are available from the chosen SBAS satellite and used in the navigation solution, the differential correction status is indicated in several output messages such as UBX-NAV-PVT, UBX-NAV-STATUS, UBX-NAV-SAT, UBX-NAV-SIG, NMEA-GGA, NMEA-GLL, NMEA-RMC, and NMEA-GNS. The UBX-NAV-SBAS message provides detailed information about the corrections available and applied. Refer to the Interface description [3] for a detailed description of the messages.

The most important SBAS feature for accuracy improvement is the ionosphere correction parameters. The measured data from regional Ranging and Integrity Monitoring Stations (RIMS) are combined to make a Total Electron Content (TEC) map. This map is transferred to the receiver via SBAS satellites to allow a correction of the ionosphere delays on each received signal.

Message type	Message content	Source
0(0/2)	Test mode	All
1	PRN mask assignment	Primary
2, 3, 4, 5	Fast corrections	Primary
6	Integrity	Primary
7	Fast correction degradation	Primary
9	Satellite navigation (ephemeris)	All
10	Degradation	Primary
12	Time offset	Primary
17	Satellite almanac	All
18	Ionosphere grid point assignment	Primary
24	Mixed fast / long-term corrections	Primary
25	Long-term corrections	Primary
26	Ionosphere delays	Primary

Table 14: Supported SBAS messages

Each satellite serves a specific region and its correction signal is only useful within that region. Planning is crucial to determine the best possible configuration, especially in areas where signals from different SBAS systems can be received:

- **Example 1 - SBAS receiver in North America:** In eastern parts of North America, make sure that EGNOS satellites do not take preference over WAAS satellites. The satellite signals from the EGNOS system should be disallowed by using the PRN scan mask (configuration key CFG-SBAS-PRNSCANMASK).
- **Example 2 - SBAS receiver in Europe:** Some WAAS satellite signals can be received in some parts of western Europe and GAGAN SBAS satellites in other parts of Europe. Therefore, it is recommended that satellites from all but the EGNOS system are disallowed using the PRN scan mask.

☞ Although u-blox receivers try to select the best available SBAS correction data, it is recommended to configure them to exclude the unwanted SBAS satellites.

To configure the SBAS functionality, use the CFG-SBAS-* configuration group.

Parameter	Description
CFG-SIGNAL-SBAS_ENA	Enabled/disabled status of the SBAS subsystem
CFG-SBAS-USE_TESTMODE	Allow/disallow SBAS usage from satellites in test mode (enable when BDSBAS is used)
CFG-SBAS-USE_IONOONLY	Allow/disallow usage of SBAS ionospheric corrections only(enabled by default)
CFG-SBAS-USE_RANGING	Use the SBAS satellites for navigation (ranging)
CFG-SBAS-USE_DIFFCORR	Combined enable/disable switch for fast, long-term, and ionosphere corrections
CFG-SBAS-USE_INTEGRITY	Apply integrity information data
CFG-SBAS-PRNSCANMASK	Allows selectively enabling/disabling SBAS satellites (BDSBAS disabled by default)

Table 15: SBAS configuration parameters

☞ When SBAS integrity data is applied, the navigation engine stops using all signals for which no integrity data is available (including all non-GPS signals). It is not recommended to enable SBAS integrity on borders of SBAS service regions in order not to inadvertently restrict the number of available signals.

☞ SBAS integrity information is required for at least five GPS satellites. If this condition is not met, SBAS integrity data will not be applied.

☞ When the receiver switches from a solution using correction data to a standard position solution, the reference frame of the output position switches as well. For an SBAS solution, the reference frame is aligned within a few centimeters of WGS84 (and modern ITRF realizations).

☞ Other SBAS systems can be enabled by adding the corresponding PRNs to the CFG-SBAS-PRNSCANMASK list.

3.1.2 QZSS SLAS

QZSS SLAS (Sub-meter Level Augmentation Service) is an augmentation technology, which provides correction data for pseudoranges of GPS, QZSS, and other major GNSS satellites. The correction stream is transmitted on the L1S signal at the L1 frequency (1575.42 MHz).

For more information on QZSS SLAS, visit qzss.go.jp/en/.

Multiple QZSS SLAS signals can be received simultaneously. When receiving QZSS SLAS correction data, MAX-M10N autonomously selects the best QZSS satellite. The selection strategy is determined by the quality of the QZSS L1S signals, the receiver configuration (test mode allowed or not), and the location of the receiver with respect to the QZSS SLAS coverage area. When outside of this coverage area, the receiver likely falls back to using SBAS corrections.

If QZSS SLAS corrections are used in the navigation solution, the differential status is indicated in several output messages such as UBX-NAV-PVT, UBX-NAV-STATUS, UBX-NAV-SAT, NMEA-GGA, NMEA-GLL, NMEA-RMC, and NMEA-GNS. The UBX-NAV-SLAS message provides detailed information about which corrections are available and applied. Refer to the Interface description [3] for a detailed description of the messages.

Message type	Message content
0	Test mode
47	Monitoring station information
48	PRN mask
49	Data issue number
50	DGPS correction
51	Satellite health

Table 16: Supported QZSS L1S SLAS messages for navigation enhancement

Use the configuration key CFG-SIGNAL-QZSS_L1S_ENA to enable QZSS L1S signal. For further QZSS SLAS functionality, use the CFG-QZSS-USE_SLAS* configuration keys.

Parameter	Description
CFG-QZSS-USE_SLAS_DGNSS	Apply QZSS SLAS corrections
CFG-QZSS-USE_SLAS_TESTMODE	Allow the correction provided by QZSS satellites that are in test mode
CFG-QZSS-USE_SLAS_RAIM_UNCORR	If this configuration is set, the receiver will try to estimate the position by using only corrected measurements; if all corrected measurements are not available, it will not use any corrections. If this configuration is not set, the receiver will mix corrected and uncorrected measurements for the navigation solution.
CFG-QZSS-SLAS_MAX_BASELINE	Maximum distance from the closest ground monitoring station (GMS) to apply the SLAS corrections. Note that due to the nature of the service, the usefulness of corrections degrades with distance. When far away from GMS, SBAS may be a better correction source.

Table 17: QZSS SLAS configuration parameters

☞ If the RAIM option is set, QZSS is the only GNSS time system that measurements can observe.

3.2 RTCM corrections

Radio Technical Commission for Maritime Services (RTCM) defines a standardized binary protocol for transmitting GNSS correction data. MAX-M10N supports RTCM as specified in RTCM 10403.4 – Differential GNSS Services, Version 3 (December 1, 2023).

RTCM corrections rely on two key components:

- Base station[2]: Collects raw GNSS measurements, converts them into RTCM format, and broadcasts correction data.
- Rover: Receives the corrections via Networked Transport of RTCM via Internet Protocol (NTRIP) or from a Virtual Reference Station (VRS) service provider operating a network of base stations.

RTCM corrections provide precise data on satellite position, atmospheric delay, and clock errors, allowing the rover to compensate for measurement deviations relative to a known reference station. This process significantly reduces positioning errors.

MAX-M10N is configured to act as a rover, applying RTCM corrections to improve position accuracy. This is particularly useful in areas without or with limited SBAS coverage, under low signal level conditions, or for short operation periods.

The u-blox portfolio includes high-precision modules that can be configured as base stations. For details on setting up a base station or accessing RTCM services, contact u-blox support.

2 The terms base, base station, reference and reference station can be used interchangeably

Alternatively, paid RTCM correction services, such as the u-blox PointPerfect service, are available globally.

The u-center 2 tool allows evaluation of this feature. For further information, refer to the u-center 2 User guide [7], section **Tools and services > NTRIP**.

3.2.1 Multiple signal messages

RTCM observation data for all GNSS constellations is standardized using Multiple Signal Messages (MSM). MSM messages provide a unified format for multi-constellation and multi-frequency GNSS corrections. They are numbered from MSM1 to MSM7, each including different levels of detail.

Table 18 summarizes the content and approximate message size for each MSM type. The size depends on the number of satellites (Nsat) and signals (Nsig) included.

MSM type	Content	Message size (bits)
MSM 1	Compact GNSS Pseudoranges	169 + Nsat*(10 + 16*Nsig)
MSM 2	Compact GNSS Phaseranges	169 + Nsat*(10 + 28*Nsig)
MSM 3	Compact GNSS Pseudoranges and Phaseranges	169 + Nsat*(10 + 43*Nsig)
MSM 4	Full GNSS Pseudoranges, Phaseranges and carrier-to-noise ratio (CNR)	169 + Nsat*(18 + 49*Nsig)
MSM 5	Full GNSS Pseudoranges, Phaseranges, PhaserangeRate and CNR	169 + Nsat*(36 + 64*Nsig)
MSM 6	Full GNSS Pseudoranges, Phaseranges and CNR with high resolution	169 + Nsat*(18 + 66*Nsig)
MSM 7	Full GNSS Pseudoranges, Phaseranges, PhaserangeRate and CNR with high resolution	169 + Nsat*(36 + 81*Nsig)

Table 18: MSM message content

MSM6 and MSM7 contain the same fields as MSM4 and MSM5, respectively, but with a higher resolution. These are suitable for high precision applications. For standard precision applications (sub-meter level), use MSM1-5 types.

Although MAX-M10N supports all MSM types, it primarily uses pseudorange measurements to compensate for atmospheric effects. These measurements are also present in the MSM1 type, which is beneficial for applications with limited data bandwidth.

3.2.2 Supported RTCM messages

The MAX-M10N supports the following RTCM input messages:

Message type	Description
RTCM 1005	Stationary RTK reference station antenna reference point (ARP)
RTCM 1006	Stationary RTK reference station ARP with antenna height
RTCM 1071	GPS MSM1
RTCM 1072	GPS MSM2
RTCM 1073	GPS MSM3
RTCM 1074	GPS MSM4
RTCM 1075	GPS MSM5
RTCM 1076	GPS MSM6
RTCM 1077	GPS MSM7
RTCM 1091	Galileo MSM1
RTCM 1092	Galileo MSM2

Message type	Description
RTCM 1093	Galileo MSM3
RTCM 1094	Galileo MSM4
RTCM 1095	Galileo MSM5
RTCM 1096	Galileo MSM6
RTCM 1097	Galileo MSM7
RTCM 1101	QZSS MSM1
RTCM 1102	QZSS MSM2
RTCM 1103	QZSS MSM3
RTCM 1104	QZSS MSM4
RTCM 1105	QZSS MSM5
RTCM 1106	QZSS MSM6
RTCM 1107	QZSS MSM7
RTCM 1111	BeiDou MSM1
RTCM 1112	BeiDou MSM2
RTCM 1113	BeiDou MSM3
RTCM 1114	BeiDou MSM4
RTCM 1115	BeiDou MSM5
RTCM 1116	BeiDou MSM6
RTCM 1117	BeiDou MSM7
RTCM 1121	SBAS MSM1
RTCM 1122	SBAS MSM2
RTCM 1123	SBAS MSM3
RTCM 1124	SBAS MSM4
RTCM 1125	SBAS MSM5
RTCM 1126	SBAS MSM6
RTCM 1127	SBAS MSM7

Table 19: MAX-M10N supported RTCM input messages

3.2.3 RTCM operation

Applying differential corrections significantly reduces position error, enabling sub-meter accuracy. The final accuracy depends on factors such as baseline length, multipath effects, and satellite visibility at both the rover and base station.

Table 20 lists messages and fields that indicate the use of RTCM differential corrections:

Message	Field and value	Comments
NMEA-GGA	`quality`=2	Quality field = 2 when using SBAS or RTCM corrections. Age of last RTCM message and station ID are reported in `diffAge` and `diffStation`.
NMEA-GLL	`status`=A, `posMode`=D	Also applicable if correction source is SBAS
NMEA-RMC	`posMode`=D	Also applicable if correction source is SBAS
NMEA-VTG	`posMode`=D	Also applicable if correction source is SBAS
UBX-NAV-PVT	`diffSoln`=1	Also applicable if correction source is SBAS. The message also includes the field `lastCorrectionAge` for RTCM message age.
UBX-NAV-SAT	`diffCorr`=1, `rtcmCorrUsed`=1	Per-satellite condition. If `diffCorr` is set but `rtcmCorrUsed` is not, the correction source is SBAS, not RTCM.

Message	Field and value	Comments
UBX-NAV-SIG	`corrSource`=4 or 5, `prCorrUsed`=1	Per-signal condition
UBX-NAV-STATUS	`diffSoln`=1, `diffCorr`=1	Also applicable if correction source is SBAS

Table 20: Messages and fields reporting differential GNSS (DGNSS) state

The RTCM corrections must match the receiver GNSS signal configuration. The receiver requires a GNSS observation message (MSM1, MSM3, MSM4, MSM5, MSM6, or MSM7) and the base station position message (RTCM 1005 or RTCM 1006) to apply the corrections and change the position fix type to differential GNSS (DGNSS). Note that this state is also reached when using SBAS corrections.

☞ RTCM corrections older than 60 seconds are not used. If the receiver still reports DGNSS, it is due to SBAS corrections.

☞ Once the corrections are applied, the receiver adopts the reference frame of the RTCM service provider.

3.3 Communication interfaces and PIOs

MAX-M10N supports communication over UART interfaces with a host CPU. UBX and NMEA protocols can be enabled simultaneously with individual interface settings, e.g. for baud rate, message rates, and so on.

3.3.1 UART

MAX-M10N supports a Universal Asynchronous Receiver/Transmitter (UART) port consisting of an RX and a TX line. The UART can be used as a host interface which supports a configurable baud rate and protocol selection.

☞ The UART interface does not support handshaking signals or hardware flow control signals.

The UART baud rate can be configured for selected speeds. Different rates than these speeds are not supported for transmission and reception.

☞ The UART RX interface is disabled when more than 100 frame errors are detected during a one-second period. This can happen if the wrong baud rate is used or the UART RX pin is grounded. An error message appears when the UART RX interface is re-enabled at the end of the one-second period.

Baud rate	Data bits	Parity	Stop bits
4800	8	none	1
9600	8	none	1
19200	8	none	1
38400	8	none	1
57600	8	none	1
115200	8	none	1
230400	8	none	1
460800	8	none	1
921600	8	none	1

Table 21: Possible UART interface configurations

Allow a short time delay of typically 100 ms between sending a baud rate change message and providing input data at the new rate. Otherwise some input characters may be ignored or the port could be disabled until the interface is able to process the new baud rate.

☞ The default baud rate is 9600 baud. Using a lower baud rate may cause buffering problems.

If there is too much data for the interface's bandwidth, the output buffer will fill up. Once the buffer space is exceeded, new messages to be sent will be dropped. To prevent message loss, the baud rate and the number of enabled messages should be selected carefully.

3.3.2 PIOs

This section describes the PIOs supported by MAX-M10N. All PIO active voltage levels are related to the V_IO supply voltage. All the inputs have internal pull-up resistors in normal operation and can be left open if unused.

☞ When assigning a different function to a PIO, ensure that the default function is disabled where applicable. For example, disable the I2C interface with the CFG-I2C-ENABLED configuration key if I2C pins are used for antenna supervisor functions.

3.3.2.1 RESET_N

MAX-M10N provides a RESET_N pin to reset the receiver. The RESET_N pin is input-only with an internal pull-up resistor to V_IO and should be left open for normal operation. Driving RESET_N low for at least 1 ms triggers a receiver reset. The RESET_N complies with the V_IO level and can be actively driven high.

☞ Use RESET_N only in critical situations to recover the receiver. RESET_N resets the receiver and clears the FW image in the code RAM and the BBR content including receiver configuration, real-time clock (RTC), and GNSS orbit data, triggering a cold start.

☞ No capacitor should be placed at RESET_N to GND, otherwise it could trigger a reset on every startup.

3.3.2.2 SAFEBOOT_N

The SAFEBOOT_N pin is for future service, updates and reconfiguration.

☞ The SAFEBOOT_N pin is internally connected to the TIMEPULSE pin through a 1 kΩ series resistor.

3.3.2.3 TIMEPULSE

MAX-M10N features one time pulse output at the TIMEPULSE pin. This can only be configured in PIO4. The details about this feature are explained in the section Time pulse.

☞ The TIMEPULSE and SAFEBOOT_N functions share the same internal IC function. If this pin is low at receiver startup, the receiver will enter safeboot mode. However, in normal operation the pin outputs the time pulse signal. Make sure this pin has no load that could pull it low at startup.

3.3.2.4 LNA_EN

The LNA_EN signal can be used to turn on and off an optional external LNA and an active antenna supply to optimize the power consumption in the backup modes and in the LEAP mode. The LNA and the active antenna supply are turned on when the LNA_EN signal is "high".

The LNA_EN signal is also used internally in MAX-M10N to control the integrated LNA. The polarity cannot be changed.

The LNA_EN signal can also be used as a part of an antenna supervisor circuit to control an active antenna power supply.

Table 22 shows the state of the LNA_EN pin in different modes of operation.

Mode/feature	Description	LNA_EN state
Normal operation		High
Software standby mode	Software command	Low
Hardware backup mode	VCC = V_IO = 0 V, V_BCKP supplied	Low
LEAP mode	Power-optimized tracking (POT) state	Duty cycling high-low
Antenna supervisor	Antenna supply power down due to short-circuit detection	Low

Table 22: LNA_EN pin state

3.3.2.5 EXTINT

MAX-M10N supports external interrupts at the EXTINT pin. The EXTINT pin has a fixed input voltage threshold with respect to V_IO. It can be used for functions such as wake-up source from Software backup mode, data batching, accurate external Frequency assistance, Time assistance, and Time mark reporting.

The EXTINT pin enables External control for host-controlled on/off operation of the receiver, and as a wake-up source for power save mode on/off operation (PSMOO). If configured for host-controlled on/off operation, the internal pull-up is disabled. Make sure the EXTINT input is always driven within the defined voltage level by the host.

The EXTINT pin can also be configured for another functionality.

☞ EXTINT functionality is only available at the EXTINT pin.

3.3.2.6 TX_READY

The receiver includes an internal message buffer that stores bytes to be sent to the host application. The TX_READY feature changes the polarity of the TX_READY pin to indicate when the buffer contains bytes ready for transmission. This allows the host application to wait for the signal instead of continuously polling the interface. The signal stays active until all of the bytes in the buffer have been transferred.

Each interface of the receiver includes a small additional transmit buffer. After the TX-ready signal deactivates, up to 16 bytes may still remain to be transferred to the host.

The TX-ready signal is enabled and configurable with the CFG-TXREADY-* configuration group.

The TX-ready signal can be assigned to any pin. Any previous configuration on the pin must be disabled before the assignment. The verification can be done with the UBX-MON-HW3 message.

The CFG-TXREADY-THRESHOLD parameter is specified in increments of 8 bytes. To configure the threshold, divide the desired byte count by 8. For instance, setting the parameter to 128 corresponds to a threshold of 1024 bytes (128 × 8 bytes). It is recommended not to exceed a threshold value of 256, equivalent to 2048 bytes, as higher values may result in delayed activation of the TX-ready signal and potential loss of messages.

3.3.2.6.1 Extended TX timeout

If the host does not communicate for more than 1.5 seconds, the receiver assumes that the host is no longer using this interface and no more packets are scheduled for this interface. This mechanism can be changed by enabling "extended TX timeouts" using the configuration key CFG-I2C-EXTENDEDTIMEOUT. When enabled, the receiver delays idling the interface until the allocated and undelivered bytes for this interface reach 4 kB.

This feature is especially useful when using the TX-ready feature with a message output rate of less than once per second, and fetching data only when available, determined by the TX_READY pin becoming active.

3.3.2.7 Geofence state

A PIO can be assigned to geofence state output on a physical pin, for example, to wake up a sleeping host when the defined geofence condition is reached. The receiver toggles the assigned pin according to the combined geofence state. The CFG-GEOFENCE-PIN configuration key is used to set the PIO number and the CFG-GEOFENCE-USE_PIO key to enable the geofence state output.

Due to hardware restrictions, the Unknown geofence state is not configurable and is always represented as HIGH. The pin state is HIGH also in the software standby mode, or in the reset state. The meaning of the LOW state is configured using the CFG-GEOFENCE-PINPOL configuration key.

☞ The CFG-GEOFENCE-PIN key refers to a PIO number and not to a physical device pin. The selected PIO must be mapped to a physical device pin.

Refer to the section Geofencing for more information.

3.4 Antenna

This section explains the antenna supervisor feature and the available implementation options.

3.4.1 Antenna supervisor

An active antenna supervisor provides the means to check the antenna for open and short circuits and to shut off the antenna supply if a short circuit is detected. Once enabled, the active antenna supervisor produces status messages that are reported in NMEA and/or UBX protocols. MAX-M10N supports two antenna supervisor variants: three-pin and two-pin implementations.

The three-pin antenna supervisor is able to detect short and open circuits and control the antenna supply. The two-pin antenna supervisor is a reduced version which is able to control the antenna supply and detect short circuits.

An overview of the two antenna supervisor variants is given in Table 23. It is recommended to make use of the full capabilities of the antenna supervisor (detect open and short circuits, and control the antenna supply).

The antenna supervisor can be configured through the CFG-HW-ANT_* configuration items. This includes enabling and disabling as well as changing the polarity of each signal. The current configuration of the active antenna supervisor can also be checked by polling the related CFG-HW_ANT_* configuration items.

The active antenna status can be determined by polling the UBX-MON-RF message or checking the NMEA notice messages. If an antenna is connected, the initial state after power-up is reported in the UBX-MON-RF message in *antStatus* and *antPower* fields. For more information, refer to Interface description [3]

Features	Three-pin	Two-pin
Short detection	Yes	Yes
Open detection	Yes	No
External components	Discrete and IC	Discrete and IC
Number of PIOs needed	Three	Two

Table 23: Antenna supervisor overview

3.4.1.1 Three-pin antenna supervisor

An active antenna supervisor circuit uses the ANT_DETECT, ANT_OFF_N, and ANT_SHORT_N signals. The ANT_OFF_N signal is already enabled and assigned to the LNA_EN pin in MAX-M10N. The ANT_DETECT and ANT_SHORT_N signals can be assigned to any unused PIOs, which may require disabling the previous function of the PIOs. For example, the open circuit detection uses the ANT_DETECT signal, "high" = Antenna detected (antenna consumes current); "low" = Antenna not detected (no current drawn). To enable the three-pin antenna supervisor, the ANT_DETECT and ANT_SHORT_N signals must be enabled in the receiver configuration. The polarity of the ANT_DETECT and ANT_SHORT_N signals must also be defined in the receiver configuration based on the design use case.

The antenna can be supplied by VCC_RF or an external supply. Note that the supply voltage must be clean, as any noise could directly couple into the RF part of the GNSS receiver which affecting the overall GNSS performance.

Refer to Reference designs for antenna supervisor examples and the required configuration.

Figure 7 presents the required three-pin antenna supervisor circuit and subsequent sections describe how to enable and monitor each feature.

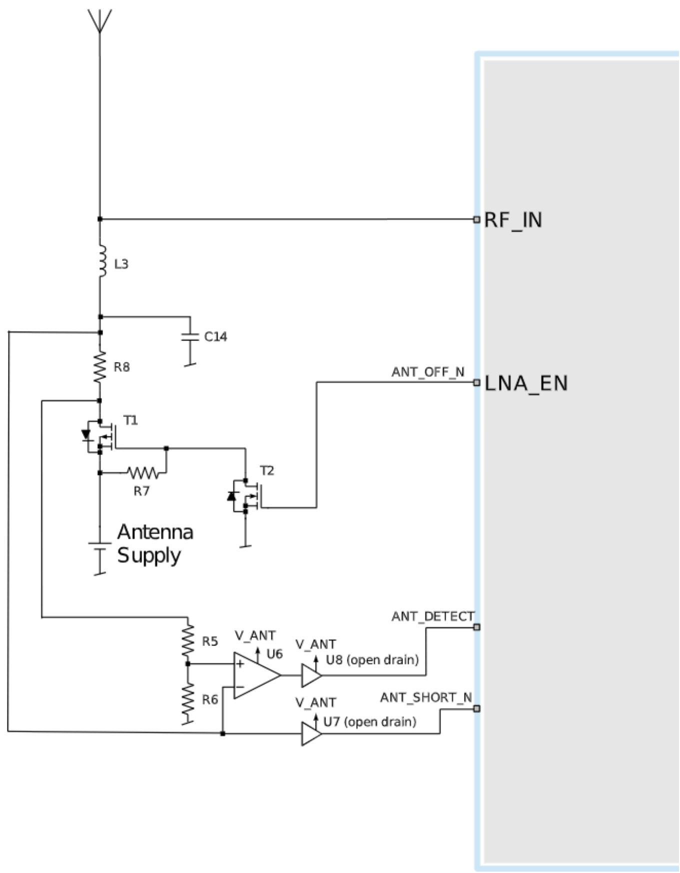


Figure 7: MAX-M10N three-pin antenna supervisor

Table 24 presents a list of the external components required for implementing the three-pin antenna supervisor design in Figure 7. Refer to External components for the recommended parts and specification.

Part	Description
C14	Filtering capacitor
L3	DC infeed inductor
T1, T2	p-channel, n-channel MOSFET acting as a switch to control the antenna supply
U6	Comparator (op-amp)
U7, U8	Open drain buffers to shift voltage levels
R7	Passive pull-up to control T1
R8	Current limiter in the event of a short circuit
R5	Defines the threshold of the comparator
R6	Defines the threshold of the comparator

Table 24: Components in antenna supervisor

The threshold voltage (V_REF) of the comparator is defined by R5 and R6. It can be calculated as: *V_REF = R6/(R6+R5)*V_ANT*.

☞ The open drain buffers shown in Figure 7 are not needed if V_ANT has the same voltage level as V_IO.

3.4.1.2 Two-pin antenna supervisor

The reduced functionality antenna supervisor circuit is connected to two signals: antenna control (ANT_OFF_N) and antenna status detection (ANT_SHORT_N). The ANT_OFF_N signal is already enabled and assigned to the LNA_EN pin in MAX-M10N and the ANT_SHORT_N signal can be assigned to any unused PIO, which may require disabling the previous function of the PIO. To enable the reduced antenna supervisor, the ANT_SHORT_N signal must be enabled in the receiver configuration. The polarity of the ANT_SHORT_N signal must also be defined in the receiver configuration based on the design use case.

The antenna can be supplied by VCC_RF or an external supply. Note that the supply voltage must be clean, as any noise could directly couple into the RF part of the GNSS receiver affecting the overall GNSS performance.

Refer to Reference designs for antenna supervisor examples and the required configuration.

Figure 8 presents the required two-pin antenna supervisor circuit and subsequent sections describe how to enable and monitor each feature.

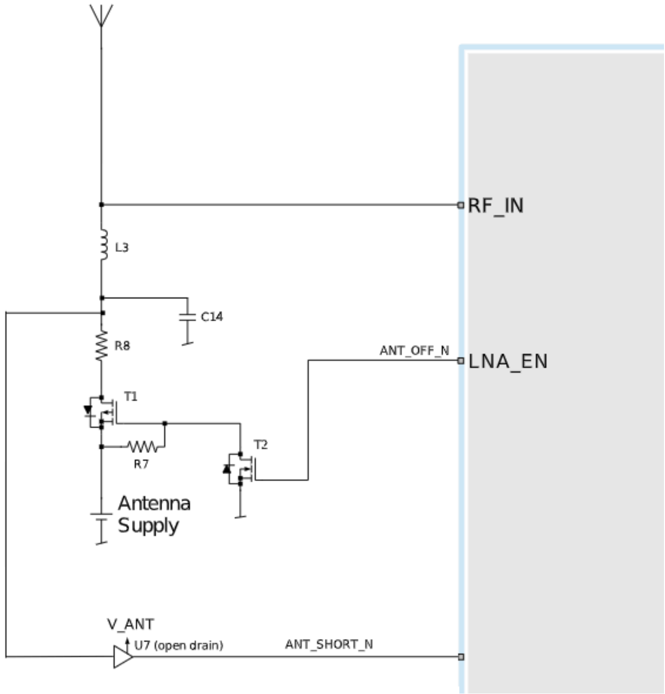


Figure 8: MAX-M10N two-pin antenna supervisor

Table 25 presents a list of the external components required for implementing the two-pin antenna supervisor design in Figure 8. Refer to External components for the recommended parts and specification.

Part	Description
C14	Filtering capacitor
L3	DC infeed inductor
T1, T2	p-channel, n-channel MOSFET acting as a switch to control the antenna supply
U7	Open drain buffers to shift voltage levels
R7	Passive pull-up to control T1
R8	Current limiter in the event of a short circuit

Table 25: Components in two-pin antenna supervisor

☞ The open drain buffer shown in Figure 8 is not needed if V_ANT is the same voltage level as V_IO.

3.4.1.3 Antenna voltage control - ANT_OFF_N

The antenna voltage control is enabled by default in MAX-M10N with the configuration item CFG-HW-ANT_CFG_VOLTCTRL set to true (1).

☞ The antenna status (as reported in UBX-MON-RF and UBX-INF-NOTICE messages) is not reported unless the antenna voltage control has been enabled.

Result:

- UBX-MON-RF: Antenna status = OK. Antenna power status = ON.
- ANT_OFF_N = active low. The pin is pulled high to enable an external antenna or LNA.

Startup message at power-up if the configuration is stored:

```
$GNTXT,01,01,02,ANTSUPERV=AC *00
```

`$GNTXT,01,01,02,ANTSTATUS=INIT*3B`

`$GNTXT,01,01,02,ANTSTATUS=OK*25`

`ANTSUPERV=AC` indicates that antenna control is activated.

3.4.1.4 Antenna short detection - ANT_SHORT_N

Enable the antenna short detection by setting the configuration item CFG-HW-ANT_CFG_SHORTDET to true (1).

Result:

- UBX-MON-RF: Antenna status = OK. Antenna power status = ON.
- ANT_OFF_N = active low to disable an external antenna. Therefore, the pin is pulled high to enable an external antenna.
- ANT_SHORT_N = active low to report a short circuit. The pin is default high (PIO pull-up enabled).

Startup message at power-up if the configuration is stored:

`$GNTXT,01,01,02,ANTSUPERV=AC SD *37`

`$GNTXT,01,01,02,ANTSTATUS=INIT*3B`

`$GNTXT,01,01,02,ANTSTATUS=OK*25`

`ANTSUPERV=AC SD` indicates that antenna control and short detection are activated.

If a short circuit is detected in the antenna (ANT_SHORT_N pulled low):

`$GNTXT,01,01,02,ANTSTATUS=SHORT*73`

- UBX-MON-RF: Antenna status = SHORT. Antenna power status = ON (automatic power-down is not enabled, CFG-HW-ANT_CFG_PWRDOWN or powerDown field in OTP memory file 0x37 is off by default).
- ANT_OFF_N = high (external antenna enabled).

☞ If CFG-HW-ANT_CFG_PWRDOWN is already enabled (set to true), the polarity of the ANT_OFF_N signal changes to power down (disable) the antenna supply when a short is detected.

☞ After a detected antenna short, the reported antenna status continues to be reported as a SHORT. If auto-recovery is enabled for the antenna short detection, the antenna status can recover after a timeout of 60 seconds. Recovering the antenna status immediately requires either a power cycle or switching the antenna short detection off and on again.

3.4.1.5 Antenna short detection auto-recovery

Enable the antenna short detection auto-recovery by setting the configuration item CFG-HW-ANT_CFG_RECOVER to true (1).

☞ To use the auto-recovery feature, enable CFG-HW-ANT_CFG_PWRDOWN which requires CFG-HW-ANT_CFG_SHORTDET and CFG-HW-ANT_CFG_VOLTCTRL to be enabled.

Result:

- UBX-MON-RF: Antenna status = OK. Antenna power status = ON.
- ANT_OFF_N = active low. The pin is pulled high to enable an external antenna.
- ANT_SHORT_N = active low. The pin is default high (PIO pull-up enabled, to be pulled low if a SHORT is detected).

Startup message at power-up if the configuration is stored:

`$GNTXT,01,01,02,ANTSUPERV=AC SD PDoS SR*3E`

`$GNTXT,01,01,02,ANTSTATUS=INIT*3B`

`$GNTXT,01,01,02,ANTSTATUS=OK*25`

`ANTSUPERV=AC SD PDoS SR` (indicates short circuit recovery added - SR)

If short circuit is detected (ANT_SHORT_N pulled low):

`$GNTXT,01,01,02,ANTSTATUS=SHORT*73`

- UBX-MON-RF: Antenna status = SHORT. Antenna power status = OFF (automatic power-down is enabled).
- ANT_OFF_N = low (external antenna disabled).

After a timeout period of 60 seconds, the receiver retests the short circuit condition by enabling the antenna (i.e. pulling ANT_OFF_N high).

If a short is not present, the receiver reports antenna condition is OK:

`$GNTXT,01,01,02,ANTSTATUS=OK*25`

UBX-MON-RF: Antenna status = OK. Antenna power status = ON.

3.4.1.6 Antenna open circuit detection - ANT_DETECT

Enable the antenna open circuit detection by setting the configuration item CFG-HW-ANT_CFG_OPENDET to true (1).

Result:

- UBX-MON-RF: Antenna status = OK. Antenna power status = ON.
- ANT_OFF_N = active low. The pin is pulled high to enable an external antenna.
- ANT_SHORT_N = active low. The pin is default high (PIO pull-up enabled, to be pulled low if a SHORT is detected).
- ANT_DETECT = active high. The pin is default high (PIO pull-up enabled, to be pulled low if the antenna is not detected).

Startup message at power-up if the configuration is stored:

`$GNTXT,01,01,02,ANTSUPERV=AC SD OD PDoS SR*15`

`$GNTXT,01,01,02,ANTSTATUS=INIT*3B`

`$GNTXT,01,01,02,ANTSTATUS=OK*25`

`ANTSUPERV=AC SD OD PDoS SR` (indicates open circuit detection added - OD)

If ANT_DETECT is pulled low to indicate no antenna connected:

`$GNTXT,01,01,02,ANTSTATUS=OPEN*35`

If ANT_DETECT is left floating or pulled high to indicate antenna connected:

`$GNTXT,01,01,02,ANTSTATUS=OK*25`

3.4.1.7 Antenna status reporting

The antenna detection and antenna power status that is available in UBX-MON-RF and NMEA notice messages, is based on the antenna's physical state. The required antenna supervisor configuration keys depend on the selected antenna supervisor implementation (three-pin or two-pin).

Table 26 and Table 27 present a summary of the antenna status that is available in the *antStatus* and *antPower* fields of the UBX-MON-RF message. Refer to the Interface description [3] for more information.

Status	Description
DON'T KNOW	Antenna status is unknown.
INIT	Antenna power control feature is initialized (if CFG-HW-ANT_VOLTCTRL is enabled).
SHORT	A short is detected from the antenna input. That is, a lot of current is drawn from the active antenna.
OPEN	Antenna is not detected. That is, little or no current is drawn from the active antenna.
OK	Antenna is detected and no short is detected.

Table 26: Available antenna detection status

Status	Description
DON'T KNOW	CFG-HW-ANT_VOLTCTRL not configured.
OFF	GNSS OFF or a short is detected and CFG-HW-ANT_PWRDOWN is enabled. Note that this status also applies when the GNSS is restarted with the CFG-RST or implicitly with the CFG-GNSS messages, when the GNSS selection is reconfigured, or when the GNSS is stopped in the software standby mode and the off state of the power save mode is on/off (PSMOO).
ON	GNSS ON and no short/open is detected from the antenna input. Similarly, when there is a short and CFG-HW-ANT_PWRDOWN is not enabled or auto-recovery is enabled.

Table 27: Available antenna power status

Table 28 shows some possible combinations of the antenna supervisor configuration and the expected antenna status based on the physical state of the antenna. Note that the short detection takes priority over the open detection and "X" in Table 28 implies an unconfigured or undetected physical state. In addition, CFG-HW-ANT_PWRDOWN requires that CFG-HW-ANT_CFG_VOLTCTRL and CFG-HW-ANT_CFG_SHORTDET are enabled. Likewise, CFG-HW-ANT_RECOVER requires CFG-HW-ANT_PWRDOWN to be enabled.

☞ The antenna supervisor is re-initialized after issuing a reset with RESET_N pin or changing any of the antenna supervisor configuration keys. Therefore, if the default configuration has changed, it is recommended to save the antenna supervisor configuration to BBR or flash to ensure that the updated configuration is applied after a reset.

Configuration keys					Physical antenna state		Reported antenna status	
VOLTCTRL	**SHORTDET**	**OPENDET**	**PWRDOWN**	**RECOVER**	**Short circuit**	**Open circuit**	**antPower**	**antStatus**
TRUE	X	X	X	X	NO	NO	ON	OK
TRUE	FALSE	X	X	X	X	NO		
TRUE	X	FALSE	X	X	NO	X		
TRUE	FALSE	FALSE	X	X	X	X		
TRUE	FALSE	TRUE	X	X	X	YES	ON	OPEN
TRUE	X	TRUE	X	X	NO	YES		
TRUE	TRUE	X	FALSE	X	YES	X	ON	SHORT
TRUE	TRUE	X	TRUE	X	YES	X	OFF	SHORT
FALSE	TRUE	X	X	X	YES	X	UNKNOWN	SHORT
FALSE	FALSE	TRUE	X	X	X	YES	UNKNOWN	OPEN

Configuration keys					Physical antenna state		Reported antenna status	
VOLTCTRL	SHORTDET	OPENDET	PWRDOWN	RECOVER	Short circuit	Open circuit	antPower	antStatus
FALSE	X	TRUE	X	X	NO	YES	UNKNOWN	OPEN

Table 28: Antenna supervisor configuration and antenna states

3.5 Forcing receiver reset

GNSS receivers typically make a distinction between cold, warm, and hot start based on the type of valid information the receiver has during the restart.

- **Cold start:** in the cold start mode, the receiver has no information from the last position (e.g. time, velocity, frequency etc.) at startup. Therefore, the receiver must search the full time and frequency space, and all possible satellite numbers. If a satellite signal is found, it is tracked to decode the ephemeris (18-36 seconds under strong signal conditions), while the other channels continue to search satellites. Once there is a sufficient number of satellites with valid ephemeris, the receiver can calculate position and velocity data. Other GNSS receiver manufacturers call this the Factory startup mode.
- **Warm start:** in the warm start mode, the receiver has approximate information for time, position, and coarse satellite position data (Almanac). In this mode, the receiver normally needs to download ephemeris after power-up before it can calculate position and velocity data. As the ephemeris data is usually outdated after 4 hours, the receiver typically starts with a warm start if it has been powered down for more than 4 hours. In this scenario, several augmentations are possible. See Multiple GNSS assistance.
- **Hot start:** in the hot start mode, the receiver has been powered down only for a short time (4 hours or less), so that its ephemeris is still valid. Since the receiver does not need to download ephemeris again, this is the fastest startup method.

Using the UBX-CFG-RST message, you can force the receiver to reset and clear data, in order to see the effects of maintaining/losing such data between restarts. For this purpose, use the `navBbrMask` field in the UBX-CFG-RST message to initiate hot, warm, and cold starts, or a combination of startup modes.

The reset type can also be specified. This is not related to GNSS, but to the way the software restarts the system.

- **Hardware reset** uses the on-chip watchdog to electrically reset the chip. This is an immediate asynchronous reset. No stop events are generated.
- **Controlled software reset** terminates all running processes in an orderly manner. Once the system is idle, restarts the receiver operation, reloads its configuration and starts to acquire and track GNSS satellites.
- **Controlled software reset (GNSS only)** only restarts the GNSS tasks, without reinitializing the full system or reloading any stored configuration.
- **Hardware reset (after shutdown)** uses the on-chip watchdog to reset the receiver after shutdown.
- **Controlled GNSS stop** stops all GNSS tasks. The receiver is not restarted, but stops any GNSS-related processing.
- **Controlled GNSS start** starts all GNSS tasks.

Table 29 below contains an overview of the different reset types and the data that is cleared.

Reset type	Clears RAM	Clears BBR	Clears Flash
0x00 - Hardware reset (immediately), 0x04 - Hardware reset (after shutdown)	Yes	Yes	No
0x01 - Controlled Software reset	Yes	No	No

Reset type	Clears RAM	Clears BBR	Clears Flash
0x02 - Controlled Software reset (GNSS only), 0x08 - Controlled GNSS stop, 0x09 - Controlled GNSS start	No	No	No
RESET_N pin	Yes	Yes	No

Table 29: Overview of the available reset types

☞ After using any reset type that clears the BBR, the TTFF is similar to performing a cold start.

3.6 Security

The security concept of MAX-M10N covers:

- The integrity of the receiver
- Communication between the receiver and the GNSS satellites

Some security functions monitor and detect threats and report them to the host system. Other functions mitigate threats and allow the receiver to operate normally.

Table 30 gives an overview about possible threats and which functionality is available to detect and/or mitigate it.

Threat	u-blox solution
GNSS receiver integrity	Secure boot Secure firmware update Receiver configuration lock
Over air signal integrity	Spoofing detection and monitoring Jamming interference detection and monitoring Jamming interference detection and monitoring

Table 30: u-blox security options

3.6.1 Configuration locking

The receiver configuration can be locked so that further changes are no longer possible. The configuration can be locked by setting the configuration item CFG-SEC-CFG_LOCK to "true". This item can be set in the same way as all other configuration items on various layers.

☞ If the configuration lock is set in BBR, it remains locked until the BBR is cleared. To test the function, set it in the RAM configuration layer. After the receiver's power cycle, the information in this layer is lost and the configuration lock is not set anymore.

3.7 Power management

u-blox receivers support different operating modes. These modes represent strategies of controlling the acquisition and tracking engines to achieve either the best possible performance or good performance with reduced power consumption.

3.7.1 Continuous mode

MAX-M10N uses dedicated signal processing engines optimized for signal acquisition and tracking. The acquisition engine actively searches for and acquires signals during cold starts or when insufficient signals are available during navigation. The tracking engine continuously tracks and downloads all the almanac data and acquires new signals as they become available during navigation. The tracking engine consumes less power than the acquisition engine.

The current consumption is lower when a valid position is obtained quickly after the start of the receiver navigation, the entire almanac has been downloaded, and the ephemeris for each satellite in view is valid. If these conditions are not met, the search for the available satellites takes more time and consumes more power.

3.7.2 LEAP mode

Low energy accurate positioning (LEAP) mode allows a reduction in system power consumption by selectively switching parts of the receiver on and off. The LEAP mode provides the best balance between current consumption and performance. It also enables automatic duty cycling of the external LNA to lower the total power consumption even further. Used in combination with multi-GNSS Assistance data, the MAX-M10N not only features fast TTFF, but also ensures minimal power consumption, since A-GNSS enables the chip to maximize its power-optimized period.

In the LEAP mode, the receiver does not shut down completely between fixes, but uses power optimized tracking (POT) instead.

The LEAP mode default navigation update rate is 1 Hz and the receiver default configuration has been tuned to this setting. The configured navigation rate has direct impact on the energy consumption. To achieve further energy savings, the navigation rate can be decreased to 0.2 Hz (one navigation epoch every 5 seconds).

☞ The maximum supported navigation rate in the LEAP mode is 2 Hz.

☞ Time pulse function is not supported in the LEAP mode and must be disabled. However, it can be enabled in full power mode (setting CFG-PM-OPERATEMODE = 0), at the cost of increased power consumption.

⚠ **CAUTION** Due to its internal duty cycle, the receiver may not receive some messages from the host. If a message is not acknowledged, resend it. Alternatively, switch to the full power mode (setting CFG-PM-OPERATEMODE = 0) before transmitting the data, and revert to the LEAP mode (CFG-PM-OPERATEMODE = 2) once the transmission is complete. This issue is limited to the UART communication interface.

3.7.2.1 Operation

LEAP is based on a state machine with three different states: acquisition, tracking and power optimized tracking (POT) state.

- Inactive states: most parts of the receiver are switched off.
- Acquisition state: the receiver actively searches for and acquires signals. This state has the highest power consumption.
- Tracking state: the receiver continuously tracks and downloads data. The power consumption is lower than it is in the acquisition state.
- POT state: the receiver repeatedly loops through a sequence of tracking (track), calculating the position fix (fix), and entering an idle period (idle). No new signal is acquired and no data is downloaded. The power consumption is much lower than in the tracking state.

A more detailed description of the LEAP mode has been outlined in the following points:

- When the receiver is switched on, it first enters the acquisition state. A larger number of signals tracked later helps the receiver to remain in the POT state if some signals get blocked and are lost. This may reduce the overall power consumption.
- If the receiver is able to acquire a valid position fix (one passing the navigation output filters) and has acquired enough satellites, it switches to the tracking state.
- Once the receiver has enough navigation data information, it enters the POT state.
- In the POT state, the receiver continues to output position fixes according to the CFG-RATE-*.

- If the signal reception is poor, some signals are lost, or the ephemeris data is out of date during POT state, the receiver enters the tracking state.
- Once the signal conditions are optimal to calculate a position fix again, the receiver re-enters the POT state.
- If the receiver cannot get a valid position fix in the tracking state, it enters the acquisition state and remains in it until a fix is possible.

3.7.2.2 Example of power consumption

At power-up, the receiver starts up in the full-power acquisition state to search for satellites. This state continues until there is a valid 3D fix and the receiver has enough available satellites. For the 3D fix, the receiver needs to receive data for the current GNSS time and information of at least four satellites (dark red points in Figure 9). The receiver continues searching for more satellites in the acquisition state (red dots in Figure 9) until it acquires enough satellites for power optimized operation. Then the receiver enters the tracking mode (orange dots in Figure 9) until it has enough information for power optimized tracking (POT). After the initial acquisition and tracking state, the receiver enters POT state (shown by the green dots in Figure 9). If the set of available satellites gets too small, the receiver re-enters the acquisition or tracking state for a short period until it has enough satellites to track. The state of the receiver is given in the psmState field in the UBX-NAV-PVT message.

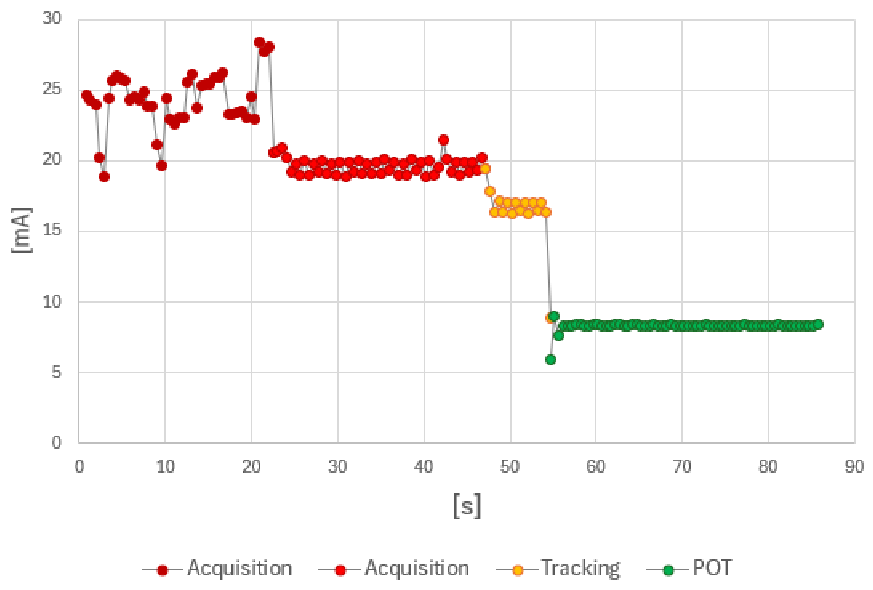


Figure 9: Example of power consumption in LEAP mode

3.7.2.3 Configuration

Config key	Default value	Description
UBX-CFG-OPERATEMODE	0 (Full power mode)	Receiver mode of operation. By default, the mode of operation is set to Full power mode UBX-CFG-OPERATEMODE = 0. It can be changed to LEAP power save mode by setting UBX-CFG-OPERATEMODE = 2.
UBX-CFG-UPDATEEPH	1 (true)	Enables periodic ephemeris update. When enabled, adds extra wakeup cycles to update the ephemeris data. The ephemeris data is updated approximately every 30 minutes. Disable to not wake up to update ephemeris data. Allowing periodic ephemeris update improves the fix quality and overall performance. For more information, refer to Satellite data download scheduler.
EXTINTWAKE	0 (false)	EXTINT "high" forces the receiver to enter full power mode. For more information, refer to External control.

Config key	Default value	Description
EXTINTBACKUP	0 (false)	EXTINT "low" forces the receiver to enter software backup mode. For more information, refer to External control.
EXTINTINACTIVE	0 (false)	EXTINT "low" forces the receiver to enter software backup mode if EXTINT pin is inactive longer time than specified by EXTINTINACTIVITY. For more information, refer to External control.
EXTINTINACTIVITY	0 (ms)	Specifies the inactivity period. Only applicable when EXTINTINACTIVE is set. For more information, refer to External control.
UBX-CFG-LIMITPEAKCURR	0 (false)	Limit peak current. When enabled, the current consumption during the acquisition state is reduced. However, this may adversely impact the speed of satellite acquisition.

Table 31: Power save mode configuration options in the CFG-PM group

3.7.2.4 External control

The external control allows switching the receiver state from its configured operation mode to the full power or the software backup mode. Operating the receiver externally through the EXTINT pin overrides internal functions that coincide with that specific operation.

If EXTINTWAKE is enabled, CFG-PM-EXTINTWAKE = 1, the receiver operation mode switches to the full power if the EXTINT pin is "high". If it is "low", the receiver continues in its configured operation mode.

If EXTINTBACKUP is enabled, CFG-PM-EXTINTBACKUP = 1, the receiver enters the software standby mode if a falling edge is detected on the EXTINT pin. Note that different wake-up sources wake the receiver in its preconfigured operation state, irrespective of the EXTINT remaining "low". The wake-up sources are:

- Rising or falling edge on the UART RX pin
- Rising or falling edge on the SPI CS pin
- Rising edge on RESET_N pin

If the receiver wakes up from one of these sources, a falling edge in the EXTINT pin returns the receiver to the software standby mode.

If both EXTINTWAKE and EXTINTBACKUP are enabled simultaneously, the receiver operation is completely under external control, and the conditions described above apply for both.

☞ Designs without an SPI flash memory must upload the FW image to the RAM memory after waking up.

The configuration key CFG-PM-EXTINTINACTIVITY sets an internal countdown to enter the software standby mode. The countdown is reset every time the EXTINT pin state changes. For example, the CFG-PM-EXTINTINACTIVE is enabled and the EXTINT pin state has not changed for the time set in EXTINTINACTIVITY. In this case, the receiver enters the software backup if EXTINT is "low", or immediately wakes up if it is "high", resetting the countdown. These two keys are independent of the previous EXTINTWAKE and EXTINTBACKUP keys.

☞ By default, the EXTINT pin is internally set "high" using a pull-up resistor.

3.7.3 Satellite data download scheduler

The receiver is not able to download time-limited satellite data (e.g. the ephemeris) while it is working in the power optimized tracking (POT) state. Therefore, it has to temporarily switch to continuous operation for the time the satellites transmit the desired data. To save power, the receiver schedules the downloads according to an internal timetable and switches to continuous operation only when data of interest is being transmitted by the satellites.

Each satellite transmits its own ephemeris data. Ephemeris data download is feasible when the corresponding satellite has been tracked with a sufficient C/N0 over a certain period of time.

When the configuration CFG-PM-UPDATEEPH = 1, the download is scheduled in a 10 to 30-minute grid or immediately when fewer than a certain number of visible satellites have valid ephemeris data. If this configuration is enabled, the receiver maximizes the number of used satellites and the time spent in the POT state.

When the configuration CFG-PM-UPDATEEPH = 0, periodic wake-ups are disabled and the receiver leaves the POT state and download data only when fewer than a certain number of visible satellites have valid ephemeris data. Without periodic wake-ups, the receiver may, over time, track fewer satellites, which can result in a deterioration of fix quality and cause the receiver to remain outside the POT state for extended periods. Therefore, it is recommended that the host system provides updated satellite data via AssistNow services.

When the satellite data is provided using the AssistNow service, the host must exit the POT state before receiving aiding data by setting the CFG-PM-OPERATEMODE message to 0. Once the data has been accepted, the host should re-enable it by setting the CFG-PM-OPERATEMODE message to 2.

Almanac, ionosphere, UTC correction, and satellite health data are transmitted by all satellites simultaneously. Therefore, these parameters can be downloaded when a single satellite is tracked with a sufficiently high C/N0.

Allowing more ephemerides to be downloaded before entering the POT state can improve the quality of the fixes and reduce the number of wake-ups needed to download ephemerides. However, this requires spending extra time in the acquisition state (only when an inadequate number of ephemerides are downloaded from tracked satellites).

3.7.4 Backup modes

A backup mode is an inactive state where the power consumption is reduced to a fraction of that in operating modes. The receiver maintains time information and navigation data to speed up the receiver restart after backup or standby mode.

MAX-M10N supports the following backup modes: hardware backup mode and software standby mode.

3.7.4.1 Hardware backup mode

The hardware backup mode allows entering a backup state and resuming operation by switching the main power supplies on and off while maintaining a V_BCKP supply via, e.g. a battery.

V_BCKP must be supplied to maintain the backup domain (BBR and RTC) to allow better TTFF, accuracy, availability and power consumption at the next startup compared with a cold start. As V_IO is not supplied, the PIOs cannot be driven by an external host processor. If driving of the PIOs cannot be avoided, buffers are required for isolating the PIOs.

3.7.4.2 Software standby mode

Software standby mode is entered using the UBX-RXM-PMREQ message. V_IO and VCC must be supplied, however VCC supply is internally disabled to save power. The V_IO supply maintains the battery-backed RAM (BBR), RTC, and PIOs.

Entering the software standby mode clears the RAM memory including the FW image, and the receiver configuration. To maintain the configuration, store it on BBR or flash layers. For more information on permanence of the stored configuration, refer to Receiver configuration.

The software standby mode can be set for a specific duration, or until the receiver is woken up by a signal at a wake-up source defined in UBX-RXM-PMREQ. The possible wake-up sources are UART RX and/or EXTINT pin. For more information on the UBX-RXM-PMREQ message, refer to the Interface description [3].

A system reset with the RESET_N signal also terminates the software standby mode, clears the BBR content and restarts the receiver.

As V_IO is supplied, the PIOs can be driven by an external host processor. No buffers are required for isolating the PIOs, which reduces cost.

☞ The LNA_EN signal is set to the "LOW" state during the software standby mode.

☞ The "force" flag must be set in UBX-RXM-PMREQ to enter software standby mode.

☞ Leave V_BCKP open if it is not used.

3.8 Time

Maintaining receiver local time and keeping it synchronized with GNSS time is essential for proper timing and positioning functionality. This section explains how the receiver maintains local time and introduces the supported GNSS time bases.

3.8.1 Receiver local time

The receiver is dependent on a local oscillator for both the operation of its radio parts and also for timing within its signal processing. No matter what nominal frequency the local oscillator has, u-blox receivers subdivide the oscillator signal to provide a 1-kHz reference clock signal, which is used to drive many of the receiver's processes. In particular, the measurement of satellite signals is arranged to be synchronized with the "ticking" of this 1-kHz clock signal.

When the receiver first starts, it has no information about how these clock ticks relate to other time systems; it can only count time in 1 millisecond steps. However, as the receiver derives information from the satellites it is tracking or from aiding messages, it estimates the time that each 1-kHz clock tick takes in the time base of the chosen GNSS system. This estimate of GNSS time based on the local 1-kHz clock is called receiver local time.

As receiver local time is a mapping of the local 1-kHz reference onto a GNSS time base, it may experience occasional discontinuities, especially when the receiver first starts up and the information it has about the time base is changing. Indeed, after a cold start, the receiver local time initially indicates the length of time that the receiver has been running. However, when the receiver obtains some credible timing information from a satellite or an aiding message, it jumps to an estimate of GNSS time.

3.8.2 GNSS time bases

GNSS receivers must handle a variety of different time bases as each GNSS has its own reference system time. What is more, although each GNSS provides a model for converting their system time into UTC, they all support a slightly different variant of UTC. So, for example, GPS supports a variant of UTC as defined by the US National Observatory, while BeiDou uses UTC from the National Time Service Center, China (NTSC). While the different UTC variants are normally closely aligned, they can differ by as much as a few hundreds of nanoseconds.

Although u-blox receivers can combine a variety of different GNSS times internally, the user must choose a single type of GNSS time and, separately, a single type of UTC for input (on EXTINT pins) and output (via the TIMEPULSE pin) and the parameters reported in corresponding messages.

The CFG-TP-TIMEGRID_TP* configuration item allows the user to choose between any of the supported GNSS (GPS, Galileo, BeiDou, etc.) time bases and UTC. Also, the CFG-NAVSPG-UTCSTANDARD configuration item allows the user to select which variant of UTC the receiver should use. This includes an "automatic" option which causes the receiver to select an appropriate UTC version itself, based on the GNSS constellation.The order of preference is:

- USNO if GPS is enabled
- NTSC if BeiDou is enabled
- NPLI if NAVIC is enabled
- NICT if QZSS is enabled
- European if Galileo is enabled

The receiver assumes that an input time pulse uses the same GNSS time base as specified for the time pulse output. So, if the user selects Galileo time for time pulse output, any time pulse input must also be aligned to the Galileo time (or to the separately chosen variant of UTC). When UTC is selected for the time pulse output, any GNSS time pulse input is assumed to be aligned with GPS time.

☞ The receiver allows users to independently choose GNSS signals used in the receiver (using CFG-SIGNAL-*) and the input/output time base (using CFG-TP-*). For example, it is possible to instruct the receiver to use GPS and Galileo satellite signals to generate BeiDou time. This practice compromises time pulse accuracy if the receiver cannot measure the timing difference between the constellations directly and is therefore not recommended.

☞ The information that allows GNSS times to be converted to the associated UTC times is only transmitted by the GNSS at relatively infrequent periods. For example, GPS transmits UTC(USNO) information only once every 12.5 minutes. Therefore, if a time pulse is configured to use a variant of UTC time, after a cold start, substantial delays can be expected before the receiver has sufficient information to start outputting the time pulse.

Each GNSS has its own time reference for which detailed and reliable information is provided in the messages listed in the table below.

Time reference	Message
GPS time	UBX-NAV-TIMEGPS
BeiDou time	UBX-NAV-TIMEBDS
Galileo time	UBX-NAV-TIMEGAL
QZSS time	UBX-NAV-TIMEQZSS
UTC time	UBX-NAV-TIMEUTC

Table 32: GNSS time messages

3.8.3 Navigation epochs

Each navigation solution is triggered by the tick of the 1 kHz clock nearest to the desired navigation solution time. This tick is referred to as a navigation epoch. If the navigation solution attempt is successful, one of the results is an accurate measurement of time in the time base of the chosen GNSS system, called GNSS system time. The difference between the calculated GNSS system time and receiver local time is called the clock bias (and the clock drift is the rate at which this bias is changing).

In practice the receiver's local oscillator is not as stable as the atomic clocks to which GNSS systems are referenced and consequently clock bias tends to accumulate. However, when selecting the next navigation epoch, the receiver always tries to use the 1 kHz clock tick which it estimates to be closest to the desired fix period as measured in GNSS system time. Consequently, the number of 1 kHz clock ticks between fixes occasionally varies. This means that when producing one fix per second, there are normally 1000 clock ticks between fixes, but sometimes, to correct drift away from the GNSS system time, there are 999 or 1001 ticks.

The GNSS system time calculated in the navigation solution is always converted to a time in both the GPS and UTC time bases for output.

Clearly when the receiver has chosen to use the GPS time base for its GNSS system time, conversion to GPS time requires no work at all, but conversion to UTC requires knowledge of the number of leap seconds since GPS time started (and other minor correction terms). The relevant GPS-to-UTC conversion parameters are transmitted periodically (every 12.5 minutes) by GPS satellites, but can also be supplied to the receiver via the UBX-MGA-GPS-UTC aiding message.

When insufficient information is available for the receiver to perform any of these time base conversions precisely, predefined default offsets are used. Consequently, plausible times are nearly always generated, but they may be wrong by a few seconds (especially shortly after receiver start). Depending on the configuration of the receiver, such "invalid" times may well be output, but with flags indicating their state (e.g. the "valid" flags in UBX-NAV-PVT).

☞ To support multiple GNSS systems concurrently, u-blox receivers employ multiple GNSS system times and/or receiver local times. For reporting GNSS system time or the receiver local time, users are recommended to use messages that report UTC time instead of using UBX messages. Other messages are retained only to support backwards compatibility.

3.8.4 iTow timestamps

The original designers of GPS chose to express time/date as an integer week number (starting with the first full week in January 1980) and a time of week (TOW) expressed in seconds. Manipulating time/date in this form is far easier for digital systems than the more conventional year/month/day, hour/minute/second representation. Therefore, most GNSS receivers use this time representation internally, and convert it to a more conventional form at external interfaces. In many UBX messages, the iTOW field provides an externally visible example of the internal time representation.

All the main UBX-NAV messages (and some other messages) contain an iTOW field to indicate the GPS time when the navigation epoch occurred. Messages with the same iTOW value can be assumed to have come from the same navigation solution, and therefore, iTOW can be used to synchronize between these UBX messages. However, the iTOW values may not be valid (i.e., they may have been generated with insufficient conversion data). Therefore, it is not recommended to use the iTOW field for any other purpose.

If reliable absolute time information is required, use the UTC time related fields in the UBX-NAV-PVT message. Additionally, the UBX-NAV-PVT message contains information about the validity and the accuracy of the provided UTC time. See the section Time validity for further information.

☞ iTOW is always referenced to GPS time, and it should not be confused with the UTC representation.

☞ The iTOW timestamps are not compensated for the Leap seconds.

3.8.5 Time validity

Information about the validity of the time solution is given in the following form:

- **Time validity**: Information about time validity is provided in the `valid` flags (e.g. `validDate` and `validTime` flags in the UBX-NAV-PVT message). If these flags are set, the time is known and considered valid for use.
- **Time validity confirmation**: Information about confirmed validity is provided in the `confirmedDate` and `confirmedTime` flags in the UBX-NAV-PVT message. If these flags are

set, the time validity can be confirmed by using an additional independent source, meaning that the probability of the time to be correct is very high. Note that information about time validity confirmation is only available if the `confirmedAvai` bit in the UBX-NAV-PVT message is set.

☞ `validDate` means that the receiver has knowledge of the current date. However, it must be noted that this date might be wrong for various reasons. Only when the `confirmedDate` flag is set, the probability of the incorrect date information drops significantly.

☞ `validTime` means that the receiver has knowledge of the current time. However, it must be noted that this time might be wrong for various reasons. Only when the `confirmedTime` flag is set, the probability of incorrect time information drops significantly.

☞ `fullyResolved` means that the UTC time is known without full seconds ambiguity. When deriving UTC time from GNSS time the number of leap seconds must be known. It might take several minutes to obtain such information from the GNSS payload. When the one second ambiguity has not been resolved, the time accuracy is usually in the range of ~20s.

3.8.6 UTC representation

UTC time is used in many NMEA and UBX messages. In NMEA messages, time is always rounded to the nearest hundredth of a second and it is normally reported with two decimal places (e.g. 124923.52). Although compatibility mode (selected using CFG-NMEA-COMPAT) requires three decimal places, rounding to the nearest hundredth of a second remains, so the extra digit is always 0.

UTC time is also reported within some UBX messages, such as UBX-NAV-TIMEUTC and UBX-NAV-PVT. In these messages date and time are separated into seven distinct integer fields. Six of these (year, month, day, hour, min. and sec.) have fairly obvious meanings and are all guaranteed to match the corresponding values in NMEA messages generated by the same navigation epoch. This facilitates simple synchronization between associated UBX and NMEA messages.

The seventh field is called nano and it contains the number of nanoseconds by which the rest of the time and date fields need to be corrected to get the precise time. So, for example, the UTC time 12:49:23.521 would be reported as: hour: 12, min: 49, sec: 23, nano: 521000000.

It is however important to note that the first six fields are the result of rounding to the nearest hundredth of a second. Consequently the nano value can range from -5000000 (i.e. -5 ms) to +994999999 (i.e. nearly 995 ms).

When the nano field is negative, the number of seconds (and maybe minutes, hours, days, months or even years) have been rounded up. Therefore, some or all of them must be adjusted to get the correct time and date. Thus in an extreme example, the UTC time 23:59:59.9993 on 31st December 2011 would be reported as: year: 2012, month: 1, day: 1, hour: 0, min: 0, sec: 0, nano: -700000.

If a resolution of one hundredth of a second is adequate, negative nano values can simply be rounded up to 0 and effectively ignored.

The UBX-NAV-TIMEUTC message gives information about the UTC time reference clock.

The preferred variant of UTC time can be specified using the CFG-NAVSPG-UTCSTANDARD configuration item. The UTC time variant configured must correspond to a GNSS that is currently enabled. Otherwise the reported UTC time is inaccurate.

3.8.7 Leap seconds

Due to the slightly uneven spin rate of the Earth, UTC time gradually moves out of alignment with the mean solar time (that is, the sun no longer appears directly overhead at 0 longitude at midday).

Occasionally, a "leap second" is announced to bring UTC back into close alignment with the mean solar time. Usually this means adding an extra second to the last minute of the year, but this can also happen on 30th June. When this happens, UTC clocks are expected to go from 23:59:59 to 23:59:60, and only then on to 00:00:00.

It is also possible to have a negative leap second, in which case there will only be 59 seconds in a minute and 23:59:58 will be followed by 00:00:00.

u-blox receivers are designed to handle leap seconds in their UTC output and consequently applications processing UTC times from either NMEA or UBX messages should be prepared to handle minutes that are either 59 or 61 seconds long.

Leap second information can be polled from the receiver with the message UBX-NAV-TIMELS.

3.8.8 Date ambiguity

Each navigation satellite transmits the current date and time in the navigation message of the signal. The time of week (TOW) indicates the number of seconds elapsed since the start of the week (midnight between Saturday and Sunday). The week number (WN) indicates the number of weeks elapsed since the particular GNSS system was started. By combining these two values, the current date and time can be determined.

When a new week begins, the time of week numbering resets to zero, and the week number increments by one. Since the week number continuously increases, it eventually reaches the maximum value and must "roll over" back to zero. Modern satellite systems, such as GPS L1C, L2C, and L5, Galileo, and BeiDou transmit sufficient information to ensure the week number remains unambiguous for the foreseeable future. The first rollover will occur in 2137 for the modern GPS, 2078 for Galileo, and 2163 for BeiDou.

Unfortunately, when the original GPS L1C/A signal was designed, only 10 bits were allocated for the week number. As a result, the transmitted week number value "rolls over" back to zero every 1024 weeks (just under 20 years). Consequently, the GPS L1 signal cannot differentiate between years like 1980, 1999, 2019, or 2038. To calculate the correct week number, GPS L1 receivers must use additional methods. The receiver can obtain additional information from its permanent configuration or from the host to resolve the ambiguity. If the receiver initially gets an ambiguous date from the GPS L1 signal, it can later acquire a modern signal to resolve the ambiguity in the GPS date.

☞ The receiver does not use date information for navigation, so rollover dates do not affect navigation calculations. The date is only used to present information in a convenient format for the host application.

☞ GPS time starts on January 6, 1980; Galileo on August 22, 1999; and BeiDou on January 1, 2006. Do not test the receiver with simulated signals before these dates.

3.8.8.1 GPS-L1-only date resolution

If the receiver is configured to use only the GPS L1C/A signal, or if the reception of other signals is delayed during startup, only the week number information from the GPS L1C/A signal will be available. In this case, the receiver establishes the date by assuming that all week numbers must be at least as large as the configured reference rollover week number. The default value for the reference rollover week number is selected at the compile time of the receiver firmware and is typically set to a value a few weeks before the software is completed. This value can be overridden by the CFG-NAVSPG-WKNROLLOVER configuration item.

The following example illustrates how this works:

Assume that the reference rollover week number set in the firmware at compile time is 2148 (which corresponds to a week in calendar year 2021, but is transmitted by the satellites as 100). In this case, if the receiver sees transmissions containing week numbers in the range of 100 to 1023, they are interpreted as week numbers 2148 to 3071 (calendar years 2021 to 2038). Transmissions with week numbers from 0 to 99 are interpreted as week numbers 3072 to 3171 (calendar years 2038 to 2040). With this method, the receiver shows the correct date for at least 19 years after the firmware creation.

☞ When supplying the receiver with simulated signals, it is important to set the reference rollover week number correctly, especially when the scenarios are in the past.

3.9 Time mark

The receiver can be used to provide an accurate measurement of the time at which a pulse was detected on the external interrupt pin. The reference time can be chosen by setting the time source parameter to UTC, GPS, BeiDou, Galileo, NAVIC or local time in the CFG-TP-* configuration group. The UTC standard can be set in the CFG-NAVSPG-* configuration group. The delay figures defined with CFG-TP-* are also applied to the results output in the UBX-TIM-TM2 message.

A UBX-TIM-TM2 message is output at the next epoch if

- The UBX-TIM-TM2 message is enabled, and
- a rising or falling edge was triggered since last epoch on the EXTINT pin.

The UBX-TIM-TM2 messages includes the time of the last time mark, new rising/falling edge indicator, time source, validity, number of marks and an accuracy estimate.

☞ Only the last rising and falling edge detected between two epochs is reported since the output rate of the UBX-TIM-TM2 message corresponds to the measurement rate configured with CFG-RATE-MEAS (see Figure 10 below).

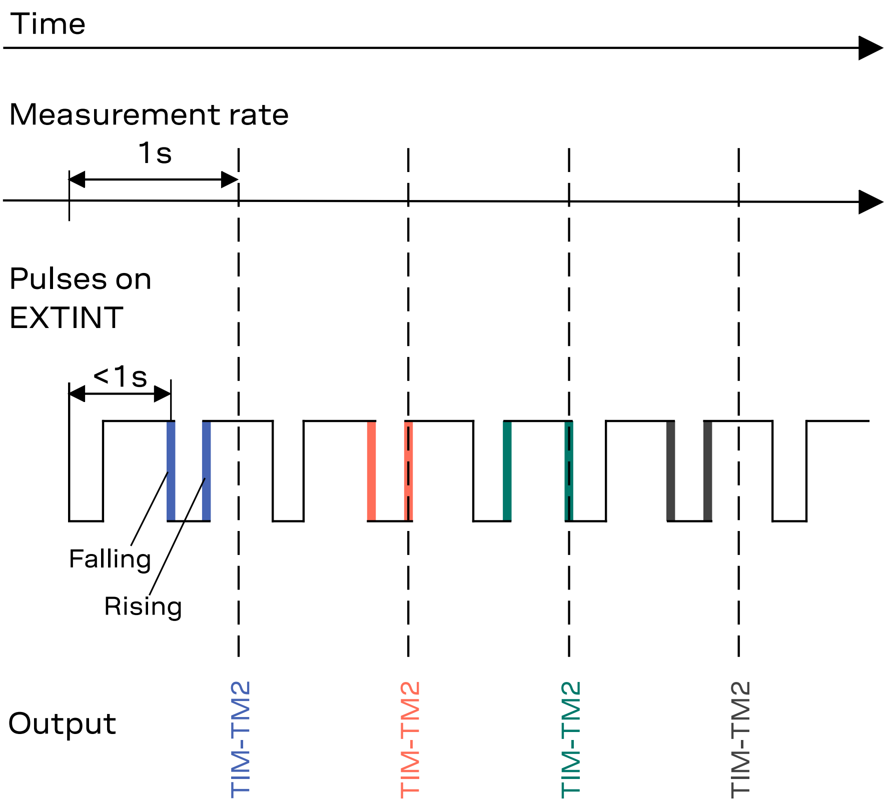


Figure 10: Time mark

3.10 Time pulse

The receiver includes a time pulse feature providing clock pulses with configurable duration and frequency. The time pulse function can be configured using the CFG-TP-* configuration group. The UBX-TIM-TP message provides time information for the next pulse and the time source.

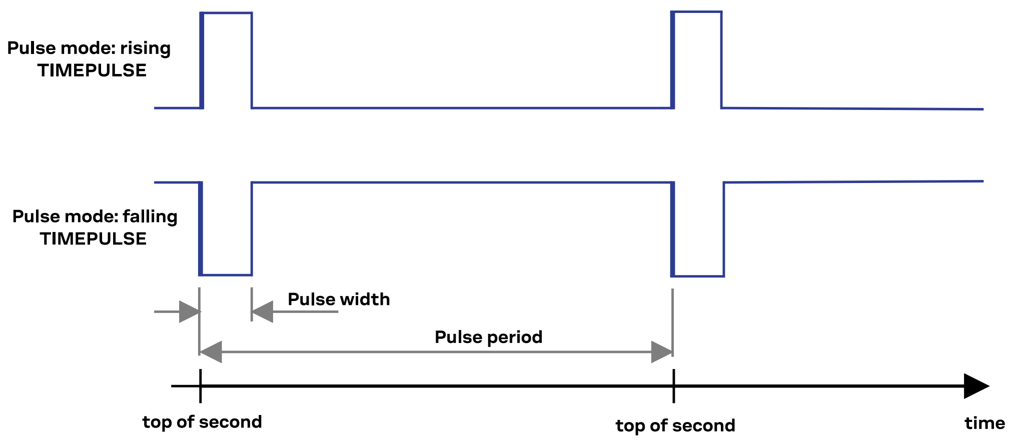


Figure 11: Time pulse

3.10.1 Recommendations

- The time pulse can be aligned to a wide variety of GNSS times or to variants of UTC derived from them. For further information, see GNSS time bases. However, it is strongly recommended that the choice of time base is aligned with the available GNSS signals (for example, to produce GPS time or UTC(USNO), ensure GPS signals are available, and for Galileo time or UTC(EU) ensure the presence of Galileo signals, etc). This involves coordinating the setting of CFG-SIGNAL-* configuration group with the choice of time pulse time base.
- When using time pulse for precision timing applications it is recommended to calibrate the antenna cable delay against a reference timing source.
- To get the best timing accuracy with the antenna, a fixed and accurate position is needed.
- If relative time accuracy between multiple receivers is required, do not mix receivers of different product families. If this is required, the receivers must be calibrated accordingly, by setting cable delay and user delay.
- The recommended configuration when using the UBX-TIM-TP message is to set both the measurement rate (CFG-RATE-MEAS) and the time pulse frequency (CFG-TP-*) to 1 Hz.

The sequential order of the signal present at the TIMEPULSE pin and the respective output message for the simple case of 1 pulse per second (1PPS) is shown in the following figure.

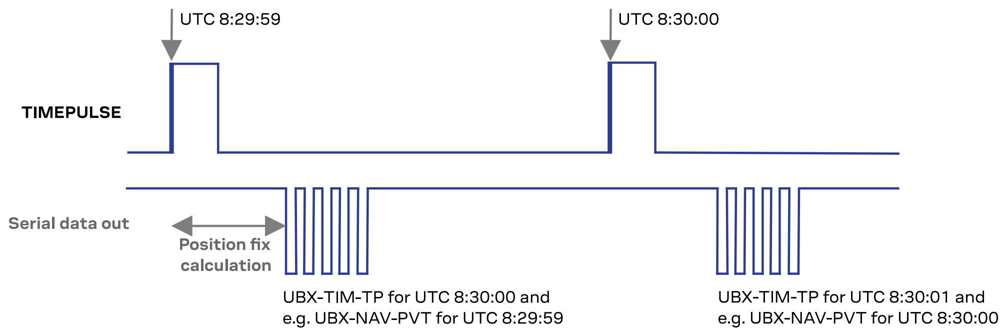


Figure 12: Time pulse and TIM-TP

3.10.2 Time pulse configuration

The time pulse (TIMEPULSE) signal has configurable pulse period, length and polarity (rising or falling edge).

It is possible to define different signal behavior (i.e. output frequency and pulse length) depending on whether or not the receiver is locked to reliable time source.

The configuration group CFG-TP-* can be used to change the time pulse settings, and includes the following parameters defining the pulse:

- **time pulse enable** - If this item is set, the time pulse is active.
- **frequency/period type** - Determines whether the time pulse is interpreted as frequency or period.
- **length/ratio type** - Determines whether the time pulse length is interpreted as length [us] or pulse ratio [%].
- **antenna cable delay** - Signal delay due to the cable between the antenna and the receiver.
- **pulse frequency/period** - Frequency or pulse time period when locked mode is not configured or not active.
- **pulse frequency/period lock** - Frequency or pulse time period for locked mode. In use as soon as the receiver has calculated a valid time from a received signal. Only used if the corresponding item is set to use another setting in locked mode.
- **pulse length/ratio** - Length or duty cycle of the generated pulse, specifies either time or ratio for the pulse to be on/off.
- **pulse length/ratio lock** - Length or duty cycle of the generated pulse for locked mode. In use as soon as the receiver has calculated a valid time from a received signal. Only used if the corresponding item is set to use another setting in locked mode.
- **user delay** - The cable delay from the receiver to the user device plus signal delay of any user application.
- **lock to GNSS freq** - If this item is set, uses the frequency gained from the GNSS signal information rather than the local oscillator's frequency.
- **locked other setting** - If this item is set, the alternative setting is used as soon as the receiver can calculate a valid time. This mode can be used, for example, to disable time pulse if the time is not locked, or to indicate a lock with different duty cycles.
- **align to TOW** - If this item is set, pulses are aligned to the top of a second.
- **polarity** - If set, the first edge of the pulse is a rising edge (pulse polarity: rising).
- **grid UTC/GNSS** - Selection between UTC (0), GPS (1), BeiDou (3), (4) Galileo and NAVIC (5) time grid. Also affects the time output by UBX-TIM-TP message.

☞ The maximum pulse length cannot exceed the pulse period.

☞ The high and the low period of the output cannot be less than 50 ns, otherwise pulses can be lost.

3.10.2.1 Example

The example below shows the 1PPS TIMEPULSE signal generated on the time pulse output according to the specific parameters of the CFG-TP-* configuration group:

- **CFG-TP-TP1_ENA** = 1
- **CFG-TP-PULSE_DEF** = 0 (PERIOD)
- **CFG-TP-PULSE_LENGTH_DEF** = 1 (LENGTH)
- **CFG-TP-PERIOD_TP1** = 1 000 000 μs
- **CFG-TP-LEN_TP1** = 100 000 μs
- **CFG-TP-TIMEGRID_TP1** = 1 (GPS)
- **CFG-TP-ALIGN_TO_TOW_TP1** = 1
- **CFG-TP-USE_LOCKED_TP1** = 1
- **CFG-TP-POL_TP1** = 1
- **CFG-TP-PERIOD_LOCK_TP1** = 1 000 000 μs

- **CFG-TP-LEN_LOCK_TP1** = 100 000 μs

The 1 Hz output is maintained whether or not the receiver is locked to GPS time. The alignment to TOW can only be maintained when GPS time is locked.

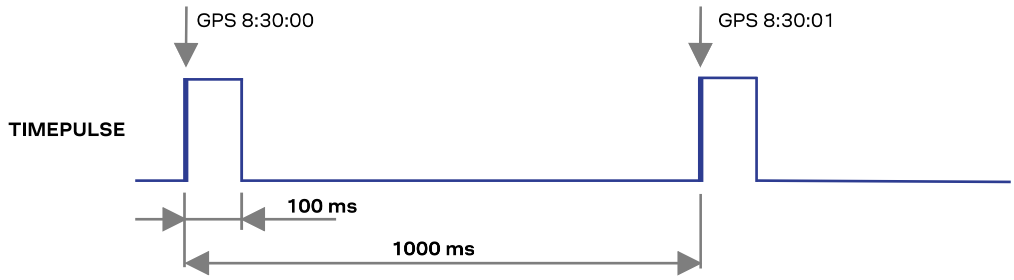


Figure 13: Time pulse signal with the example parameters

3.11 Time maintenance

Maintaining accurate time can improve the speed and performance of the receiver restart. Estimate of GNSS time can be maintained by a real-time clock, or it can be provided to the receiver by the host. Estimate of the clock drift of the receiver local oscillator or an external reference frequency can also be provided to improve the startup performance.

3.11.1 Real-time clock

The receiver contains a real-time clock (RTC). The RTC section is located in the backup domain and can keep time while the receiver is otherwise powered off. When the receiver powers up, it attempts to use the RTC to initialize receiver local time and in most cases this leads to considerably faster and more accurate first fixes.

In the case of RTC absence, the MAX-M10N provides an improved time to first fix (TTFF) when booting back from the backup mode. The receiver uses the time of the last position that is stored in the BBR before the receiver is powered off to estimate the current time at the next startup. This improved TTFF only occurs with C/N0 levels around 40 dBHz and for off periods in the range of a few minutes.

3.11.2 Time assistance

The host can deliver time assistance to the receiver using UBX-MGA-INI-TIME_UTC or UBX-MGA-INI-TIME_GNSS for better startup performance.

The current GNSS time can be supplied to the receiver as a coarse value via the standard communication interfaces. This method suffers from communication latency and unpredictable delays so the accuracy of the supplied time is poor. The time aiding is also described in the Transferring assistance data to the receiver.

Accuracy of the supplied time can be improved greatly if the host system has a very good sense of the current time and can deliver an exactly timed pulse to the EXTINT pin. This pulse informs the receiver when the supplied time assistance data is to be applied.

UTC time leap seconds and GPS-to-UTC conversion parameters are transmitted periodically by GPS satellites, but that happens only every 12.5 minutes. The receiver can normally calculate the correct leap seconds value from other GNSS systems immediately, but in some situations that is not possible. If the leap seconds information or the difference of time between GPS and UTC system is important for the host application, the information can be supplied to the receiver via the UBX-MGA-GPS-UTC aiding message.

3.11.3 Frequency assistance

Frequency assistance can improve the cold start speed in crystal-based designs. For TCXO-based designs, the frequency assistance has only minimal impact as the receiver is quick to acquire accurate frequency from satellite transmissions. A stable external reference frequency can be used to speed up receiver testing in production test setup.

To supply hardware frequency assistance, connect a periodic rectangular signal with a frequency of up to 500 kHz to the EXTINT pin. The frequency can have an arbitrary duty cycle but the low/high phase duration must not be shorter than 50 ns. The applied frequency value must be submitted to the receiver using the UBX-MGA-INI-FREQ message.

3.11.4 Clock drift assistance

Estimate of the clock drift of the external local oscillator (TCXO or XTAL) can also be fetched from the receiver using the UBX-NAV-CLOCK message. This estimate can then be sent back to the receiver using the UBX-MGA-INI-CLKD message.

3.12 AssistNow GNSS assistance

GNSS receivers require information on the exact time as well as satellite orbits for several satellites to obtain a position fix. At start-up, the receiver finds the satellite signals and decodes the information from the signals. This typically takes 20–30 s under good signal conditions. Under adverse signal conditions, finding signals and decoding the data may take several minutes or even completely fail.

Assistance data enables the receiver to find satellites faster and even compute the position without the need to download data from satellites. This significantly reduces the time-to-first-fix (TTFF), improves fix accuracy, and increases satellite availability, even under poor signal conditions.

The AssistNow service is a u-blox proprietary assisted GNSS (A-GNSS) service compatible with the u-blox GNSS receivers. The AssistNow Live Orbits and the AssistNow Predicted Orbits services are accessed over the HTTP or HTTPS protocols.

The AssistNow Autonomous feature runs locally on the receiver and generates mid- and long-term predicted orbits without an internet connection. The orbit prediction is based on broadcast ephemerides.

For more information on the AssistNow services, refer to the AssistNow service documentation [6].

Table 33 summarizes the AssistNow feature and services.

	AssistNow Live Orbits	AssistNow Predictive Orbits	AssistNow Autonomous
Type of feature	Service	Service	Stand-alone feature
Internet access	Always available	Sporadic	Not needed
Data types[3]	EPH, ALM, TIME, AUX	ANO, ALM	ANO, ALM
Data validity	2–4 hours	1–14 days	3–6 days
Data transfer size	Medium	High	None (local)
Flash memory	Not required	Optional, receiver or host	Optional, receiver or host

3 Data types: ephemeris (EPH), almanac (ALM), predictive orbits (ANO), auxiliary data (AUX), and time (TIME)

	AssistNow Live Orbits	AssistNow Predictive Orbits	AssistNow Autonomous
TTFF performance	Best	Good	Good/fair

Table 33: AssistNow service overview

3.12.1 AssistNow Live Orbits

AssistNow Live Orbits is used in GNSS receiver systems with direct internet access. At start-up, the host downloads the satellite ephemerides, time aiding, and other optional data from the service and sends it to the receiver. The assistance data can reduce the TTFF down to 1–2 s under good signal conditions.

For the supported GNSS constellations and signals, refer to the MAX-M10N Data sheet [1].

3.12.1.1 Live Orbits data types

Fast signal acquisition requires time information as well as satellite and orbital data. Current time is also needed to determine the validity and use other assistance data.

The time assistance message UBX-MGA-INI-TIME_UTC contains the current time and the accuracy for the time provided. Time accuracy is set by default to 10 s to consider the network latency and any processing delay on the host. The ephemerides contain precise short-term satellite and orbital data. The ephemerides are valid approximately 2–4 hours depending on the GNSS system.

Almanac contains a reduced-precision subset of the clock and ephemeris parameters. It is valid for a few weeks and enables the receiver to find the currently visible satellites faster. AssistNow Live Orbits provides also auxiliary information on satellite health status, ionospheric corrections to improve position accuracy, and time information on different GNSS systems.

The total AssistNow Live Orbits data size is approximately 2–4 kB per GNSS constellation. For detailed information on the supported GNSS constellations and satellites, data types, validity period, data size, and the full list of messages, refer to the AssistNow service documentation [6].

Table 34 summarizes the AssistNow Live Orbits assistance data and the related messages.

Type of data	Message(s)	Description
Time assistance (TIME)	UBX-MGA-INI-TIME_UTC	Current time (coarse)
Ephemeris (EPH)[4]	UBX-MGA-XXX-EPH	Current satellite and orbital data for fix calculation
Almanac (ALM)[4]	UBX-MGA-XXX-ALM	Coarse orbital information
Auxiliary data (AUX)[4]	UBX-MGA-XXX-IONO UBX-MGA-XXX-HEALTH UBX-MGA-XXX-UTC UBX-MGA-XXX-TIMEOFFSET	Simplified ionospheric corrections (Klobuchar) Satellite health information UTC time information Time offset relative to the GPS time

Table 34: AssistNow Live Orbits assistance data

3.12.1.2 Host operation

The host operation consists of the following steps:

- Start up the receiver
- Download the AssistNow Live Orbits assistance data from the service
- Transfer the Assistance data to the receiver immediately at start-up

The AssistNow C Client Toolkit [6] helps in implementing the AssistNow data download and transfer in the host application.

4 The string XXX stands for the GNSS constellations GPS, GAL, BDS, GLO, or QZSS. Almanac and auxiliary data are not available for all GNSS constellations.

Downloading and storing assistance data
The host downloads the AssistNow Live Orbits assistance data from the service. The received data consists of UBX messages starting with time assistance followed by ephemerides, almanac, and auxiliary data.

The host can store the assistance data in the memory and use it for subsequent receiver start-ups until it expires. In addition, the receiver can maintain the assistance data and other information in its battery-backed RAM (BBR) memory.

Data download from the service is not required if valid assistance data is already available on the host system or on the receiver. However, it is recommended to replace ephemerides close to expiry with fresh data from the service.

For more information on maintaining data on the receiver, refer to the section Preserving AssistNow and operational data during power-off.

Transferring assistance data to the receiver
The assistance data must be sent to the receiver immediately at the receiver start-up. If downloading data from the service, send the messages directly after receiving them in the same order they were received: time assistance followed by ephemerides, almanac, and auxiliary data.

☞ Send the assistance data immediately at the receiver start-up

If valid assistance data stored on the host is used, the host must construct the UBX-MGA-INI-TIME_UTC message with current time and accuracy for it. The time accuracy provided must not be better than the actual accuracy of the host time.

☞ Do not provide too optimistic value for the time accuracy. This may degrade the start-up performance.

The internal time maintained in the receiver is generally significantly more accurate than the assisted time. If the receiver already has time information available, it therefore ignores the time assistance and continues to use its internal time. To force the receiver use the assisted time, clear the real-time-clock (RTC) time in the BBR memory before sending time assistance.

Apart from time assistance, sending new assistance data from the host clears the existing data in the receiver. Valid assistance data is stored in the receiver's BBR memory and used accordingly. However, expired assistance data is rejected resulting in a loss of assistance data.

☞ Do not send ephemerides close to expiring to the receiver. This may degrade the start-up performance.

⚠ **CAUTION** Due to its internal duty cycle in the LEAP power save mode, the receiver may not receive some messages from the host. If a message is not acknowledged, resend it. Alternatively, switch to the full power mode (setting CFG-PM-OPERATEMODE = 0) before transmitting the assistance data, and revert to the LEAP mode (CFG-PM-OPERATEMODE = 2) once the transmission is complete. This issue is limited to the UART communication interface.

3.12.2 AssistNow Predictive Orbits

AssistNow Predictive Orbits is used in GNSS receiver applications with infrequent internet access. The host downloads assistance data from the service and stores it on the host side to be available for use at every start-up. The data can also be stored on a flash memory connected to the receiver. The assistance data can reduce the TTFF to well below 10 s under good signal conditions.

For the supported GNSS constellations and signals, refer to the MAX-M10N Data sheet [1].

3.12.2.1 Predictive Orbits data types

AssistNow Predictive Orbits provides medium-term assistance data for fast signal acquision. The data can be used for several receiver start-ups over a long period of time.

The time assistance message UBX-MGA-INI-TIME_UTC containing current time is only provided with Galileo or GLONASS almanac. The time accuracy is set by default to 10 s to consider the network latency and any processing delay on the host. Time assistance is only valid for an receiver start-up directly after the data download.

The UBX-MGA-ANO messages contain medium-term orbital data valid up to 14 days. There are one or more UBX-MGA-ANO messages for each satellite depending on the selected validity period. Each UBX-MGA-ANO message contains information on the GNSS constellation (gnssId), satellite ID (svId), and date (year, month, day). Time (hour) is always set to UTC 12:00. The data is most accurate at the time indicated.

Almanac contains a reduced-precision subset of the clock and ephemeris parameters. It is valid for a few weeks and enables the receiver to find the currently visible satellites faster. Refer to the Interface description for more information on the UBX-MGA-* messages.

The total AssistNow Predictive Orbits data size is of the order of 2 - 3 kB per one day and 35 kB per 14 days per GNSS constellation. For detailed information on the supported GNSS constellations and satellites, data types, validity period, data size, and the full list of messages, refer to the AssistNow service documentation [6].

Table 35 summarizes the AssistNow Predictive Orbits assistance data and the related messages.

Type of data	Message(s)	Description
Time assistance (TIME)	UBX-MGA-INI-TIME_UTC	Current time (coarse). Provided only with Galileo and GLONASS almanac.
Predictive Orbits (ANO)	UBX-MGA-ANO	Medium-term satellite and orbital data. Valid for 1 – 14 days.
Almanac (ALM)[5]	UBX-MGA-XXX-ALM	Coarse orbital information

Table 35: AssistNow Predictive Orbits assistance data

3.12.2.2 Host operation

The host has two main tasks:

- Download the AssistNow Predictive Orbits assistance data from the service
- Transfer the assistance data to the receiver. There are two options available: the flash-based operation and the host-based operation.

⚠ **CAUTION** Due to its internal duty cycle in the LEAP power save mode, the receiver may not receive some messages from the host. If a message is not acknowledged, resend it. Alternatively, switch to the full power mode (setting CFG-PM-OPERATEMODE = 0) before transmitting the assistance data, and revert to the LEAP mode (CFG-PM-OPERATEMODE = 2) once the transmission is complete. This issue is limited to the UART communication interface.

The AssistNow C Client Toolkit [6] helps in implementing the AssistNow data download and transfer in the host application.

Downloading assistance data

The host downloads the AssistNow Predictive Orbits assistance data from the service. The received data consists of UBX messages starting with time assistance (optional) followed by almanac and

5 The string XXX stands for the GNSS constellations GPS, GAL, BDS, GLO, or QZSS. Almanac is not available for all GNSS constellations.

Predictive Orbits data. The Predictive Orbits data is chronologically ordered starting with the earliest date.

The Predictive Orbits data is typically valid for several receiver start-ups. Data download from the service is not required if valid assistance data is already available on the host system or on the receiver.

Flash-based operation

The flash-based operation requires a flash memory connected to the receiver. The host downloads the assistance data from the service and sends it to the receiver to store in the flash memory. At receiver start-up, the receiver automatically applies valid assistance data.

The host sends the data to the receiver using the UBX-MGA-FLASH-DATA message. Due to the large amount of data and the slow write process to flash, the host must split the assistance data into 512-byte blocks. The last data block can be smaller than 512 bytes.

Typical steps to download assistance data and transfer it to the flash memory include:

- Download Predictive Orbits data from the service
- Split the data into 512-byte blocks
- Send each block with the UBX-MGA-FLASH-DATA message
- After each block sent, wait for the UBX-MGA-FLASH-ACK message and take action based on the message content: send next block, resend previous block, or end the process
- Send the UBX-MGA-FLASH-STOP message once the data transfer is completed
- Wait for the final UBX-MGA-FLASH-ACK message. The receiver processes the assistance data for use, so there can be several seconds delay before the receiver sends the acknowledgement.

Any existing AssistNow Predictive Orbits data in the flash memory is deleted as soon as the first block of new data arrives. Useful data becomes available after the data transfer is complete.

Each block of data has a sequence number, starting from zero for the first block. To prevent invalid partial data downloads, the receiver does not accept blocks that are out of sequence.

At receiver start-up, the receiver automatically applies valid assistance data available in the flash memory. To assess the validity of the data, the receiver needs time information. If time is not maintained in the receiver, send the time aiding message UBX-MGA-INI-TIME_UTC with the current time immediately at the receiver start-up.

For more information on maintaining data on the receiver, refer to the section Preserving AssistNow and operational data during power-off.

☞ Time information is required at receiver start-up to use the assistance data.

Host-based operation

In host-based operation, the host downloads the assistance data from the service and stores it on the host system. At receiver start-up, the host selects valid assistance data and sends it to the receiver.

Typical steps in the host-based operation include:

- Download Predictive Orbits data from the service
- Select the UBX-MGA-ANO messages based on the GNSS constellation, satellite ID, and time that best matches the current time. There is only one UBX-MGA-XXX-ALM message for each satellite.
- At receiver start-up, construct and send the time aiding message UBX-MGA-INI-TIME_UTC (optional) followed by the Predictive Orbits and almanac data to the receiver.

☞ Do not provide too optimistic value for the time accuracy. This may degrade the start-up performance.

The internal time maintained in the receiver is generally significantly more accurate than the assisted time. If the receiver already has time information available, it therefore ignores the time assistance and continues to use its internal time. To force the receiver use the assisted time, clear the real-time-clock (RTC) time in the BBR memory before sending time assistance.

Apart from time assistance, sending new assistance data from the host clears the existing data in the receiver. Valid assistance data is stored in the receiver's BBR memory and used accordingly. However, expired assistance data is rejected resulting in a loss of assistance data.

For more information on maintaining data on the receiver, refer to the section Preserving AssistNow and operational data during power-off.

3.12.3 Preserving AssistNow and operational data during power-off

The receiver stores the assistance data from the host in the navigation database. During operation, the receiver collects satellite broadcast data and updates the database when new data becomes available. Also time and position are continuously updated.

The receiver automatically copies the satellite data to the BBR to maintain it during the receiver power-off. Alternatively, the host can store satellite data on the host side or in the flash memory connected to the receiver.

The following mechanisms can be used instead or in parallel to the AssistNow feature. The maintained assistance data can be used for subsequent receiver start-ups until it expires.

☞ The quality of the assistance data degrades over time resulting in longer TTFF and reduced initial position accuracy. Provide fresh assistance data for best start-up performance.

⚠ **CAUTION** Due to its internal duty cycle in the LEAP power save mode, the receiver may not receive some messages from the host. If a message is not acknowledged, resend it. Alternatively, switch to the full power mode (setting CFG-PM-OPERATEMODE = 0) before transmitting the assistance data, and revert to the LEAP mode (CFG-PM-OPERATEMODE = 2) once the transmission is complete. This issue is limited to the UART communication interface.

Maintain BBR and RTC time

The recommended way to preserve satellite and operational data during power-off is to maintain the BBR and RTC time. The satellite data, last position, user configuration, and calibration data are all stored in the BBR. The RTC must be present to maintain an estimate of time. The receiver automatically applies the data at the next startup.

☞ The backup domain must be supplied to maintain BBR and RTC time.

Database dump

The database dump is used to store the receiver's navigation database on the host. Before power-off, the host reads the navigation database by polling the UBX-MGA-DBD message and stores the received UBX-MGA-DBD messages for later use. At the next startup, the host sends time assistance and the UBX-MGA-DBD messages back to the receiver. For more information on the UBX-MGA-DBD message, refer to the Interface description [3].

Save-on-shutdown (SOS)

The save-on-shutdown (SOS) feature is used to store the BBR contents to a flash memory connected to the receiver. Before power-off, the host instructs the receiver to store the BBR contents in the flash memory. The data is automatically restored from the flash at the next startup. For more information on the SOS feature, refer to the section Save-on-shutdown.

3.12.4 AssistNow Autonomous

The *AssistNow Autonomous* feature provides a functionality similar to *AssistNow Predictive Orbits* without the need for a host and a connection. Based on a broadcast ephemeris downloaded from the satellite (or obtained by *AssistNow Live Orbits*), the receiver can autonomously (i.e. without any host interaction or online connection) generate an accurate satellite orbit representation ("AssistNow Autonomous data") that is usable for navigation much longer than the underlying broadcast ephemeris was intended for. This makes downloading new ephemeris or aiding data for the first fix unnecessary for subsequent startups of the receiver.

☞ The *AssistNow Autonomous* feature is disabled by default.

3.12.4.1 Concept

The *AssistNow Autonomous orbit* is a feature designed to extend the usefulness of satellite orbit data beyond the limited validity of broadcast ephemerides. A broadcast ephemeris, which is downloaded directly from a satellite, provides a highly accurate snapshot of the satellite's orbit—typically valid for around four hours in the case of GPS. However, once this period expires, the data diverges significantly from the satellite's true trajectory and can no longer be used reliably for positioning.

To overcome this limitation, *AssistNow Autonomous* generates long-term orbit predictions by extending one or more broadcast ephemerides. Although these predictions are not perfectly precise, they are accurate enough to support reliable navigation over multiple satellite revolutions. The predictive orbit data is created automatically and stored in the receiver's battery-backed memory (BBR), and optionally in external flash memory or on the host device. The number of satellites for which data can be stored depends on the receiver's configuration and may vary during operation.

This entire process is transparent to the user and does not interfere with the normal functioning of the receiver. All calculations are performed in the background. If no broadcast ephemeris is available at the time of navigation, *AssistNow Autonomous* uses the stored data to generate the necessary orbit segments for positioning. The system also ensures that the data remains current, minimizing the time needed for orbit calculations when navigation is initiated.

To maintain accuracy, *AssistNow Autonomous* automatically invalidates orbit data that has become too old and could lead to significant positioning errors. This expiration threshold can be configured using the key CFG-ANA-ORBMAXERR. The quality of orbit predictions improves when a satellite has been observed multiple times, though this enhancement requires the presence of flash memory. Better prediction quality also extends the period during which the data remains usable.

AssistNow Autonomous supports satellites from GPS, GLONASS, Galileo, and BeiDou systems. However, it excludes satellites with high orbital eccentricity, specifically those with eccentricity greater than 0.05, such as Galileo E18.

The Figure 14 illustrates the *AssistNow Autonomous* concept in a graphical way.

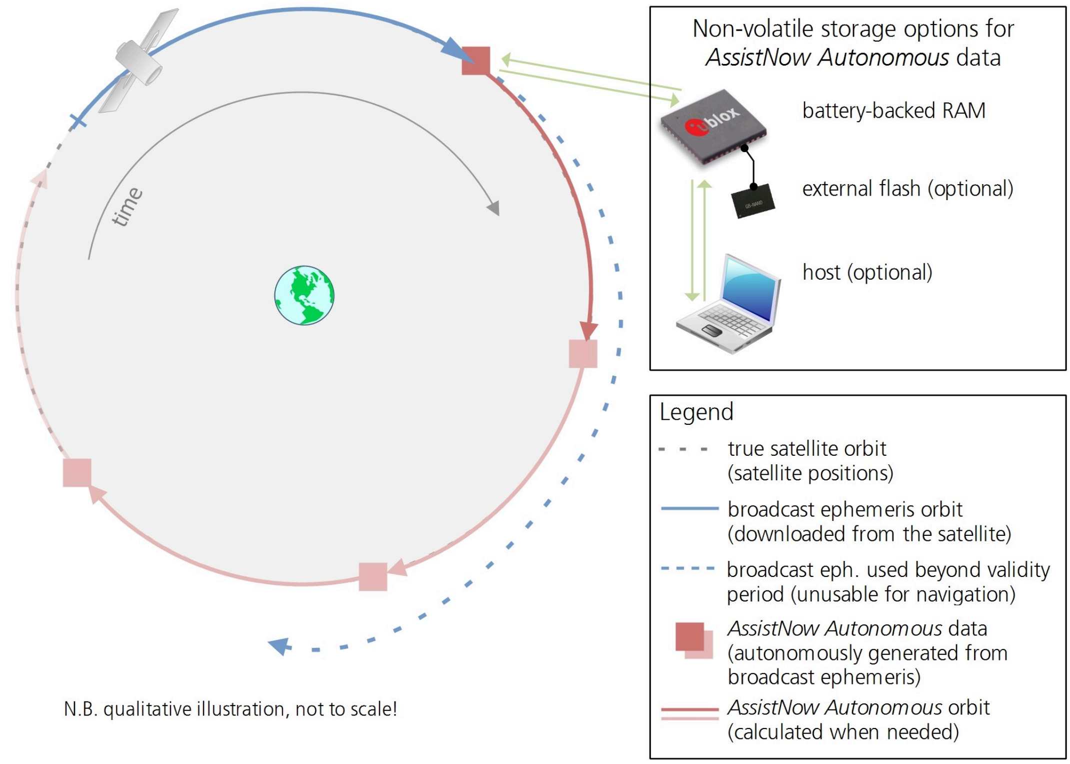


Figure 14: AssistNow Autonomous illustrative concept

3.12.4.2 Interface

The configuration keys and interface messages related to the *AssistNow Autonomous* are listed in Table 36.

Message name	Message Type	Description
CFG-ANA-USE_ANA	Configuration key	Used to enable or disable the *AssistNow Autonomous* feature. ☞ If the receiver uses flash memory, disabling the *AssistNow Autonomous* feature will delete all previously collected satellite observation data from the flash memory.
CFG-ANA-ORBMAXERR	Configuration key	Allows changing «maximum acceptable modeled orbit error» (in meters). Note that this number does not reflect the true orbit error introduced by extending the ephemeris. It is a statistical value that represents a certain expected upper limit based on a number of parameters. A rough approximation that relates the maximum extension time to this setting is: *maxError [m] = maxAge [d] * f*, where the factor *f* is 30 for data derived from satellites seen once and 16 for data derived for satellites seen multiple times during a long enough time period. It is recommended to use the firmware default value that corresponds to a default orbit data validity of approximately three days (for GPS satellites observed once) and up to six days (for satellites observed multiple times over a period of at least half a day).
UBX-NAV-AOPSTATUS	Output	Provides information on the current state of the *AssistNow Autonomous* subsystem. The status indicates whether the *AssistNow Autonomous* subsystem is currently idle (or not enabled) or busy generating data or orbits. Hosts should monitor this information and only power off the receiver when the subsystem is idle (that is, when the `status` field shows a steady zero).
UBX-NAV-SAT	Output	Indicates the use of *AssistNow Autonomous* orbits for individual satellites.
UBX-NAV-ORB	Output	Indicates the availability of *AssistNow Autonomous* orbits for individual satellites.

Message name	Message Type	Description
UBX-MGA-DBD	Input/Output	Provides a means to retrieve the *AssistNow Autonomous* data from the receiver to preserve the data in power-off mode where no battery backup is available. Note that the receiver requires the absolute time (i.e. full date and time) to calculate *AssistNow Autonomous* orbits. For the best performance, it is therefore recommended to supply this information to the receiver using the UBX-MGA-INI-TIME_UTC message in this scenario.

Table 36: AssistNow Autonomous related messages

The Save-on-Shutdown (SOS) feature preserves the *AssistNow Autonomous* data. For more information about this feature, see the interface description [3].

3.12.4.3 Benefits and drawbacks

AssistNow Autonomous can provide quicker startup times by lowering the TTFF, provided that data is available for enough visible satellites. This is particularly true under weak signal conditions where it might not be possible to download broadcast ephemerides at all and therefore, no fix would be possible without *AssistNow Autonomous* (or A-GNSS). It is however required that the receiver roughly knows the absolute time, either from an RTC or from time-aiding and that it knows which satellites are visible, either from the almanac or from tracking the respective signals.

The *AssistNow Autonomous* orbit (satellite position) accuracy depends on various factors, such as the particular type of satellite, the accuracy of the underlying broadcast ephemeris, or the orbital phase of the satellite and Earth, and the age of the data (errors add up over time).

There is no direct relation between (true and statistical) orbit accuracy and positioning accuracy. The positioning accuracy depends on various factors, such as the satellite position accuracy, the number of visible satellites, and the geometry (DOP) of the visible satellites. Position fixes that include *AssistNow Autonomous* orbit information may be significantly worse than fixes using only broadcast ephemerides. Therefore, it might be necessary to adjust the limits of the navigation output filters (CFG-NAVSPG-OUTFIL_*).

Unknown future events form a fundamental deficiency of any system and can prevent precise satellite orbit predictions. Hence, the receiver will not be able to know about satellites that will have become unhealthy, have undergone a clock swap, or have had a maneuver. This means that the navigation engine might rarely mistake a wrong satellite position as the true satellite position. However, provided that there are enough other good satellites, the navigation algorithms will eventually eliminate a defective orbit from the navigation solution.

The repeatability of the satellite constellation is a potential pitfall for the use of the *AssistNow Autonomous* feature. For a given location on Earth, the (GPS) constellation (geometry of visible satellites) repeats every 24 hours. Hence, when the receiver «learned» about a number of satellites at some point in time, the same satellites will in most places *not* be visible 12 hours later, and the available *AssistNow Autonomous* data will not be of any help. However, after another 12 hours, usable data would be available because it was generated 24 hours ago.

The longer a receiver observes the sky, the more satellites it will see. At the equator, and with full sky view, approximately ten (GPS) satellites will show up in a one-hour window. After four hours of observation approximately 16 satellites (i.e. half the constellation), after 10 hours approximately 24 satellites (2/3rd of the constellation), and after approximately 16 hours the full constellation will have been observed (and *AssistNow Autonomous* data generated). Lower sky visibility reduces these figures (i.e. the number of satellites seen). Further away from the equator, the numbers improve because the satellites can be seen twice a day. For example, at 47 degrees north, the full constellation can be observed in approximately 12 hours with full sky view.

The calculations required for *AssistNow Autonomous* are carried out on the receiver. This requires energy and therefore, users may occasionally see increased power consumption during short periods (several seconds, rarely more than 60 seconds) when such calculations are running. Ongoing calculations will automatically prevent the power save mode from entering the power-off state. The power-down will be delayed until all calculations are done.

☞ AssistNow Autonomous should be enabled if the system has sporadic access to the AssistNow Predictive Orbits service. In this case, the receiver chooses intelligently the more reliable orbit predictions for each satellite. This way the autonomous prediction can provide performance improvements if the data becomes old or gets outdated.

3.13 Save-on-shutdown

The save-on-shutdown feature (SOS) enables the u-blox receiver to store the contents of the battery-backed RAM (BBR) to an external flash memory and restore it upon startup. This allows the u-blox receiver to preserve some of the features available only with a battery backup (preserving configuration and satellite orbit knowledge) without having a battery backup supply present. However, the receiver does not preserve any kind of time knowledge. Save-on-shutdown must be commanded by the host. Refer to the MAX-M10N Interface description [3] for more information on the required UBX-UPD-SOS messages. The restoring of data on startup is automatically done if the corresponding data is present in the flash. Data expiration is not checked.

The following outlines the suggested shutdown procedure when using the SOS feature:

- With the UBX-CFG-RST message, the host commands the u-blox receiver to stop, specifying reset mode 0x08 ("Controlled GNSS stop") and a BBR mask of 0 ("Hotstart").
- The host commands the saving of the contents of BBR to the flash memory using the UBX-UPD-SOS-BACKUP message.
- For a valid request the u-blox receiver reports on the success of the backup operation with a UBX-UPD-SOS-ACK message.
- The host powers off the u-blox receiver.

The startup procedure is as follows:

- The host powers on the u-blox receiver.
- The u-blox receiver detects the previously stored data in the flash. It restores the corresponding memory and reports the success of the operation with a UBX-UPD-SOS-RESTORED message on the port on which it had received the save command message (if the output protocol filter on that port allows it). It does not report anything if no stored data has been detected.
- Additionally the u-blox receiver outputs a UBX-INF-NOTICE and/or a NMEA-TXT message with the contents `RESTORED` in the boot screen (depends on the configuration of the port and information messages) upon success.
- Optionally the host can deliver coarse time assistance using UBX-MGA-INI-TIME_UTC for better startup performance.

☞ It is recommended to delete the stored data using a UBX-UPD-SOS-CLEAR message once the u-blox receiver has started up. The u-blox receiver responds with a UBX-ACK-ACK / UBX-ACK-NAK message.

3.14 Data logging

The data logging feature allows position fixes and arbitrary byte strings from the host to be logged in the receiver's attached flash memory. Logging the position fixes happens independently of the host system, and can continue while the host is powered down.

Table 37 lists all the logging related messages.

Message	Type	Description
UBX-LOG-CREATE	Input	Creates the log file. Note that only one log at a time can be created. The logging only starts if CFG-LOGFILTER-RECORD_ENA = 1.
UBX-LOG-ERASE	Input	Erases the log file
UBX-LOG-INFO	Output	Provides information about the log file and logging activity
UBX-LOG-STRING	Input	Stores a string of bytes to a log entry. The message must be sent once the logging starts and the string is applied to a specific entry.
UBX-LOG-RETRIEVE	Input	Requests to retrieve the logged data
UBX-LOG-RETRIEVESTRING	Output	Reponse to UBX-LOG-RETRIEVE. Returns the index entries with a written string (if any).
UBX-LOG-RETRIEVEPOS	Output	The message is output after the request UBX-LOG-RETRIEVE is sent to the receiver. Each logged entry contains: • Entry index. The first entry is numbered as zero. • Latitude and longitude to a precision of one millionth of a degree. Depending on position on Earth this is a precision in the order of 0.1 m. • Altitude (height above mean sea level) determined with 0.1 m accuracy • Horizontal accuracy estimate serves as an indicator of the fix quality. Any accuracy less than 0.7 m is logged as 0.7 m and any value above 1 km is logged as 1 km. Within these limits, the logged accuracy is always greater than the actual fix accuracy (by up to 40%). • Ground speed with 1 cm/s accuracy • Heading with one degree accuracy • The fix type (only successful fix types are recorded) • The time and date of the fix with one second accuracy • The number of satellites used • Odometer distance data (if odometer is enabled)
UBX-LOG-RETRIEVEPOSEXTRA	Output	Reponse to UBX-LOG-RETRIEVE. This message is only output if the odometer is enabled and reports the odometer value.
UBX-LOG-FINDTIME	Input/ Output	UBX-LOG-FINDTIME-RESPONSE returns the index of the entry closer to a given date and time (defined by UBX-LOG-FINDTIME-REQUEST)

Table 37: Data logging related messages

3.14.1 Data logging configuration

Logging configuration, including start and stop control, is managed through the CFG-LOGFILTER configuration group. The configuration keys are listed in Table 38.

Configuration message	Description
CFG-LOGFILTER-RECORD_ENA	Set to 1 to start recording. Set to 0 to stop recording.
CFG-LOGFILTER-APPLY_ALL_FILTERS	Applies all logging-related filters
CFG-LOGFILTER-ONCE_PER_WAKE_UP_ENA	• Records only one single position after a wake up from PSMOO • The value set in this message takes effect only when CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled
CFG-LOGFILTER-TIME_THRS	• Minimum time between two consecutive log entries • The value set in this message takes effect only when CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled
CFG-LOGFILTER-MIN_INTERVAL	• Minimum time interval between two entries. If 0, it is not set. • This is only applied in combination with the speed and/or position thresholds • If both MIN_INTERVAL and TIME_THRS are set, the value of MIN_INTERVAL must be less than or equal to TIME_THRS • The value set in this message takes effect only when CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled

Configuration message	Description
CFG-LOGFILTER-SPEED_THRS	• It logs the entry if the current speed is greater than the configured speed threshold. If 0, it is not set. • CFG-LOGFILTER-MIN_INTERVAL also applies • The value set in this message takes effect only when CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled
CFG-LOGFILTER-POSITION_THRS	• It logs the entry if the distance between the current position and the previous entry is greater than the set threshold. If 0, it is not set. • CFG-LOGFILTER-MIN_INTERVAL also applies • The value set in this message takes effect only when CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled

Table 38: Data logging configuration messages

3.14.2 Logging operation

Log file creation

UBX-LOG-CREATE message creates an empty log file in the flash memory. Users can specify the log size in bytes using predefined values. Due to internal overhead from both the logging subsystem and the receiver's file storage, the actual space available for log entries is slightly less than the defined size.

Only one log file can exist at a time. An attempt to create a second log triggers a UBX-ACK-NAK response from the receiver.

This message also supports the creation of circular logs. In a circular log, older entries are overwritten by new ones once the memory log is full. In contrast, non-circular logs reject new entries when the memory limit is reached.

The logging activity only starts if CFG_LOGFILTER-RECORD_ENA = 1. If a logging task is active and the receiver is powered off, logging automatically resumes upon power-up or recovery from backup mode.

Log entry characteristics

Log entries are compressed and include housekeeping metadata, making their exact size variable. Typical entry sizes are:

- Minimum: 9 bytes
- Typical: 10-11 bytes
- Maximum: 24 bytes
- If odometer is enabled: +3 bytes per entry

Each log file also includes a fixed overhead, depending on its type:

- Circular: up to 40 kB
- Non-circular: up to 8 kB

To estimate the maximum number of entries that can be stored:

Approximate number of entries = (flash size available for logging - log overhead) / typical entry size

For example, if 1500 kB of flash memory is available for logging (after accounting for firmware and other uses) a non-circular log can store approximately:

*((1500*1024)-(8*1024))/11 = 138891* entries.

Only valid position fixes are logged. Invalid or failed fixes are excluded. To optimize compression, fix values are rounded, which may result in slight differences between logged and real-time data.

Monitoring and managing

UBX-LOG-INFO reports current logging status, including number of entries, timestamp of the first entry, and log size. This helps determine what data to retrieve and when.

Only valid position fixes are logged. Invalid or failed fixes are excluded. To optimize compression, fix values are rounded, which may result in slight differences between the logged and real-time data.

UBX-LOG-STRING allows the host application to embed a byte string into the next log entry for later identification.

Log data retrieval

To retrieve data, send the UBX-LOG-RETRIEVE message. The number of entries and the starting index must be specified. The receiver responds with one or more of the following messages:

- UBX-LOG-RETRIEVEPOS: One message per entry, containing detailed position fix data (see Table 37).
- UBX-LOG-RETRIEVEPOSEXTRA: Includes odometer data per entry (only if odometer is enabled).
- UBX-LOG-RETRIEVESTRING: Returns entries with stored byte strings, including index, timestamp, and string stored.

Use UBX-LOG-FINDTIME to locate the entry index closest to a specific timestamp. If no exact match is found, the message returns the last entry before the requested time. Once the index is known, fetch the data with the UBX-LOG-RETRIEVE message.

The maximum number of entries per retrieval request is 256. To retrieve more, send multiple requests with updated startNumber indexes.

For faster data transfer, increase the baud rate. Additionally, it is recommended to stop GNSS operation (UBX-CFG-RST) during data retrieval to prevent potential overwriting of entries in circular log configurations.

Stopping and erasing logs

Logging can be stopped at any time by setting CFG_LOGFILTER-RECORD_ENA = 0. The log file remains stored in flash memory until erased using the UBX-LOG-ERASE message.

3.15 Data batching

3.15.1 Introduction

The data batching feature allows position fixes to be stored in the RAM of the receiver to be retrieved later in one batch. Batching of position fixes happens independently of the host system, and can continue while the host is powered down.

Table 39 lists all the batching-related messages:

Message	Description
UBX-MON-BATCH	Provides information about the buffer fill level and dropped data due to overrun
UBX-LOG-RETRIEVEBATCH	Starts the batch retrieval process
UBX-LOG-BATCH	A batch entry returned by the receiver

Table 39: Batching-related messages

3.15.2 Setting up the data batching

Data batching is disabled per default and it has to be configured before use via the CFG-BATCH-* configuration group.

The feature must be enabled and the buffer size must be set to greater than 0. It is possible to set up a PIO as a flag that indicates when the buffer is close to filling up. The fill level when this PIO is asserted can be set by the user separately from the buffer size. The notification fill level must not be larger than the buffer size.

If the host does not retrieve the batched fixes before the buffer fills up, the oldest fix will be dropped and replaced with the newest.

The RAM available in the chip limits the size of the buffer. To make the best use of the available space, users can select what data they want to batch. When batching is enabled, a basic set of data is stored and the configuration flags EXTRAPVT and EXTRAODO can be used to store more detailed information about the position fixes. However, enabling the EXTRAPVT and EXTRAODO flags reduces the number of fixes that can be batched.

The receiver will reject the configuration if it cannot allocate the required buffer memory. To ensure robust operation of the receiver the limits in Table 40 are enforced:

EXTRAPVT	EXTRAODO	Maximum number of epochs
0	0	300
0	1	221
1	0	154
1	1	130

Table 40: Maximum number of batched epochs

☞ It is recommended to disable all periodic output messages when using data batching. This improves system robustness and also helps ensure that the output of batched data is not delayed by other messages.

☞ The buffer size is set up in terms of navigation epochs. This means that the time that can be covered with a certain buffer depends on the navigation rate. This rate can be set via the "CFG-RATE-" configuration group.

3.15.3 Retrieval

UBX-LOG-RETRIEVEBATCH message starts the process which allows the receiver to output batch entries. Batching must not be stopped for readout because all batched data is lost when the feature is disabled.

Batched fixes are always retrieved starting with the oldest fix in the buffer and progressing towards newer ones. There is no way to skip certain fixes during retrieval.

When a UBX-LOG-RETRIEVEBATCH message is sent the receiver transmits all batched fixes. It is recommended to send a retrieval request with `sendMonFirst` set. This way the receiver will send a UBX-MON-BATCH message first that contains the number of fixes in the batching buffer. This information can be used to detect when the u-blox receiver finishes sending data.

Once retrieval has started, the receiver will first send UBX-MON-BATCH message if `sendMonFirst` option was selected in the UBX-LOG-RETRIEVEBATCH message. After that, it will send UBX-LOG-BATCH messages with the batched fixes.

To maximize the speed of transfer, it is recommended that a high communication data rate is used.

☞ The receiver will discard retrieval request while processing a previous UBX-LOG-RETRIEVEBATCH message.

☞ The receiver does **not** acknowledge the reception of UBX-LOG-RETRIEVEBATCH message. The receiver responds with UBX-MON-BATCH (optional) and UBX-LOG-BATCH messages.

3.16 Geofencing

The geofencing feature allows for the configuration of up to four circular areas (geofences) on the Earth's surface (Figure 15). The receiver evaluates whether the current position lies within each of these areas, and signals the state via UBX messaging and PIO toggling (optional).

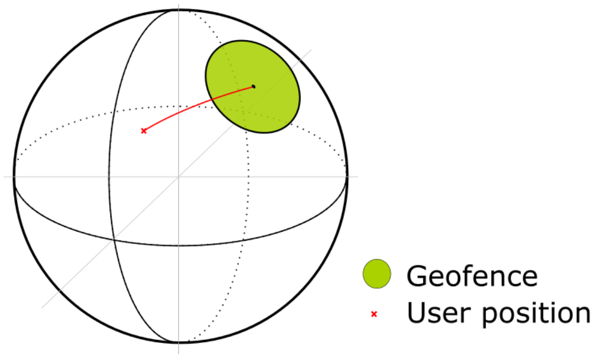


Figure 15: Geofence

3.16.1 Geofence interface

Geofencing can be configured using the CFG-GEOFENCE-* configuration group. The geofence evaluation is active whenever there is at least one geofence configured.

The receiver outputs the current state of each geofence and the combined state of the different geofence areas in the UBX-NAV-GEOFENCE message with every navigation epoch. Optionally, the combined geofence state can be assigned to a PIO to make the output available on a physical pin.

3.16.2 Geofence state evaluation

With every navigation epoch, the receiver evaluates the current solution's position against the configured geofences. There are three possible outcomes for each geofence:

- *Inside* - The position is inside the geofence with the configured confidence level
- *Outside* - The position lies outside of the geofence with the configured confidence level
- *Unknown* - There is no valid position solution or the position uncertainty does not allow for unambiguous state evaluation

The position solution uncertainty (standard deviation) is multiplied with the configured confidence sigma level and taken into account when evaluating the geofence state (red circle in Figure 16).

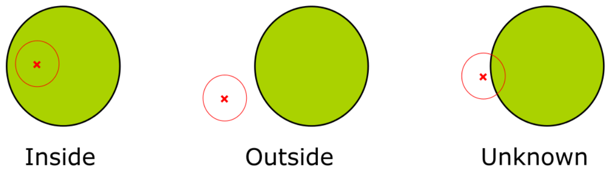


Figure 16: Geofence states

The combined state for all geofences is evaluated as the combination (union) of all geofences:

- *Inside* - The position lies inside of at least one geofence
- *Outside* - The position lies outside of all geofences
- *Unknown* - All remaining states

A pin can be configured to indicate the geofence status, allowing the host system to enter standby mode and wait for a wake-up signal from the receiver. For more information, see Geofence state.

3.17 CloudLocate

In CloudLocate setup, the host processor of the customer application fetches a set of satellite signal measurements from the receiver and sends those to the u-blox CloudLocate service for position calculation. The CloudLocate service uses these measurements and current assistance data to calculate the receiver position. This data is then provided to the customer enterprise cloud for further use. Power saving up to 90% is possible compared to a cold start scenario.

The receiver starts to collect measurements as soon as it finds any satellite signals. It does not need to wait for a position fix for this. Collecting the measurements takes only a short time, so the application can quickly turn off the receiver or put it into a backup state.

3.17.1 CloudLocate measurements

The satellite signal measurements can be requested from the receiver either as a complete or compact raw measurement message.

The complete raw measurement message (UBX-RXM-MEASX) provides measurements for all visible satellites. The customer application can wait until the number of satellites in the message is sufficient and then send the message to the CloudLocate service. The amount of data in a MEASX message for five satellites is about 170 bytes. This increases by 24 bytes for each additional satellite.

Compact raw messages (UBX-RXM-MEAS50, UBX-RXM-MEAS20, UBX-RXM-MEAS12C, and UBX-RXM-MEAS12D) can be used when the amount of data to be sent to the cloud needs to be minimized. The data can be reduced to 50, 20, or even 12 Bytes, and the time for the receiver to stay on shall be limited to the minimum as well. These messages contain the measurement data in compressed format and use satellites only from a limited set of GNSS constellations. With the default settings, these messages contain measurement data only for a small number of satellites.

The raw measurement messages are enabled with the configuration keys in the CFG-MSGOUT configuration group. For example, setting the configuration key CFG-MSGOUT-UBX_RXM_MEAS50_UART1 to value 1 with UBX-CFG-VALSET message enables output of the UBX-RXM-MEAS50 message in the UART1 port for each navigation epoch.

The UBX-RXM-MEASX message can be sent with UBX header and checksum, and the message can be either in binary format or as encoded text. With the compact raw measurement messages, only the payload portion of the message is sent. Encoding the data would increase the size, so the compact messages should be sent in binary format.

In adverse conditions, satellite signal reception may take a long time or may not be possible. In such a situation, the payload of a compact raw measurement message is empty. The host application can wait until the payload contains data and only then switch off the receiver. In the case of using UBX-RXM-MEASX, the payload always has some data. The host application must check if the message contains enough information to calculate a position. For example, the application can wait for a confirmation from the customer enterprise cloud.

For more information on the raw measurement messages and on using the CloudLocate setup, see the Interface description [3] and the u-blox website CloudLocate documentation.

3.18 Firmware update

u-blox may release updated firmware images containing, for example, security fixes, enhancements, bug fixes, etc. Therefore, it is important to implement a firmware update mechanism in the host system.

Contact u-blox for more information on firmware update.

3.19 Collecting debug logfiles: design-in guidance

The ability to collect logfiles with additional UBX protocol messages is essential throughout the MAX-M10N receiver's lifecycle: development, integration, validation, and field support. Enabling and collecting logfiles makes the integration and troubleshooting faster, more precise, and cost-effective, making it easier to resolve issues promptly, even if they occur in the field.

The additional messages offer deep insight into:

- Signal acquisition, tracking and navigation performance
- Protocol-level communication and interfaces
- Internal status and diagnostic data
- Unexpected behavior

To avoid support delays and ensure robust integration, incorporate logfile collection in both the hardware interface design and host software capabilities during the design-in phase.

3.19.1 Checklist for designing debug logfile collection

To ensure a successful and reliable integration of UBX message output within your system, follow the steps below. These cover both hardware and host configuration requirements.

Hardware interface

- Ensure at least one GNSS receiver interface is available for UBX message output.
- Provide access to an interface for external connection (for example to PC or data logger).
- Design the interface to handle the bandwidth required for UBX message output.
- Configure a baud rate that supports the expected debug message volume.

Host and configuration

- Enable the host to send UBX configuration strings to activate message output on the correct interface.
- Choose the message set based on the intended use case or guidance from u-blox technical support.
- Set up the host to store or stream the collected logfiles as needed.

Support and flexibility

- Request message definitions and estimated output payload sizes from u-blox technical support.
- Ensure updates or modifications to the message set are possible without firmware changes.

3.19.2 Hardware interface options for collecting debug logfiles

The receiver supports UBX message output over all available communication interfaces. When collecting logfiles, consider the scenarios in Table 41.

Scenario	Hardware design guidance
Logs collected by host	Use a communication interface (e.g. UART) connected to the host processor. The host enables messages and stores the logfile locally.
Logs collected by external device	Expose a communication interface via a debug header or test points to allow connection to a PC or external logger.

Table 41: Logging interface options by scenario

Interfaces must be available and accessible at runtime. Messages are explicitly enabled per interface using configuration commands.

Enabling UBX output messages on a specific interface:

By default, the GNSS receiver does not output UBX messages used for in-depth analysis. Select the required messages based on the specific application or follow guidance from u-blox's technical support. Activate the messages on the appropriate interface using configuration messages.

- u-blox technical support provides a tailored set of binary CFG-MSGOUT configuration strings to enable the required messages on the selected interface.
- They also provide an estimated output payload size to help you assess the interface bandwidth requirements.
- These UBX configuration strings are sent to the MAX-M10N either by the host application or an external PC and the messages are output on the selected interface.

3.19.3 Host application and configuration requirements for collecting debug logfiles

The host must:

- Transmit the provided UBX configuration to the GNSS receiver
- Manage the interface on which the messages are enabled (host-connected or externally exposed)
- Log, buffer or stream the collected messages as required
- Provide a means to enter a debug mode and switch message sets based on a use case or support guidance
- Monitor interface health using MON-TXBUF to detect potential overflows

Message throughput depends on the message set and GNSS activity. Select appropriate baud rates and buffering strategies to avoid data loss.

3.19.4 Recommended message groups by use case

Table 42 presents a preliminary set of a messages which can be enabled for logfile collection. These messages can assist technical support during the initial troubleshooting and help refine the list of debug messages to be provided for more detailed analysis.

Use case	Purpose	Typical UBX messages	Estimated interface load
General debug	OS-level issues, system exceptions	MON-HW, MON-COMMS, MON-SYS, MON-EXCEPT, INF-*	Low
RF / interference	Signal quality, jamming, spoofing analysis	SEC-SIG, SEC-SIGLOG, MON-RF, MON-SPAN, NAV-SIG	High
Tracking	Satellite tracking and signal acquisition	RXM-RAWX, RXM-MEASX, RXM-SFRBX	High
Navigation	Standard precision GNSS positioning solution	NAV-PVT, NAV-STATUS, NAV-SIG, NAV-SAT, NAV-CLOCK	Medium
Timing	GNSS time synchronization and pulse accuracy	TIM-TP, NAV-TIMEGPS, NAV-TIMEUTC, TIM-SVIN	Low

Table 42: Recommended message groups per use case

☞ Only enable the message sets required for the specific use case. This keeps the system easier to manage and helps avoid unnecessary bandwidth usage. This is especially important in applications with limited buffer space or constrained network capacity. Selective message logging ensures optimal performance and cleaner diagnostic data.

4 Hardware integration

This chapter explains how the receiver can be integrated into an application design.

4.1 Power supply

MAX-M10N has the following power supply pins: VCC, V_IO and V_BCKP.

Power supply at VCC and V_IO must be present for normal operation. These two pins can either be connected together or supplied independently by the application. Power supply at V_BCKP is optional. If present, it enables the hardware backup mode when V_IO and VCC supplies are off.

The 3.3 V and 1.8 V operating voltages allow different combinations for the power supply design. These are listed in Supply design examples.

Refer to the MAX-M10N Data sheet [1] for absolute maximum ratings, operating conditions, and power requirements.

4.1.1 VCC

VCC provides power to the core and RF domains and must be supplied during normal operation. A filtered VCC supply is available on the VCC_RF pin. The VCC_RF output voltage is derived from the VCC supply and is available whenever VCC is supplied.

☞ Do not add series resistance greater than 0.2 Ω on the supply line to avoid voltage ripple due to the dynamic current conditions.

4.1.2 V_IO

V_IO supplies all the digital IOs, clock, and the backup domain. The current drawn at V_IO depends on the activity and loading of the PIOs and the main oscillator.

A power interruption at V_IO will erase the battery-backed RAM (BBR) unless there is an external supply connected to V_BCKP.

V_IO allows two voltage ranges, 1.8 V or 3.3 V operation. For 1.8 V designs, the VIO_SEL pin must be connected to GND. For 3.3 V designs, it must be left open.

4.1.3 V_BCKP

Power supply at V_BCKP is optional. If the power supply at V_IO is interrupted, but the V_BCKP pin is supplied, the receiver enters the hardware backup mode. In this mode, the RTC time and the GNSS orbit data in the BBR are maintained. Valid time and GNSS orbit data at startup improves positioning performance by enabling hot starts, warm starts, and AssistNow Autonomous. This ensures faster TTFF when V_IO is supplied again. To make these features available, connect an independent power supply to V_BCKP to ensure backup domain supply when V_IO is not supplied.

Designs using an external battery as a power source at the V_BCKP pin must consider the battery capacity. That is, the GNSS satellite ephemeris data is typically valid for up to 4 hours for hot starts. Furthermore, for products supporting AssistNow Offline and Autonomous, the assistance data is valid up to few days for warm starts .

☞ Avoid high resistance on the V_BCKP line. During the switch to V_BCKP supply, a short current adjustment peak may cause a high voltage drop at the pin.

☞ If the hardware backup mode is not used, leave the V_BCKP pin open.

4.1.4 Supply design examples

The two voltage ranges for V_IO allow several combinations when designing the receiver power supply. Depending on the chosen combination, there are certain requirements to be considered. These are summarized in Table 43.

Option	Nominal supply (V)		Design case / Requirements
	V_IO	VCC	
1	3.3	3.3	3.3 V design where VCC and V_IO are connected together. See Figure 17 for designs using the hardware backup mode, and Figure 18 for designs without backup supply. • VIO_SEL pin left open. • Voltage at VCC_RF pin = VCC - 0.1 V.
2	1.8	1.8	1.8 V design with VCC and V_IO connected together. This design requires an accurate supply. See Figure 17 for designs using the hardware backup mode, and Figure 18 for designs without backup supply. • VIO_SEL is grounded. • Note that the voltage output at VCC_RF = VCC - 0.1 V. • Note that the maximum supply tolerance is 1.8 V ± 2 %.
3	1.8 / 3.3	1.8 / 3.3 / 5.5	VCC and V_IO are supplied independently. V_IO is supplied with a 1.8 V or 3.3 V supply. VCC can be either supplied with 1.8 V, 3.3 V, or 5.5 V. For designs using the hardware backup mode, see Figure 19, and for designs without backup supply, see Figure 20. • VIO_SEL pin is grounded (V_IO = 1.8 V) or open (V_IO = 3.3 V). • Voltage at VCC_RF pin = VCC - 0.1 V. • For V_IO = 1.8 V, the maximum V_IO supply tolerance is ± 2 %.

Table 43: Voltage supply options

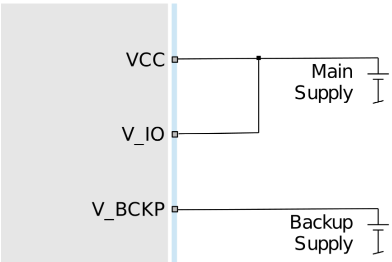


Figure 17: VCC and V_IO connected to the main supply, and external power supply at V_BCKP

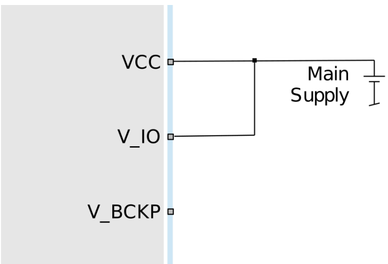


Figure 18: VCC and V_IO connected to the main supply. No external power supply at V_BCKP.

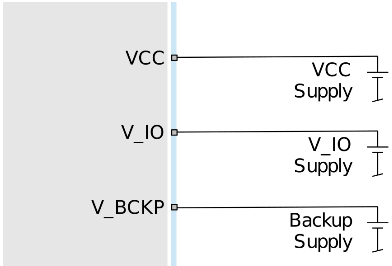


Figure 19: VCC and V_IO supplied by separate supplies, and external power supply at V_BCKP

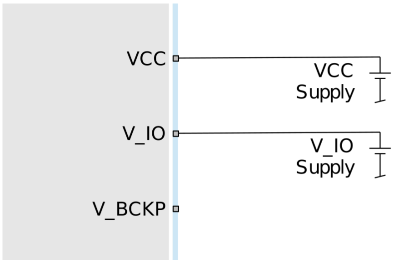


Figure 20: VCC and V_IO supplied by separate supplies. No external power supply at V_BCKP.

4.2 RF interference

The GNSS signal power received at the antenna is very low compared to other wireless communication signals. The received nominal –130 dBm GNSS signal strength makes the GNSS receiver susceptible to interference from any kind of nearby RF sources.

As an example, cellular applications emit signals with power levels of approximately +30 dBm, while the GNSS signal is less than –130 dBm when reaching the antenna. By simply comparing these numbers, it is obvious that interference issues must be seriously considered during the design phase.

4.2.1 In-band interference

Although the radio communications standards prevent intentional RF signal sources from interfering the GNSS frequencies, many devices emit RF power into the GNSS band at levels much higher than the GNSS signal itself.

One reason is that the frequency band above 1 GHz is not well regulated with regards to EMI, and even if permitted, signal levels are much higher than the GNSS signal power. In particular, all types of digital equipment, such as PCs, digital cameras, LCD screens, etc. tend to emit a broad frequency spectrum up to several GHz of frequency. Also wireless transmitters may generate spurious emissions that fall into the GNSS band.

☞ The Layout section defines measures against in-band interference during the design phase of the application.

4.2.2 Out-of-band interference

Out-of-band interference is caused by signal frequencies that are different from the GNSS carrier frequency. The main sources are wireless communication systems such as LTE, GSM, CDMA, WCDMA, Wi-Fi, BT, etc. Typically, these systems may emit their specified maximum transmit power in close proximity to the GNSS receiving antenna, especially if such a system is integrated with the GNSS receiver. Even at reasonable antenna selectivity, destructive power levels may reach the RF input of the GNSS receiver. In addition, larger signal interferers may generate intermodulation products inside the GNSS receiver front-end that fall into the GNSS band and contribute to in-band interference.

Measures against out-of-band interference include maintaining a good grounding concept in the design and adding a GNSS band-pass filter into the antenna input line to the receiver.

☞ The sections Out-of-band blocking immunity and Out-of-band rejection provide more information about the RF immunity of the MAX-M10N module and mitigating out-of-band interference.

4.2.3 Spectrum analyzer

The UBX-MON-SPAN message can be enabled in u-center 2 to provide a low-resolution spectrum analyzer sufficient to identify noise or jammers in the reception band. Once enabled, u-center 2 includes a real-time chart that is updated once per second with the message data. See Figure 21 for an example.

The design or device environment can generate interference at the in-band that can be analyzed from the spectrum in the UBX-MON-SPAN message. Hence, the shape of the spectrum as well as visible peaks help to identify in-band interference. Out-of-band interference can also cause peaks that appear in the in-band. However, there can be out-of-band interference that is not visible within the span of the spectrum. The presence of out-of-band interference may be seen as reduction in the PGA value.

The vertical axis compares the power level in dB for each frequency. A good spectrum shape is characterized by an even noise floor along with the GNSS band. For example, if any unwanted interference peak stands out, the vertical axis gives a rough approximation of the power level in dB compared to the noise floor.

Next to the chart, the center frequency, span, and resolution values set for the spectrum, and the PGA value are also displayed. The PGA value represents the internal gain set by the receiver, which depends on the external amplification of the GNSS input signal.

The vertical discontinuous lines in the chart area represent the offset to the center frequency in MHz. This helps to estimate the frequency of any spurious emission seen.

In addition, u-center 2 includes three functions commonly found in any spectrum analyzer. These features support the RF front-end design and help to spot out any jammer present during the application operation.

- Hold: if selected, the current spectrum shape freezes in a colored line. This allows for a comparison between the time the spectrum was frozen and the real-time spectrum. This is particularly helpful in assessing the impact of running other onboard components.
- Average: if selected, a colored line shows the averaged spectrum for each frequency. This supports the analysis over time and obtaining a less noisy shape.
- Max hold: if selected, a colored line shows the maximum amplitude measured at each frequency. This option helps to spot out any jammer over a period of time.

Figure 21 shows the spectrum view in u-center 2 with the hold, average and max hold options selected. The green, yellow and red lines represent the frozen hold, average and max hold spectrums, while the blue line represents the current continuous spectrum.

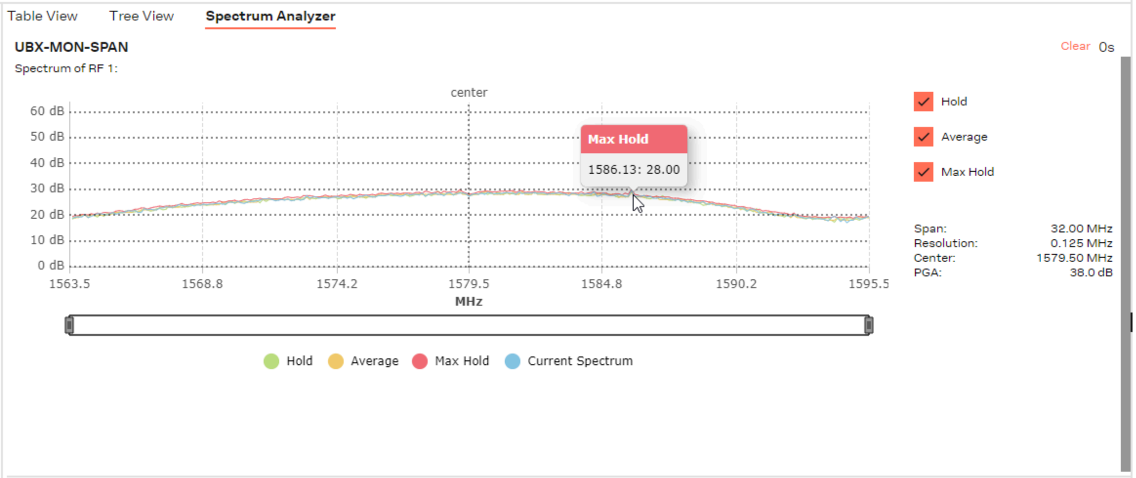


Figure 21: Spectrum analyzer view in u-center 2

☞ By changing the enabled GNSS constellations, the span widens or narrows. This has a direct impact on the spectrum resolution, as the number of measured values is fixed to 256. For further details about this message and how to calculate each frequency, see the Interface description [3].

☞ A peak may be visible around the center frequency. The signal comes internally from the receiver and it does not cause any degradation in the performance.

4.3 RF front-end

GNSS receivers operate with very low signal levels, ranging from –130 dBm to approximately –167 dBm. This alone is a challenge for the GNSS application design. Out-of-band sources of interference such as GSM, CDMA, WCDMA, LTE, Wi-Fi, or Bluetooth wireless systems with a much higher signal level require additional specific measures. The goal of the RF front-end design is to receive the in-band signal with minimum loss and added noise while suppressing the out-of-band interference.

The MAX-M10N module has two RF front-end variants, MAX-M10N-**00B** (LNA-SAW) and MAX-M10N-**10B** (SAW-LNA-SAW). Select the appropriate variant based on the design and application requirements.

MAX-M10N-**00B** RF front-end is designed for the highest sensitivity. The integrated RF circuit includes an LNA and a SAW filter on the RF path, allowing the use of passive antenna. The RF input is matched to 50 Ω and includes a built-in DC block.

MAX-M10N-**10B** RF front-end is designed for the highest out-of-band immunity performance and it is suitable for cellular applications. The integrated RF circuit includes a SAW filter, LTE Band 13 notch filter, an LNA, and a SAW filter. The RF input is matched to 50 Ω and includes a built-in DC block.

Refer to the Block diagram for an overview of the RF front-end.

4.3.1 Internal LNA modes

In addition to the integrated LNA in the RF front-end circuit in MAX-M10N, there is also an internal LNA in the u-blox M10 receiver.

The receiver's internal LNA has three operating modes: normal gain, low gain, and bypass mode.

- By default, the internal LNA is configured for the low-gain mode for optimized sensitivity and immunity against RF interference.
- For RF front-end designs with 10 - 15 dB or higher total external gain, bypass mode is recommended to improve immunity. The power consumption is also slightly reduced in bypass mode.
- Normal-gain mode is not recommended for MAX-M10N.

The internal LNA mode can be configured at run time in the RAM, BBR and flash layers using the CFG-HW-RF_LNA_MODE configuration item and applying a reset. Once in production, the flash layer can be configured to automatically apply the configuration at every startup.

For information on RF parameters, refer to the MAX-M10N Data sheet [1].

4.3.2 Out-of-band blocking immunity

Out-of-band RF interference may degrade the quality and availability of the navigation solution. Out-of-band immunity limit describes the maximum power allowed at the receiver RF input with no degradation in performance. Minor violation of the immunity limit may reduce C/N0 of the received signals but does not necessarily affect the overall receiver performance. However, a significant violation may reduce receiver sensitivity or cause a complete loss of signal reception. The severity of the interference depends on the repetition rate, frequency, signal level, modulation, and bandwidth of the signal.

Figure 22 shows a typical out-of-band immunity level at the MAX-M10N RF input. The internal LNA is in the low-gain mode (default). The measurement has been done at room temperature using a test signal with 64QAM modulation and 10 MHz bandwidth similar to an LTE signal.

In MAX-M10N-00B, the out-of-band compression point of the LNA at the module RF input largely determines the immunity. In addition, the immunity typically reduces toward the receiver's in-band. At 500 MHz and 800 MHz ranges, the harmonic multiples generated at the integrated LNA input falling at the receiver's in-band reduce the immunity.

In MAX-M10N-10B, the additional SAW filter protects the integrated LNA. This both prevents the LNA to be driven in saturation and suppresses the harmonic generation. The Band 13 notch filter further suppresses the harmonic generation at the 800 MHz region. The additional filtering significantly improves the overall immunity of the receiver.

☞ If the out-of-band immunity limit is exceeded, it is recommended to verify that the receiver performance is not affected or is at an acceptable level in the presence of interference.

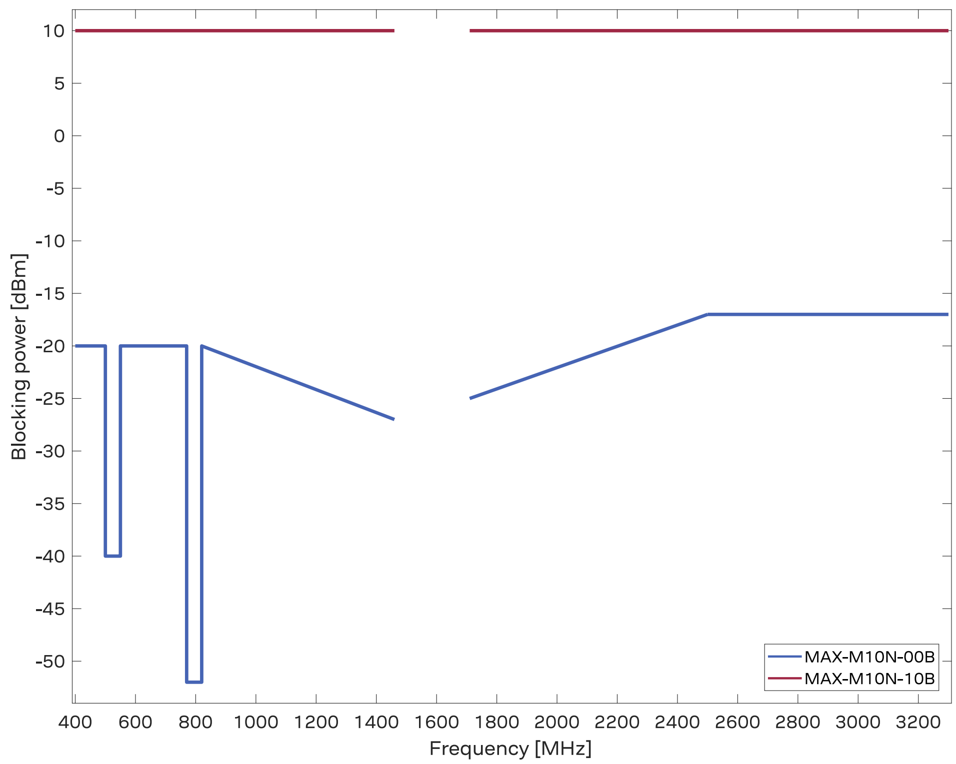


Figure 22: MAX-M10N-00B and MAX-M10N-10B out-of-band immunity level at 400–1460 MHz and 1710–3300 MHz for the low-gain mode (default).

Table 44 shows tabulated values for out-of-band immunity at selected cellular and Wi-Fi frequencies.

Parameter									
Frequency (MHz)	699	785	915	1710	1880	1980	2350	2440	2690
Immunity level (dBm)									
MAX-M10N-00B	−20	−52	−21	−25	−23	−22	−19	−18	−17
MAX-M10N-10B	+10	+10	+10	+10	+10	+10	+10	+10	+10

Table 44: MAX-M10N out-of-band immunity for the low-gain mode at selected frequencies.

4.3.3 Out-of-band rejection

RF interference is typically first coupled into the antenna and subsequently conducted into the receiver input. Typical out-of-band interference sources include transmitting antennas of other radio systems.

Estimation of the RF interference level coupled into the receiver antenna is a starting point for RF front-end design. For designs with other radio systems, the maximum power coupled into the antenna can be estimated from the maximum transmission power and the isolation between the antennas. Practical values for antenna isolation can range from 15 - 20 dB down to 6 - 10 dB for very small devices. RF interference may also couple from external sources such as nearby mobile devices or base stations.

☞ A simplified test board can be used to estimate the isolation between two antennas. The size of the board and the placement of the antennas must match the final design. Connect the RF cables to the antenna inputs and measure S21 over the frequency band of interest with a vector network analyzer (VNA).

The required out-of-band rejection or isolation is the difference of the maximum power coupled into the antenna input terminal and the immunity level of the receiver RF input. Select the appropriate MAX-M10N variant based on the required isolation in the design and application.

RF interference from other parts of the design is more difficult to estimate. One option is to measure the interference level at the receiver input using a spectrum analyzer. Interference within the design is primarily a problem at the receiver in-band, where it cannot be addressed by filtering on the RF path. Outside the GNSS band, the SAW-LNA-SAW variant (MAX-M10N-**10B**) provides better out-of-band rejection compared to the LNA-SAW variant (MAX-M10N-**00B**) variant.

4.3.4 Antenna power supply

Figure 23 shows an active antenna supply network to connect the antenna supply to the RF signal line. The inductance L3 connects the antenna power supply to the RF signal line. The capacitance C14 filters out high-frequency interference from the power supply and the resistor R8 limits the short-circuit current.

The type and value of L3 is selected to have a resonance peak at GNSS frequencies. This provides a high series impedance above 500 Ω at GNSS L1 frequencies, creating an impedance mismatch with respect to the 50 Ω RF signal line. This minimizes the effect of the feed point on the RF signal line, and isolates the antenna supply from the RF signal line at GNSS frequencies. Both R8 and L3 must have sufficient current and power rating to withstand the short-circuit current. Example component values for the antenna supply network are given in Standard resistors, Standard capacitors, and Inductors.

The VCC_RF pin can be used to supply an active antenna. VCC_RF is a RF filtered supply voltage derived from the VCC supply. Refer to the Data sheet[1] for the VCC_RF specification.

The Antenna supervisor can be used to detect open and short circuits on the antenna supply network and disconnect the antenna supply if a short circuit is detected.

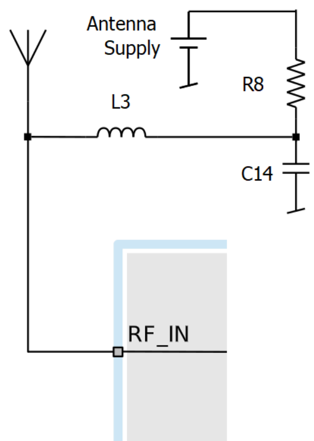


Figure 23: Antenna supply network

4.4 Layout

GNSS signals on the surface of the earth have a very low signal strength and are about 15 dB below the thermal noise floor. When integrating a GNSS receiver into a PCB, the placement of the components, as well as grounding, shielding, and interference from other digital devices are crucial issues that need to be considered very carefully.

An important factor in achieving high GNSS performance is the placement of the receiver with respect to other components on the PCB.

To minimize signal loss on the RF connection from the antenna to the receiver input and to avoid possible coupled interference, the connection to the antenna must be kept short while keeping some distance from the antenna to other electronic components.

The RF section should not be subject to noisy digital supply currents running through its GND plane. Make sure that critical RF circuits are clearly separated from any other digital circuits on the system board. To achieve this, position the receiver digital part towards the digital section of the system PCB and place the RF section and antenna as far away as possible from the other digital circuits on the board. Keep at least a 5 mm distance to any RF component and ensure proper grounding.

☞ For applications using cellular antennas, increase the distance between both antennas as much as possible.

Another very important factor in GNSS applications is the grounding concept. Ensure good ground reference to the host ground by increasing the number of GND vias. The GND vias will improve the GND reference between all the layers, and the pads will serve as thermal relief.

Any stubs at the ground planes must be avoided or ended with a via to the reference ground. Otherwise, they could pick up and propagate interference.

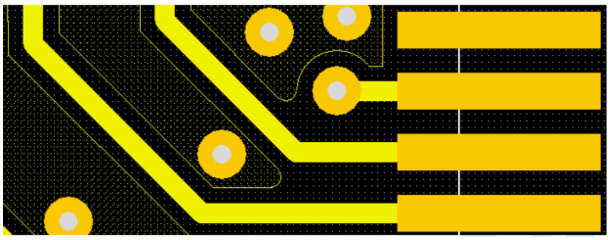


Figure 24: GND stub ended with a via

It is recommended to ground the area below the module, on the top and second layer. Avoid signal lines crossing below the module at these two layers.

For the RF signal line, it is best to use the co-planar waveguide with ground on the second layer. All the RF parts need a solid GND plane underneath in order to achieve the targeted impedance in the RF signal line.

The length and geometry in the RF signal line must be carefully analyzed. The impedance of the RF signal line must be 50 Ω. Select the stack-up, copper, and dielectric properties of the PCB accordingly to fulfill this condition. The RF signal line should be as short as possible and the ground plane around should be filled with GND vias.

Avoid placing the receiver in the proximity of cooling fans or heat-emitting components, such as power devices. Temperature-sensitive components, such as the receiver oscillator, are sensitive to sudden changes in ambient temperature, which can adversely impact satellite signal tracking. Heating and cooling sources can include co-located power devices, cooling fans or thermal conduction via the PCB.

The GND planes can conduct heat to other elements, but they can act as heat dissipators as well. Increasing the number of GND vias helps to decrease sudden temperature changes.

⚠ If needed, shield the receiver with temperature-sensitive components to reduce air convection and improve thermal stability.

4.4.1 Package footprint, copper and solder mask

The mechanical specification is available in the data sheet [1].

Figure 25 and Table 45 describe the footprint. Figure 26 and Table 46 provide recommendations for the paste mask.

☞ The copper and solder masks have the same size and position.

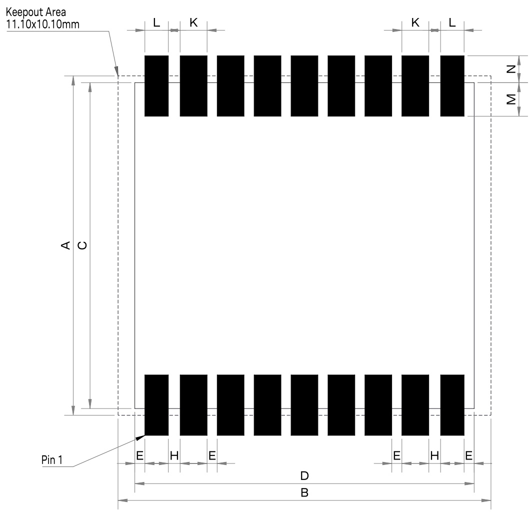


Figure 25: Recommended copper land and solder mask opening for MAX-M10N

Symbol	Dimension (mm)
A	10.1
B	11.1
C	9.7
D	10.1
E	0.3
H	0.35
K	0.8
L	0.7
M	1.0
N	0.8

Table 45: MAX-M10N footprint dimensions

To improve the wetting of the half vias, reduce the amount of solder paste under the module and increase it outside of the module by defining the dimensions of the paste mask to form a T-shape (or equivalent) extending beyond the copper mask.

Recommended stencil thickness is 150 µm.

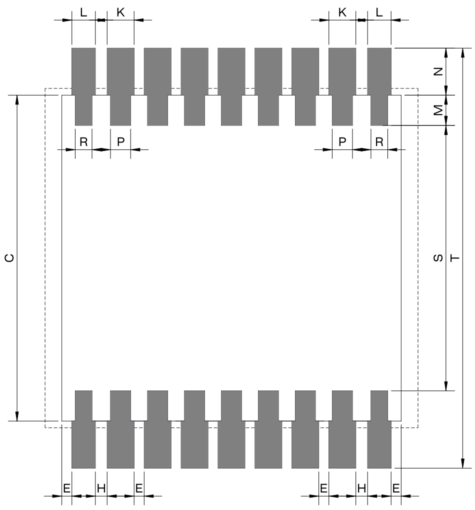


Figure 26: Recommended paste mask pattern for MAX-M10N

Symbol	Dimension (mm)
C	9.7
E	0.3
H	0.35
K	0.8
L	0.7
M	0.9
N	1.4
P	0.6
R	0.5
S	7.9
T	12.5

Table 46: MAX-M10N paste mask dimensions

☞ These are only recommendations and not specifications. The exact geometry, distances, stencil thicknesses and solder paste volumes must be adapted to the customer's specific production processes (for example, soldering).

5 Product handling

5.1 Safety

5.1.1 ESD precautions

⚠ CAUTION! Risk of electrostatic discharge (ESD) damage. u-blox chips and modules are electrostatic sensitive devices containing highly sensitive electronic circuitry. A discharge of static electricity may damage the device or reduce the life expectancy of the device. To avoid ESD damage, adhere to the standard guidelines for handling ESD devices.

Consider the following:

Preventing electrostatic discharge

- Keep components in their original packages during transport.
- Open the package within an ESD-protected area (EPA), as in Figure 27.
- At a workstation, store components in an EPA.
- Place ESD sensitive devices inside of shielding packaging or containers when transported outside of an EPA.
- Use protective clothing and proper personnel grounding at all necessary points when touching electrostatic sensitive device or assembly. For instance, wear ESD-safe clothing and shoes and wear an ESD wrist strap connected to a groundedworkstation. Use heel straps when standing on conductive floors or dissipating floor mats.
- Hold the devices by the edges and avoid touching component contacts, pins, or circuitry

Product handling

- When handling RF transceivers and patch antennas, work in an EPA.
- When connecting test equipment or any other electronics to the module (as a standalone or PCB-mounted device), the first point of contact must always be between the local ground and the PCB ground.
- Before mounting a ceramic patch antenna, connect the device to ground.
- When handling the RF pin, do not touch any charged capacitors. Be especially careful when handling materials like patch antennas (~10 pF), coaxial cables (~50-80 pF/m), soldering irons, or any other materials that can develop charges.
- If there is any risk of touching an exposed antenna area in a non-ESD protected work area, implement proper ESD protection measures in the design.
- When soldering RF connectors and patch antennas to the receiver's RF pin, use an ESD-safe soldering iron (tip)

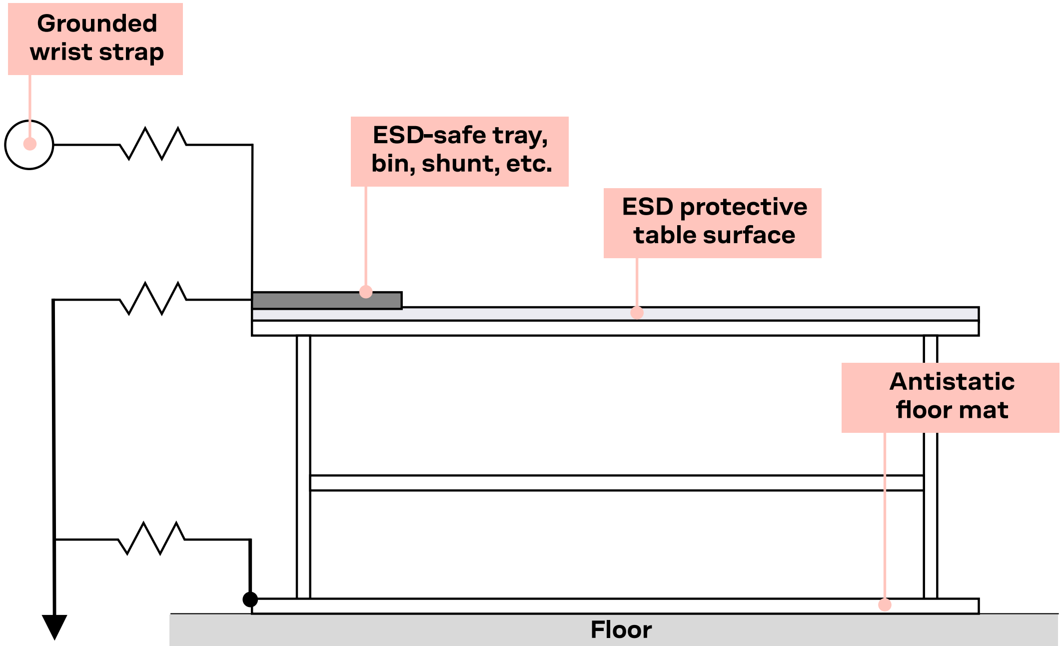


Figure 27: Standard workstation setup for safe handling of ESD-sensitive devices

5.1.2 Safety precautions

The MAX-M10N modules must be supplied by an external limited power source in compliance with the clause 2.5 of the standard IEC 60950-1. In addition to external limited power source, only Separated or Safety Extra-Low Voltage (SELV) circuits are to be connected to the module including interfaces and antennas.

☞ For more information about SELV circuits see section 2.2 in Safety standard IEC 60950-1.

5.2 Soldering

Reflow soldering procedures are described in the IPC/JEDEC J-STD-020 standard [5].

☞ When populating the modules, make sure that the pick and place machine is aligned to the copper pins of the module instead of the module edge.

Soldering paste

Use of “no clean” soldering paste is highly recommended, as it does not require cleaning after the soldering process. For instance, the following paste meets these criteria.

- Soldering paste: OM338 SAC405 / Nr.143714 (Cookson Electronics)
- Alloy specification: Sn 95.5/ Ag 4/ Cu 0.5 (95.5% tin/ 4% silver/ 0.5% copper)
- Melting temperature: 217 °C
- Stencil: The exact geometry, distances, stencil thicknesses and solder paste volumes must be adapted to the customer's specific production processes.

Reflow soldering

⚠ CAUTION. Risk of device damage. Exceeding the peak temperature of the recommended soldering profile may permanently damage the device.

The final soldering temperature chosen at the factory depends on additional external factors such as the choice of soldering paste, size, thickness and properties of the base board, etc.

As a reference, see “IPC-7530 Guidelines for temperature profiling for mass soldering (reflow and wave) processes”, published in 2001.

A convection-type soldering oven is highly recommended over the infrared-type radiation oven. Convection-heated ovens allow precise control of the temperature, and all parts will heat up evenly, regardless of material properties, thickness of components and surface color.

⚠ CAUTION. Risk of device damage. Modules must not be soldered with a damp heat process.

☞ To avoid falling off, the modules should be placed on the topside of the board during soldering.

For the recommended soldering profile and conditions, see Figure 28 and Table 47

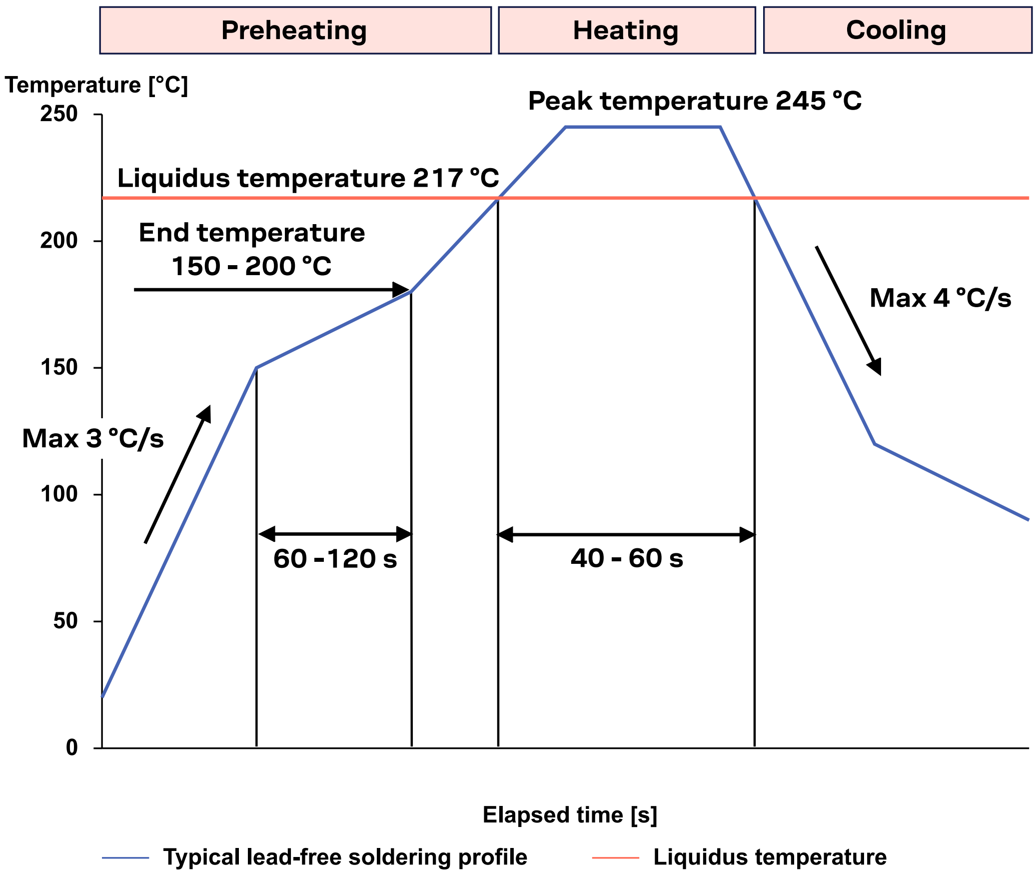


Figure 28: Recommended soldering profile

Phase	Value	Details
Preheating		During the initial heating of component leads and balls, residual humidity is dried out. Note that the preheating phase does not replace prior baking procedures.
Temperature rise rate	Max 3 °C/s	If the temperature rise is too rapid in the preheat phase, excessive slumping may be caused.
Time	60 – 120 s	If the preheating is insufficient, rather large solder balls tend to be generated. Conversely, if performed excessively, fine balls and large balls will be generated in clusters.
End temperature	150 – 200 °C	If the temperature is too low, non-melting tends to be caused in areas containing large heat capacity.
Heating - reflow		

Phase	Value	Details
Time limit above 217 °C liquidus temperature	40 – 60 s	The temperature rises above the liquidus temperature of 217 °C. Avoid a sudden rise in temperature as the slump of the paste could become worse.
Peak reflow temperature	245 °C	
Cooling		
Temperature fall rate	Max 4 °C/s	A controlled cooling prevents negative metallurgical effects of the solder (solder becomes more brittle) and possible mechanical tensions in the products. Controlled cooling helps to achieve bright solder fillets with a good shape and low contact angle.

Table 47: Recommended conditions for reflow soldering

Optical inspection

After soldering the module, consider optical inspection.

Cleaning

☞ Do not clean with water, solvent, or ultrasonic cleaner:

- Cleaning with water leads to capillary effects where water is absorbed into the gap between the baseboard and the module. The combination of residues of soldering flux and encapsulated water leads to short circuits or resistor-like interconnections between neighboring pins.
- Cleaning with alcohol or other organic solvents can result in soldering flux residues flowing underneath the module into areas that are not accessible for post-cleaning inspections. The solvent also damages the sticker and the printed text on the module.

⚠ CAUTION. Risk of device damage. Ultrasonic cleaning permanently damages the module, in particular the quartz oscillators.

The best approach is to use a “no clean” soldering paste to eliminate the cleaning step after the soldering.

Repeated reflow soldering

☞ Repeated reflow soldering processes or soldering the module upside down are not recommended.

A board that is populated with components on both sides may require more than one reflow soldering cycle. In such a case, the process should ensure the module is only placed on the board submitted for a single final upright reflow cycle. A module placed on the underside of the board may detach during a reflow soldering cycle due to lack of adhesion.

The module can also tolerate an additional reflow cycle for rework purposes.

Wave soldering

Base boards with combined through-hole technology (THT) components and surface-mount technology (SMT) devices require wave soldering to solder the THT components. Only a single wave soldering process is encouraged for boards populated with modules.

Rework

⚠ CAUTION. Risk of device damage. Using a hot air gun is an uncontrolled process. It can lead to overheating and severely damage the module. Always avoid overheating the module.

After the module is removed from the oven, clean the pins before reapplying the solder paste, placing the module in the oven and proceeding with the reflow soldering of a new module.

☞ Never attempt to alter the module itself, e.g. by replacing individual components. Such actions immediately void the warranty.

Conformal coating

Certain applications employ a conformal coating of the PCB using HumiSeal® or other related coating products. These materials affect the RF properties of the GNSS module and it is important to prevent them from flowing into the module. The RF shields do not provide 100% protection for the module from coating liquids with low viscosity. Apply the coating carefully.

☞ Conformal coating of the module voids the warranty.

Casting

If casting is required, use viscose or another type of silicon pottant. The OEM is strongly advised to qualify that such processes are suitable for the module before implementing them in the production.

☞ Casting voids the warranty.

Grounding metal covers

Attempts to improve grounding by soldering ground cables, wick or other forms of metal strips directly onto the EMI covers is done at customer's own risk. The numerous ground pins should be sufficient to provide optimum immunity to interferences and noise.

☞ u-blox provides no warranty for damages to the module caused by soldering metal cables or any other forms of metal strips directly onto the EMI covers.

Use of ultrasonic processes

Some components on the module are sensitive to ultrasonic waves.

⚠ CAUTION. Risk of device damage. Use of any ultrasonic processes (cleaning, welding etc.) may cause damage to the receiver.

☞ u-blox provides no warranty against damages to the module caused by ultrasonic processes.

Appendix

A Migration

u-blox is committed to ensuring that products in the same form factor are backwards compatible over several technology generations. This section describes important differences to consider when migrating from the u-blox 7/8/M8/M10 SPG 5.10 to u-blox M10 SPG 5.30.

A.1 Hardware changes

Migration from MAX-M10S

Table 48 lists the key **hardware-related changes** between MAX-M10N and MAX-M10S.

Feature	Change	Action needed / Remarks
VCC supply range	MAX-M10N: • Absolute maximum rating: 6.0 V • Operating conditions: 1.76 V - 5.5 V MAX-M10S: • Absolute maximum rating: 3.6 V • Operating conditions: 1.76 V - 3.6 V	Extended VCC supply range in MAX-M10N.
Software standby current	The software standby current into VCC: MAX-M10N: 3.6 µA MAX-M10S: 120 nA	Check if the changes in current affects target design/ application requirements.
I2C	Not supported	A hardware change is required if I2C is used. The I2C pins 16 and 17 are reserved and have no internal connection.
RF input power, maximum rating	MAX-M10N-10B: • +15 dBm (outband), 0 dBm (inband) MAX-M10N-00B and MAX-M10S: • 0 dBm	
RF front-end	MAX-M10N-10B: • B13 notch-SAW-LNA-SAW • Noise figure 3 dB MAX-M10N-00B: • LNA-SAW • Noise figure 1.5 dB MAX-M10S: • B13 notch-LNA-SAW • Noise figure 1.5 dB	MAX-M10S design with no external SAW filter: • Select MAX-M10N-00B (drop-in replacement) MAX-M10S design with an external SAW filter, B13 protection not required: • Select MAX-M10N-10B, remove external SAW filter • Select MAX-M10N-00B (drop-in replacement) MAX-M10S design with an external SAW filter, B13 protection required: • Select MAX-M10N-10B, remove external SAW filter • Select MAX-M10N-00B, keep external SAW filter and add external B13 notch filter
LNA_EN	The LNA_EN pin is internally connected to the V_IO supply voltage through a transistor switch. LNA_EN supports duty cycling in the LEAP mode.	The drive strength and voltage level of the LNA_EN signal is not identical to the PIOs.
Pin-out	Pin-to-pin compatible with MAX-M10S (I2C pins are reserved).	

Table 48: MAX-M10N hardware features compared to MAX-M10S

Migration from u-blox 7/8/M8

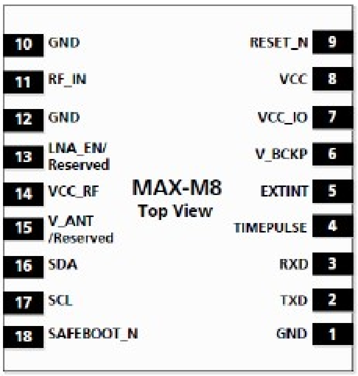
Table 49 lists the key **hardware-related changes** between MAX-M10N and MAX-7/8/M8 modules.

Feature	Change	Action needed / Remarks
Absolute maximum ratings	The absolute maximum voltage for PIOs changed to V_IO + 0.3 V in MAX-M10N.	A HW change is required if main supply voltage source is not within the specified range.
VCC, V_IO	The V_IO voltage range is selected with the VIO_SEL pin.	For designs with 1.8 V supply at V_IO, switch off the V_IO supply 100 ms before VCC when transitioning to the hardware backup mode. Alternatively, put the receiver in the software standby mode by sending the UBX-RXM-PMREQ message before switching off V_IO and VCC. Refer to the minimum and maximum V_IO ramp requirements in the MAX-M10N data sheet [1].
VCC and V_IO supply range	MAX-M10N VCC: 1.76 V - 5.5 V MAX-M10N V_IO: 1.76 V - 1.98 V, 2.7 V - 3.6 V MAX-7C/8C/M8C: 1.65 V - 3.6 V	A HW change is required when migrating from a MAX-7C/8C/M8C design that uses less than 2.7 V VCC and V_IO supply because VIO_SEL (pin 15) needs to be connected to GND in MAX-M10N and the voltage range for 1.8 V designs is different.
V_BCKP supply range	MAX-M10N: 1.65 V - 3.6 V MAX-7/8/M8: 1.4 V - 3.6 V	A HW change is required when migrating from a MAX-7/8/M8 design that uses less than 1.65 V V_BCKP supply.
Hardware backup current	The hardware backup current into V_BCKP in MAX-M10N is higher than in MAX-7Q/7W/8Q/M8Q/W modules. The hardware backup current in MAX-M10N is significantly lower than in MAX-7C/8C/M8C modules.	Check the backup battery capacity if it is still suitable for the target design requirements.
Software standby current	The software standby current into V_IO in MAX-M10N is higher than in MAX-7Q/8Q/M8Q/W modules. MAX-7W has higher software backup current than MAX-M10N. The software standby current in MAX-M10N is significantly lower than in MAX-7C/8C/M8C.	Check if the changes in current affects target design/ application requirements.
I2C	Not supported	A hardware change is required if I2C is used.
Pin-out	1:1 pin-out mapping with MAX-7Q/7C/8Q/8C modules (excluding I2C in MAX-M10N), but not with MAX-7W/M8W.	A HW change is required for MAX-7W/M8W designs that use V_ANT (pin 15) to supply an active antenna, which is VIO_SEL in MAX-M10N.
RF front-end	MAX-M10N-**00B**: LNA-SAW, which includes an LNA and a SAW filter on the RF path, allowing the use of passive antenna. MAX-M10N-**10B**: SAW-LNA-SAW, which includes a SAW filter, band 13 notch filter, an LNA, and a SAW filter, allowing improved out-of-band immunity performance. MAX-M10N-**00B** noise figure: 1.5 dB MAX-M10N-**10B** noise figure: 3 dB MAX-7/8/M8 noise figure: 3.5 dB	External BOM savings are possible if the same components have been added externally on the RF path of the MAX-7/8/M8 design. Use the internal LNA of the receiver in bypass mode when an active antenna is used. The internal LNA mode of the receiver can be configured to low gain and bypass modes because the MAX-M10N module has an LNA on the RF path. This allows flexible optimization of the receiver performance and power consumption with respect to the selected RF front-end/ antenna. The default internal LNA mode on MAX-M10N is set to low gain. Select the appropriate variant based on the target design and application requirements. Check that the external RF path (antenna and filtering) is suitable for all of the GNSS signals in use.

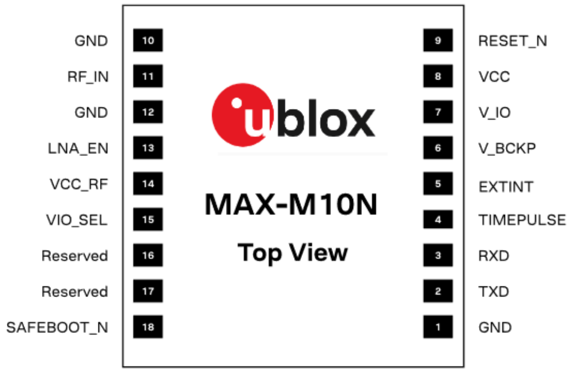
Feature	Change	Action needed / Remarks
Absolute max RF input power	MAX-M10N-**00B**: 0 dBm MAX-M10N-**10B**: +15 dBm for outband; 0 dBm for inband MAX-M8 modules: +15 dBm MAX-7 modules: +13 dBm	Use the SAW-LNA-SAW variant (MAX-M10N-**10B**) for cellular applications.
SAFEBOOT_N, TIMEPULSE	The SAFEBOOT_N pin is internally connected to TIMEPULSE pin through a 1 kΩ series resistor .	Do not drive the TIMEPULSE pin low at startup because it will put the receiver in safeboot mode.
RESET_N	MAX-M10N: clears the RTC time and BBR contents (GNSS orbit data). MAX-7/8Q/M8: clears only the RTC time.	After resetting MAX-M10N with the RESET_N pin, the TTFF is similar to performing a cold start. After resetting the MAX-7/8Q/M8 modules, the TTFF is typically around 6 s in open-sky conditions.
LNA_EN	The LNA_EN pin is internally connected to the V_IO supply voltage through a transistor switch, which serves as a buffer.	The drive strength and voltage level of the LNA_EN signal is not identical to the PIOs.
Digital IO	External IO isolation is required. The output current of the digital IOs is 2 mA compared to 4 mA in MAX-7 and MAX-8Q/M8 modules, except for the TIMEPULSE pin, which still provides 4 mA drive strength.	**Caution!** Risk of receiver damage. If the IO pins are not isolated, the GNSS receiver may get permanently damaged when high current is drawn from the IO pins. Use external IO isolation to avoid receiver damage. Do not drive IOs when VCC and V_IO are not supplied. Removing the VCC or V_IO supply does not isolate the module IO pins from the the host device IO pins. Additional external pull-up resistors may be required to increase the drive strength of the digital IO output pin or adjust the load accordingly.

Table 49: MAX-M10N hardware features compared to MAX-7/8/M8 modules

Migrating from MAX-7W/M8W to MAX-M10N requires special care and may require some re-design because there are changes in the assignment of pins 13 and 15, as shown in Figure 29. Consequently, pin 15 should be left open (i.e. not connected) when migrating from MAX-7W/M8W because there is no built-in antenna supervisor support in MAX-M10N. Therefore, in MAX-M10N, the active antenna supply and antenna supervisor circuitry (if used) need to be connected externally as shown in Antenna supervisor designs.



13	MAX-M8C/Q	LNA_EN	16	O	Antenna control
	MAX-M8W	Reserved	-	-	Reserved
14	All	VCC_RF	-		Output Voltage RF section
15	MAX-M8W	V_ANT	-		Active Antenna Supply Voltage
	MAX-M8C/Q	Reserved	-	-	Reserved



13	LNA_EN	-	O	On/Off external LNA or active antenna
14	VCC_RF	-	O	Output voltage RF section
15	Reserved	-	-	Reserved

Figure 29: MAX-M8 vs. MAX-M10N comparison (pin 13 - 15)

Refer to the product Data sheets for details.

A.2 Firmware changes

Migragtion from u-blox M10 SPG 5.10

Table 51 presents a summary of the key **firmware-related changes** between u-blox M10 SPG 5.30 and u-blox M10 SPG 5.10, as well as actions required during the migration.

Feature	Change	Action needed / Remarks
Signals		
Default GNSS configuration	MAX-M10N: GPS, Galileo, BeiDou B1I, QZSS and SBAS. ROM SPG 5.10: GPS, Galileo, QZSS and SBAS. MAX-M10S: GPS, Galileo, BeiDou B1I, QZSS and SBAS.	Code change (optional)
GLONASS	Not supported	Code change (optional)
SBAS	New SBAS systems supported: BDSBAS (PRN 130, 143-144), KASS (PRN 122), and SouthPan (PRN 134) Default SBAS PRN selection: 122-123, 127-129, 131, 133-137.	Code change (optional)

Feature	Change	Action needed / Remarks
Protocols		
RTCM	RTCM supported, protocol version RTCM 3.4. Supported GNSS systems: GPS, Galileo, BeiDou, QZSS, SBAS.	Code change (optional)
General		
Power-save mode (PSM)	Low energy accurate positioning (LEAP) replaces the power-save mode cyclic tracking (PSMCT) operation. Power-save mode on-off (PSMOO) operation is not supported.	Code change Disable time pulse to enable the LEAP mode.
Ultra-high sensitivity mode	Ultra-high sensitivity mode is disabled by default excluding the weakest signals from the navigation solution. The mode can be enabled.	Code change (optional)
Navigation update rate	Navigation update rate up to 25 Hz for single GNSS, 20 Hz for 2 GNSS, and 10 Hz for 3 GNSS constellations.	Code change (optional)
Features		
AssistNow	AssistNow Online and Offline services are replaced by AssistNow Live Orbits and Predictive Orbits. Simultaneous operation of AssistNow Live Orbits, Predictive Orbits, and Autonomous.	Code change (optional)
Geofecing	Up to 4 circular geofences can be defined.	Code change (optional)
Data logging	Position fixes and arbitrary data from the host can be stored in the flash memory connected to the receiver.	Code change (optional)
Location batching	The maximum number of epochs reduced to 300 (5 min at 1 Hz navigation update rate).	Code change (optional)
Protection level	Protection level is not supported.	Code change (optional)
Security		
Jamming and spoofing detection	Improved jamming and spoofing detection. New interface with the UBX-SEC-SIG and UBX-SEC-SIGLOG messages.	Code change (optional)

Table 50: MAX-M10N (u-blox M10 SPG 5.30) firmware features

Migragtion from u-blox 7/8/M8

Table 51 presents a summary of the key **firmware-related changes** between u-blox M10 SPG 5.30 and u-blox 7/8/M8, as well as actions required during the migration.

Feature	Change	Action needed / Remarks
Signals		
Default GNSS configuration	MAX-M10N: GPS, Galileo, BeiDou B1I, QZSS and SBAS. ROM SPG 5.10: GPS, Galileo, QZSS and SBAS. MAX-M10S: GPS, Galileo, BeiDou B1I, QZSS and SBAS.	Code change (optional) HW change (optional): extend antenna and RF front-end support for the BeiDou B1I signal at 1561.098 MHz.
BeiDou B1I	Supports BeiDou satellite IDs up to 63.	
BeiDou B1C	New signal. Supports BeiDou satellite IDs up to 63. Cannot be used simultaneously with BeiDou B1I. AssistNow is not supported.	Code change (optional)
SBAS	New SBAS systems supported: BDSBAS (PRN 130, 143-144), KASS (PRN 122), and SouthPan (PRN 134). SPG 5.30 default SBAS PRNs: 122-123, 127-129, 131, 133-137.	Code change (optional)
QZSS L1S	SLAS corrections are now applied for navigation.	Code change (optional)
QZSS IMES	Not supported.	Code change (optional)
Protocols		

Feature	Change	Action needed / Remarks
NMEA	Supports NMEA V4.11, V4.10, V4.0, V2.3, and V2.1. NMEA V4.11 is enabled by default. In MAX-M10N, the NMEA GSV messages for zero signal C/N0 level are grouped separately as compared to u-blox 7/8/M8. If the signal's C/N0 level is 0, the FW treats it as an unknown signal for not being able to track it and sets the signal ID to zero. The firmware creates two separate GSV message groups. The first group contains GSV messages for the tracked signals with non-zero C/N0 and the second group contains GSV messages for the untracked signals with zero C/N0. In the untracked signals message group, the C/N0 field is blank (i.e. , ,). There is no configuration to revert this to the u-blox 7/8/M8 format.	Code change (optional)
RTCM	New protocol version RTCM 3.4. Supported GNSS systems: GPS, Galileo, BeiDou, QZSS, SBAS.	Code change (optional)
General		
Configuration concept	New configuration scheme using UBX-CFG-VALSET and UBX-CFG-VALGET messages.	Code change
Navigation update rate	Navigation update rate up to 25 Hz for single GNSS, 20 Hz for 2 GNSS, and 10 Hz for 3 GNSS constellations.	Code change (optional)
Super-S	New feature. Improves performance under weak signal conditions. Enabled by default.	
Power-save mode (PSM)	Low energy accurate positioning (LEAP) replaces the power-save mode cyclic tracking (PSMCT) operation. LEAP period is configured with CFG-RATE configuration group. Time pulse must be disabled to enable the LEAP mode. Power-save mode on-off (PSMOO) operation is not supported.	Code change (new configuration messages)
Altitude limit	The maximum altitude limit increased to 80 000 m.	-
Features		
AssistNow	AssistNow Online and Offline services are replaced by AssistNow Live Orbits and Predictive Orbits. Simultaneous operation of AssistNow Live Orbits, Predictive Orbits, and Autonomous. New AssistNow data download options are available.	Code change (optional)
CloudLocate	New feature. Position calculation in the cloud provided by the u-blox CloudLocate service. Extends the life of energy-constrained IoT applications. Low payload messages supported.	Code change (optional)
Galileo return link message	Galileo search and rescue (SAR) return link message UBX-RXM-RLM.	Code change (optional)
RF spectrum view	New message. UBX-MON-SPAN shows in-band RF spectrum around the GNSS band. This can be used to identify potential in-band RF interference sources in the design.	Code change (optional)
Location batching	New feature that can be used to reduce power consumption by saving navigation solutions on the receiver for up to 5 minutes (at 1 Hz) and polling the solutions afterwards to a host MCU.	Code change (optional)
Security		
Message integrity	New feature. Authentication of data output based on private/public key pair.	Code change (optional)
Unique chip identifier	A unique chip identifier output in boot screen and in the UBX-SEC-UNIQID message.	Code change (optional)
Configuration lock	New security feature that is enabled with CFG-SEC-CFG_LOCK message for locking the receiver configuration.	Code change (optional)

Feature	Change	Action needed / Remarks
Additional changes for u-blox-7 migration	Support for multiple GNSS and new signals: Galileo E1, BeiDou B1I, and BeiDou B1C. Multiple GNSS assistance (MGA) with AssistNow services. Geofencing support. UBX message integrity mechanism added. Spoofing detection added. Power save modes support. Wrist dynamic model added. Odometer to measure traveled distance with support for different user profiles. Time pulse and time mark reference changed from GPS to UTC.	Code change (optional)

Table 51: MAX-M10N firmware features vs. MAX-7/8/M8

For more information on the supported features and messages in the u-blox M10 SPG 5.30 receiver, refer to the SPG 5.30 Release note and Interface description [2], [3].

B Reference designs

The External components section provides the specification and recommendations for the external components that are shown in each reference design.

This section provides some reference designs for typical and antenna supervisor design cases.

B.1 Typical design

Here are the key features for a MAX-M10N typical design:

- VCC and V_IO are connected together to a single supply. In designs with 3.3 V supply, the VIO_SEL pin must be left open, as shown in Figure 30. In designs with 1.8 V supply, the VIO_SEL pin must be connected to GND, as shown in Figure 31.
- V_BCKP supply is optional. If present, the hardware backup mode is supported. This mode maintains the time and GNSS orbit data in the battery-backed RAM memory if the main supply is switched off.
  If there is no backup supply, time aiding with the UBX-MGA-INI-TIME_UTC message (optionally with a timing signal at the EXTINT pin) and the GNSS orbit data from the AssistNow services or stored on the host controller can be used to reduce the TTFF.
- Both passive and active antennas are supported. The integrated LNA provides sufficient gain for passive antennas. An optional active antenna can be supplied from VCC_RF or from an external supply.
- MAX-M10N-**00B** has an integrated RF front-end which includes a low-noise amplifier (LNA) and a SAW filter. It is well suited for designs which require maximum sensitivity.
- MAX-M10N-**10B** has an integrated RF front-end which includes a SAW filter, band 13 notch filter followed by a low-noise amplifier (LNA), and a SAW filter. The SAW-LNA-SAW RF front-end provides improved out-of-band immunity against RF interference, making the MAX-M10N-**10B** module particularly suitable for cellular applications.
- Only UART communication interface is available.
- For an absolute minimum design using UART, other PIOs (RESET_N, EXTINT, TIMEPULSE, SAFEBOOT_N) can be left open.

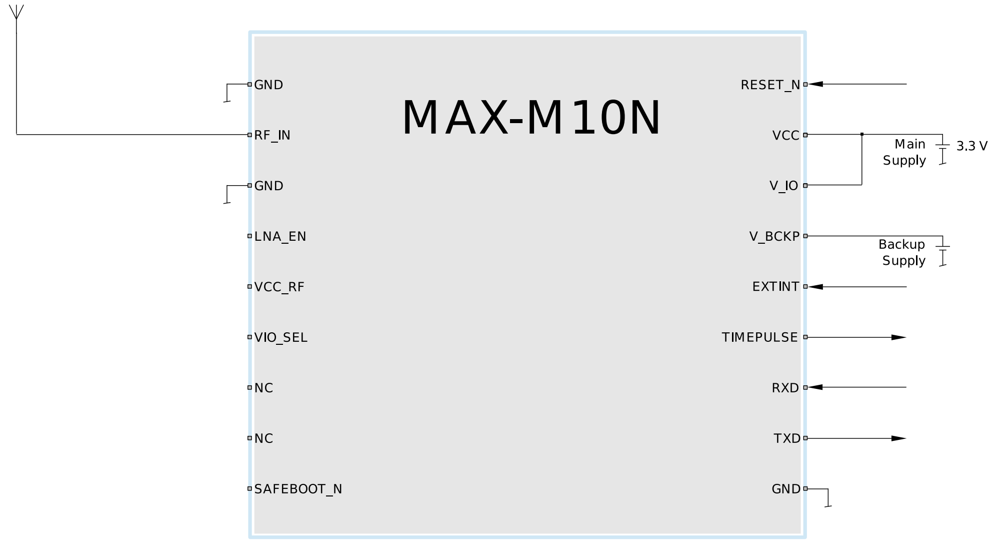


Figure 30: Typical 3.3 V design

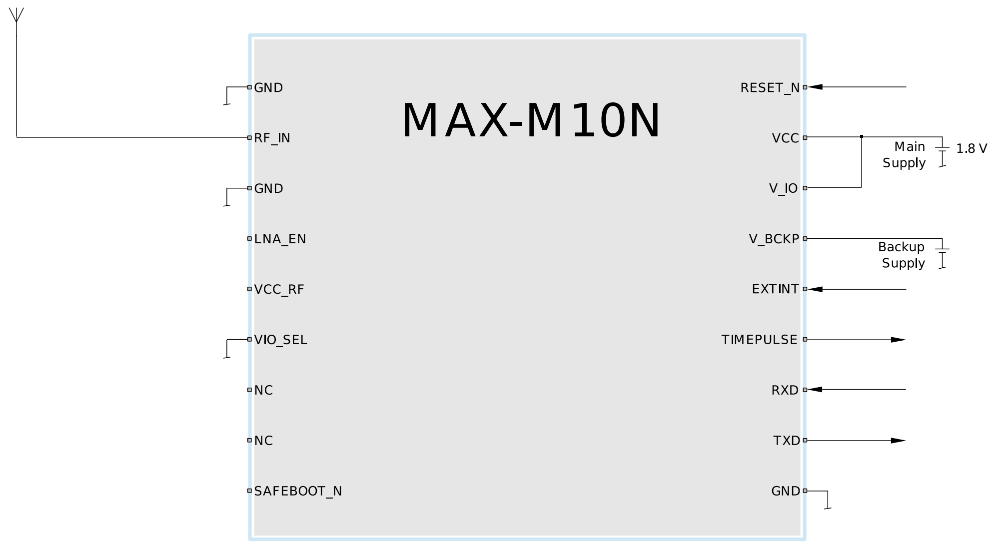


Figure 31: Typical 1.8 V design

B.2 Antenna supervisor designs

Figure 32 and Figure 33 show a reference design for a 2-pin and 3-pin antenna supervisor design respectively. Here are the key features:

- VCC and V_IO are connected together to a single supply.
- Supply at V_BCKP is optional. If present, the hardware backup mode is supported. This mode maintains the time and GNSS orbit data in the battery-backed RAM memory if the main supply is switched off.
  If there is no backup supply, time aiding with the UBX-MGA-INI-TIME_UTC message (optionally with a timing signal at the EXTINT pin) and the GNSS orbit data from the AssistNow services or stored on the host controller can be used to reduce the TTFF.
- MAX-M10N-**00B** includes a LNA-SAW RF front-end for maximum sensitivity designs.
- MAX-M10N-**10B** includes a SAW-LNA-SAW RF front-end for maximum immunity designs.
- An active antenna can be supplied with the VCC_RF output from MAX-M10N or from an external supply. VCC_RF is a filtered output voltage supply, which outputs VCC - 0.1 V. In addition, the active antenna supply can be turned on/off by the LNA_EN signal, which also controls the internal LNA of MAX-M10N.
- External open drain buffers and operational amplifiers are also needed depending on whether a 2-pin or 3-pin antenna supervisor design is used.
- Only UART communication interface is available.

☞ Disable TIMEPULSE function with CFG-TP-TP1_ENA configuration key to be able to use the TIMEPULSE pin for antenna supervisor functions.

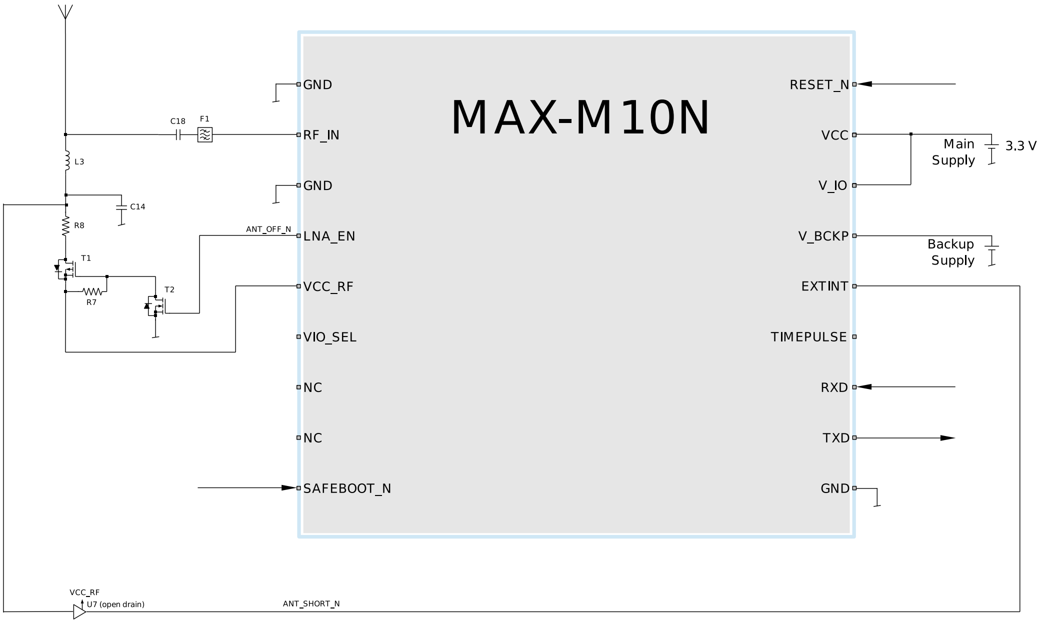


Figure 32: 2-pin antenna supervisor design

Table 52 lists the configuration settings required for the 2-pin antenna supervisor reference design given in Figure 32.

☞ The antenna supervisor switch pin and power down control polarity are pre-configured in the MAX-M10N module.

Configuration key	Value
CFG-HW-ANT_CFG_VOLTCTRL	1 (true), default (no configuration required)
CFG-HW-ANT_CFG_SHORTDET	1 (true)
CFG-HW-ANT_CFG_SHORTDET_POL	1 (true), default (no configuration required)
CFG-HW-ANT_SUP_SHORT_PIN	5
CFG-HW-ANT_CFG_PWRDOWN	1 (true)
CFG-HW-ANT_CFG_RECOVER	1 (true)

Table 52: Configuration for the 2-pin antenna supervisor design

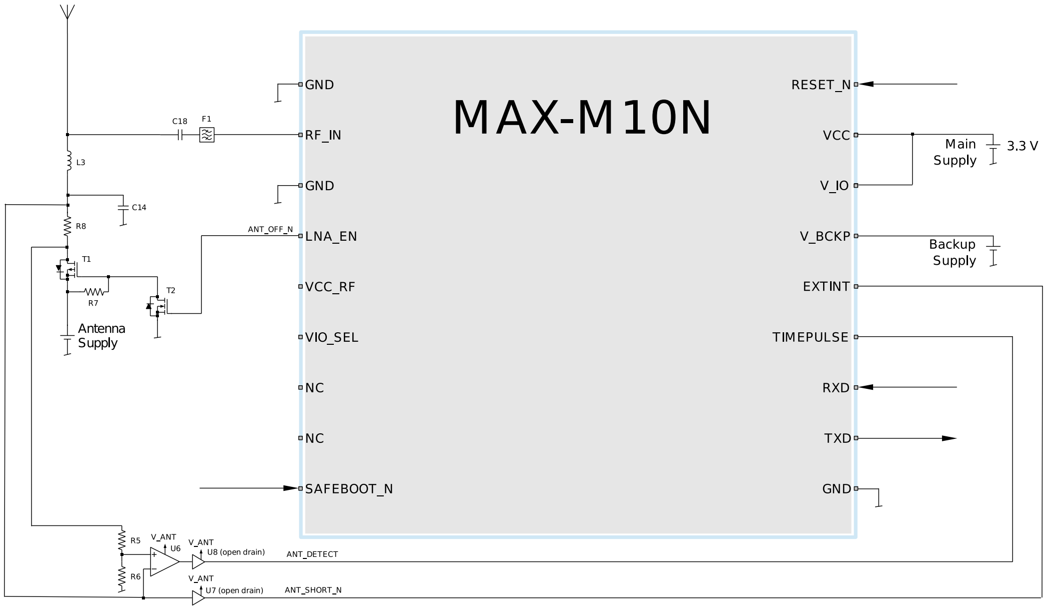


Figure 33: 3-pin antenna supervisor design

Table 53 lists the configuration settings required for the 3-pin antenna supervisor reference design given in Figure 33.

☞ The antenna supervisor switch pin and power down control polarity are pre-configured in the MAX-M10N module.

Configuration key	Value
CFG-TP-TP1_ENA	0 (false)
CFG-HW-ANT_CFG_VOLTCTRL	1 (true), default (no configuration required)
CFG-HW-ANT_CFG_SHORTDET	1 (true)
CFG-HW-ANT_CFG_SHORTDET_POL	1 (true), default (no configuration required)
CFG-HW-ANT_SUP_SHORT_PIN	5
CFG-HW-ANT_CFG_OPENDET	1 (true)
CFG-HW-ANT_CFG_OPENDET_POL	1 (true), default (no configuration required)
CFG-HW-ANT_SUP_OPEN_PIN	4
CFG-HW-ANT_CFG_PWRDOWN	1 (true)
CFG-HW-ANT_CFG_RECOVER	1 (true)

Table 53: Configuration for the 3-pin antenna supervisor design

C External components

This section lists the recommended values for the external components in the reference designs.

C.1 Antenna

Examples of suitable GNSS antennas for u-blox M10 platform are listed in Table 54.

Manufacturer	Order no.	Comments
Taoglas	CGGBP.18.4.A.02	GPS/Galileo/BeiDou/GLONASS 18 x 18 x 4 mm^3 passive
Taoglas	GP.1575.18.4.A.02	GPS/Galileo 18 x 18 x 4 mm^3 passive
Amotech	YDRA-A18-1575	GPS/Galileo 18x18x4 mm^3 passive
Amotech	AGA363913-S0-A1	GPS/Galileo/BeiDou/GLONASS 5 V / 14 mA active
INPAQ	PA1575MZ50J4G	GPS/Galileo 18.4 x 18.4 x 4 mm^3 passive
INPAQ	B3G02G-S3-XX-A	GPS/Galileo/BeiDou/GLONASS 2.7 to 3.9 V / 10 mA active

Table 54: L1 GNSS antennas

C.2 Standard capacitors

Table 55 presents the recommended capacitor values for MAX-M10N.

Name	Use	Type / Value
C14	RF Bias-T capacitor	10 nF, 10%, 16 V, X7R

Table 55: Standard capacitors

C.3 Standard resistors

Table 56 presents the recommended resistor values for MAX-M10N.

Name	Use	Type / Value
R5	Antenna supervisor voltage divider	560 Ω, 5%, 0.1 W
R6	Antenna supervisor voltage divider	100 kΩ, 5%, 0.1 W
R7	Pull-up resistor at antenna supervisor transistor	100 kΩ, 5%, 0.1 W
R8	Antenna supply current limiter/shunt resistor	10 Ω, 5%, 0.25 W

Table 56: Standard resistors

C.4 Inductors

Table 57 presents the recommended inductor values for MAX-M10N.

Name	Use	Type / Value	Recommended component
L3	RF Bias-T inductor	27 nH, 5%	Murata LQG15H, LQW15A series Johanson Technology L-07W series Any other inductor with impedance >500 Ω at GNSS frequency and current rating above 300 mA.

Table 57: Recommended inductors

C.5 Operational amplifier

Name	Manufacturer	Order no.
U6	Linear Technology	LT6000, LT6003

Table 58: Recommended parts list for the operational amplifier

C.6 Open drain buffers

Name	Manufacturer	Order no.
U7, U8	Fairchild	NC7WZ07P6X

Table 59: Recommended parts list for the open drain buffers

C.7 Switch transistors for antenna supervisor

Table 60 presents the recommended switch transistors for MAX-M10N.

Name	Manufacturer	Order no.	Comments
T1, T2	Vishay	Si1016X-T1-GE3	p-channel, n-channel MOSFET

Table 60: Recommended parts list for the antenna supervisor switch transistors

Related documents

[1] MAX-M10N-00B Data sheet, UBXDOC-963802114-13143
MAX-M10N-10B Data sheet, UBXDOC-304424225-18248

[2] u-blox M10 SPG 5.30 Release note, UBXDOC-304424225-20393

[3] u-blox M10 SPG 5.30 Interface description, UBXDOC-304424225-20395

[4] Product packaging reference guide, UBX-14001652

[5] Joint IPC/JEDEC standard, www.jedec.org

[6] AssistNow service, https://support.thingstream.io/

[7] u-center 2 User guide, https://www.u-blox.com/en/info/u-center-2-user-guide

☞ For regular updates to u-blox documentation and to receive product change notifications please register on our homepage https://www.u-blox.com.

Revision history

Revision	Date	Status / comments
R01	10-Oct-2025	Initial release
R02	04-Feb-2026	Content correction on the communication interfaces
R03	23-Apr-2026	Added sections • Optimizing navigation performance in common GNSS challenges • RTCM corrections • Collecting debug logfiles: design-in guidance

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